

Module 1: Body structure, organization and function

★ Clinical anatomy of upper limb and back

• ROLL OF ROTATOR CUFF IN SHOULDER DISLOCATION

- Four of the scapulohumeral muscles (intrinsic shoulder muscles)—supraspinatus, infraspinatus, teres minor, and subscapularis (referred to as the SITS muscles)—are called **rotator cuff muscles** because they form a **musculotendinous rotator cuff** around the **glenohumeral joint**. All except the supraspinatus are **rotators** of the humerus; the supraspinatus, besides being part of the rotator cuff, initiates and assists the deltoid in the first 15° of **abduction** of the arm.
- The tendons of the SITS muscles protect the glenohumeral joint and give it stability. The tonic contraction of the contributing muscles holds the relatively large head of the humerus in the small, shallow glenoid cavity of the scapula during arm movements.
- The **coracoid process of scapula** is used as a bony **landmark** when performing a **brachial plexus block**, and its position is of importance in diagnosing **shoulder dislocations**.
- It may be damaged by injury or disease, resulting in instability of the shoulder joint.
- Trauma or injury of the shoulder produces a fracture—dislocation of the proximal humeral epiphysis because the joint capsule of the glenohumeral joint, reinforced by the rotator cuff, is **stronger** than the epiphysal plate. In severe fractures, the shaft of the humerus is displaced, but the humeral head retains its normal relationship with the glenoid cavity of the scapula.
- **The main stability of the shoulder joint is provided by the musculotendinous rotator cuff**, it strengthens the shoulder joint everywhere, except inferiorly. Consequently, if a person falls when the humerus is abducted, the head of this bone may be levered out of the glenoid fossa of the scapula, producing a **dislocation of the shoulder joint**.
- The **supraspinatus tendon** is the most **commonly** torn part of the rotator cuff, this injury is common in **baseball pitchers**.
- The supraspinatus tendon does not rupture very often in young people because their tendons are usually so strong that they will tear away (avulse) the tip of the greater tubercle of the humerus rather than rupture the tendon.
- The tendon of the supraspinatus is separated from the coracoacromial ligament, the acromion, and the deltoid muscle by the **subacromial bursa**. When this bursa is inflamed (**subacromial bursitis**), **abduction** of the arm is painful.
- **Degenerative tendonitis of the musculotendinous rotator cuff** is a common disease, especially in **older** people.
- **Arterial periarticular anastomosis in the elbow joint for collateral circulation**
 - The vasculature of the hand is characterized by multiple anastomoses between both **radial and ulnar vessels and palmar and dorsal vessels**. Functionally and clinically important periarticular arterial anastomoses that surround the elbow are generally available to all parts of the hand in **all positions** as well as while **performing functions** (gripping or pressing) that might otherwise compromise especially the palmar structures.

- The resulting collateral circulation allows blood to reach the forearm when flexion of the elbow compromises flow through the terminal part of the brachial artery.
- The **proximal radio-ulnar joint** is supplied by the radial portion of the periarticular arterial anastomosis of the elbow joint, blood supply to **carpometacarpal** and **intermetacarpal** are supplied by periarticular arterial anastomoses.
- **ULNAR ARTERY**:-Branches of the ulnar artery arising in the forearm participate in the periarticular anastomoses of the elbow and supply muscles of the medial and central forearm, the common flexor sheath, and the ulnar and median nerves. The **anterior and posterior ulnar recurrent arteries** anastomose with the inferior and superior ulnar collateral arteries, respectively, thereby participating in the periarticular arterial anastomoses of the elbow.
- **RADIAL ARTERY**:- The **radial recurrent artery** participates in the periarticular arterial anastomoses around the elbow by anastomosing with the **radial collateral artery**, a branch of the deep artery of the arm.
 - The **palmar and dorsal carpal branches of the radial artery** participate in the periarticular arterial anastomosis around the wrist by anastomosing with the corresponding branches of the ulnar artery and terminal branches of the anterior and posterior interosseous arteries, forming the palmar and dorsal carpal arches.
- **BRACHIAL ARTERY**:- The main named branches of the brachial artery arising from its medial aspect are the
 - deep artery of the arm
 - superior and inferior ulnar collateral arteries.
- **Deep artery branch of brachial artery** accompanies **radial nerve** along radial groove of humerus, supplying posterior compartment of arm and participating in periarticular arterial anastomosis around elbow joint. The deep artery of the arm is the largest branch of the brachial artery and has the most superior origin. The deep artery terminates by dividing into **middle and radial collateral arteries**, which participate in the periarticular arterial anastomoses around the elbow
- The **superior ulnar collateral artery** arises from the medial aspect of the brachial artery near the middle of the arm and accompanies the **ulnar nerve** posterior to the medial epicondyle of the humerus. Here it anastomoses with the **posterior ulnar recurrent artery and the inferior ulnar collateral artery**, participating in the periarticular arterial anastomoses of the elbow.
- **Difficulties in identifying fracture of carpal bones and their complications.**
 - **Fracture of scaphoid**:-The scaphoid is the most frequently fractured carpal bone.
 - It often results from a fall on the palm when the hand is **abducted**, the fracture occur across the **narrow part of the scaphoid**.
 - Pain occurs primarily on the **lateral** side of the wrist, especially during **dorsiflexion and abduction** of the hand.
 - Initial radiographs of the wrist may not reveal a fracture; often this injury is **(mis)diagnosed** as a **severely sprained wrist**.
 - Radiographs taken 10-14 days later reveal a fracture because bone resorption has occurred there. Owing to the **poor blood supply** to the proximal part of the

scaphoid, union of the fractured parts may take at least 3 months. **Avascular necrosis** of the proximal fragment of the scaphoid may occur and produce degenerative joint disease of the wrist.

- necessary to fuse the carpals surgically (**arthrodesis**).
- **Fracture of the hamate:-** may result in non-union of the fractured bony parts because of the traction produced by the attached muscles.
 - The **ulnar nerve** is close to the hook of the hamate, the nerve may be injured by this fracture, causing **decreased grip strength** of the hand.
 - The **ulnar artery** may also be damaged when the hamate is fractured.
- **Bone related nerves of the upper limb**
 - **Axillary nerve around surgical neck of humerus:-** The deltoid atrophies when the axillary nerve (C5 and C6) is severely damaged. Because it passes inferior to the humeral head and winds around the surgical neck of the humerus, the axillary nerve is usually injured during fracture of this part of the humerus. It may also be damaged during dislocation of the glenohumeral joint and by compression from the incorrect use of crutches.
 - The deltoid is a common site for the intramuscular injection of drugs. The axillary nerve runs transversely under cover of the deltoid at the level of the surgical neck of the humerus. Awareness of its location also avoids injury to it during surgical approaches to the shoulder.
 - The axillary nerve may be injured when the glenohumeral joint dislocates because of its close relation to the inferior part of the joint capsule. Axillary nerve injury is indicated by
 - paralysis of the deltoid--manifest as an inability to abduct the arm to or above the horizontal level.
 - As the deltoid atrophies, the rounded contour of the shoulder is flattened compared to the uninjured side. This gives the shoulder a **flattened appearance** and produces a slight **hollow** inferior to the acromion.
 - loss of sensation may occur over the lateral side of the proximal part of the arm, the area supplied by the superior lateral cutaneous nerve of the arm, the cutaneous branch of the axillary nerve.
 - **Deltoid tuberosity related with radial nerve:-** The shaft of the humerus has two prominent features: the **deltoid tuberosity** laterally, for attachment of the deltoid muscle, and the oblique **radial groove** (groove for radial nerve, spiral groove) posteriorly, in which the radial nerve and deep artery of the arm lie as they pass anterior to the long head and between the medial and the lateral heads of the triceps brachii muscle.
 - Anterior to the lateral epicondyle, the radial nerve then divides into deep and superficial branches.
 - The **deep branch** of the radial nerve is entirely muscular and articular in its distribution.
 - The **superficial branch** of the radial nerve is entirely cutaneous in its distribution, supplying sensation to the dorsum of the hand and fingers.

- **Injury to Radial Nerve in Arm**
 - Injury to the radial nerve **superior to the origin** of its branches to the **triceps brachii** results in **paralysis** of the triceps, brachioradialis, supinator, and extensor muscles of the wrist and fingers. **Loss of sensation** in areas of skin supplied by this nerve also occurs.
 - **Midhumeral fracture** when the nerve is **injured in the radial groove in humeral shaft**, the triceps is usually not completely paralyzed but only weakened because of high origin of the nerves to two of its three heads of triceps **only the medial head is affected**; however, the muscles in the posterior compartment of the forearm that are supplied by more distal branches of the nerve are paralyzed.
 - The **characteristic clinical sign of radial nerve injury** is **wrist-drop**—inability to extend the wrist and the fingers at the metacarpophalangeal joints. Instead, the relaxed wrist assumes a partly flexed position owing to unopposed tonus of flexor muscles and gravity
- **Medial epicondyle of humerus related with ulnar nerve:-** The ulnar nerve passes posterior to the medial epicondyle and medial to the olecranon to enter the forearm. Posterior to the medial epicondyle, where the ulnar nerve is referred to in lay terms as the “funny bone,” it is superficial, easily palpable, and vulnerable to injury.
 - More than 27% of nerve lesions of the upper limb affect the ulnar nerve. Ulnar nerve injuries usually occur in four places: (1) posterior to the medial epicondyle of the humerus, (2) in the cubital tunnel, (3) at the wrist, and (4) in the hand.
 - Ulnar nerve injury occurs **most commonly** where the nerve passes posterior to the **medial epicondyle** of the humerus. The injury results when the medial part of the elbow hits a hard surface, fracturing the medial epicondyle (“funny bone”).
 - Any lesion **superior to the medial epicondyle** will produce paresthesia (numbness and tingling) of the median part of the dorsum of the hand. Check by Plucking your ulnar nerve at the posterior aspect of your elbow
 - Ulnar nerve injury can result in **extensive motor and sensory loss to the hand**. An injury to the nerve in the distal part of the forearm denervates most intrinsic hand muscles.
 - Power of wrist **adduction** is impaired,
 - when an attempt is made to **flex** the wrist joint, the hand is drawn to the **lateral** side by the FCR (supplied by the median nerve) in the **absence of the “balance”** provided by the FCU.
 - **difficulty making a fist** because, in the absence of opposition, the metacarpophalangeal joints become hyperextended, and he or she cannot flex the 4th and 5th digits at the distal interphalangeal joints when trying to make a fist.
 - the person **cannot extend** the interphalangeal joints when trying to straighten the fingers. This characteristic appearance of the hand, resulting from a distal lesion of the ulnar nerve, is known as **claw hand**. The deformity results from

atrophy of the interosseous muscles of the hand supplied by the ulnar nerve.

The claw is produced by the unopposed action of the extensors and FDP.

- Compression of the ulnar nerve at the elbow (cubital tunnel syndrome) is also common. Uncommonly, the ulnar nerve is compressed as it passes through the ulnar groove.

Cubital Tunnel Syndrome

- The ulnar nerve may be compressed in the cubital tunnel formed by the tendinous arch joining the humeral and ulnar heads of attachment of the FCU . The signs and symptoms of cubital tunnel syndrome are the same as an ulnar nerve lesion in the **ulnar groove on the posterior aspect of the medial epicondyle of the humerus**.
- **Carpal tunnel related with median nerve:-**median nerve passes deep to flexor retinaculum (transverse carpal ligament) as it passes through carpal tunnel to hand.
 - A **cystic swelling** of the **common flexor synovial sheath** on the anterior aspect of the wrist can enlarge enough to produce compression of the median nerve by **narrowing the carpal tunnel** (carpal tunnel syndrome). This syndrome produces **pain and paresthesia** in the sensory distribution of the median nerve and clumsiness of finger movements
 - When the median nerve is severed in the **elbow region**,
 - flexion of the proximal interphalangeal joints of the 1st-3rd digits is lost and flexion of the 4th and 5th digits is weakened. Flexion of the distal interphalangeal joints of the 2nd and 3rd digits is also lost.
 - Flexion of the distal interphalangeal joints of the 4th and 5th digits is not affected because the medial part of the FDP, which produces these movements, is supplied by the ulnar nerve.
 - The ability to flex the metacarpophalangeal joints of the 2nd and 3rd digits is affected because the digital branches of the median nerve supply the 1st and 2nd lumbricals. Thus, when the person attempts to make a fist, the 2nd and 3rd fingers remain partially extended (“**hand of benediction**”).
 - Thenar muscle function (function of the muscles at the base of the thumb) is also lost, as in carpal tunnel syndrome.

The Brachial Plexus

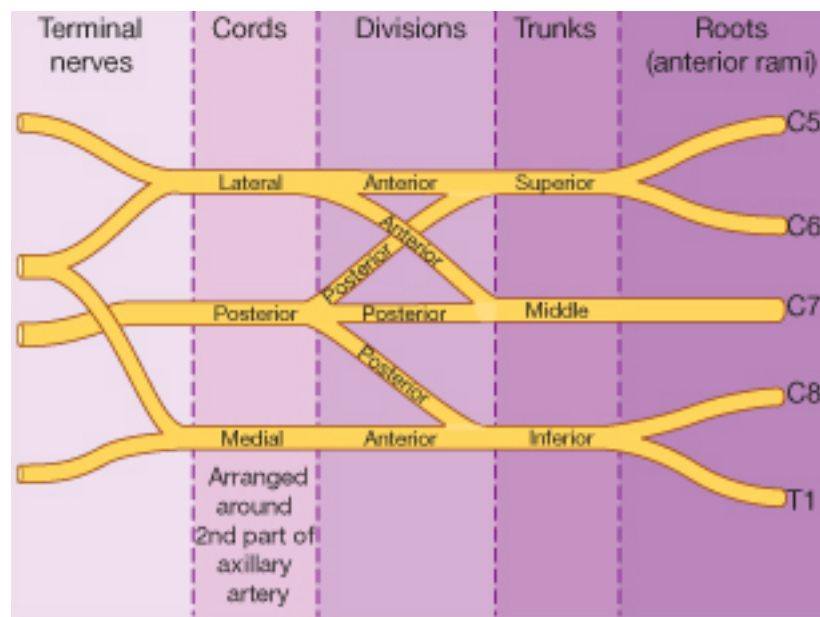
The brachial plexus is a large and very important plexus of nerves.

- It is situated partly in the neck and partly in the axilla.
- The **supraclavicular part** (rami and trunks with their branches) of the brachial plexus is in the posterior triangle of the neck, and its **infraclavicular part** (cords and their branches) is in the axilla.
- The brachial plexus is formed by the **union of the ventral rami of nerves C5 to C8** and the **greater part of T1 ventral ramus**.

- The rami that form the brachial plexus are often referred to as the "**roots of the brachial plexus**".
- They lie between the scalenus anterior and scalenusmedius muscles.

Plan of the Brachial Plexus

- The ventral rami from **C5 and C6** unite to form the **superior (upper) trunk**.
- **C7** continues as the **middle trunk**.
- **C8 and T1** unite to form the **inferior (lower) trunk**.
- Each of the 3 trunks divides into **anterior and posterior** divisions, posterior to the clavicle.
- These divisions are of fundamental significance as the **anterior division supply anterior (flexor) parts** and the **posterior division supply posterior (extensor) parts** of the upper limb.
- The **3 posterior divisions** unite to form the **posterior cord**.
- The **anterior divisions of the superior and middle trunks** unite to form the **lateral cord**.
- The **anterior division of the inferior trunk** continues as the **medial cord**.
- The cords of the brachial plexus bear their name in relation to their position to the axillary artery.
- Each cord of the brachial plexus **divides into two terminal branches**.
- Lateral cord: musculocutaneous nerve and lateral root of median nerve.
- Medial cord: ulnar nerve and medial root of median nerve.
- Posterior cord: axillary and radial nerves.
- The three nerves (musculocutaneous, median and ulnar) are arranged like the limbs of a capital **M**.



Branches of the Brachial Plexus

Supraclavicular Branches of the Brachial Plexus

- These arise from the rami and trunks of the brachial plexus and are approachable through the posterior triangle of the neck.

The Dorsal Scapular Nerve

- This is formed at the posterior aspect of the ventral ramus of C5 with frequent contributions from C4.
- It pierces the scalenus medius muscle, and runs deep to the levator scapulae muscle, which it helps to supply.
- It then enters the deep surface of the **rhomboid muscles**, which it supplies.

The Long Thoracic Nerve

- This is formed from the posterior aspect of the ventral rami of C5, C6 C7.
- It passes through the apex of the axilla posterior to the other components of the brachial plexus.
- It supplies the serratus anterior muscle.
- The roots of C5 and C6 pierce the scalenus anterior muscle and the root of C7 passes anterior to this muscle.

The nerve to the Subclavius Muscle

- This nerve is from the anterior aspect of the **superior trunk**, from C5, with occasional additions from C4 and C6.
- It descends posterior to the clavicle and anterior to the brachial plexus to supply the **subclavius muscle**.

The Suprascapular Nerve

- This nerve arises from the posterior aspect of the **superior trunk**, fibres from the ventral rami of C5 and C6 and often C4 (50% of people).
- It supplies the supraspinatus and infraspinatus muscles and the shoulder joint.
- It passes laterally across the posterior triangle of the neck, superior to the brachial plexus and then through the **scapular notch**.

Infraclavicular Branches of the Brachial Plexus

- These arise from the cords of the plexus and are approachable through the axilla.

Branches/es of the Lateral Cord

- The lateral cord has **3 branches**; **one side branch**, the lateral pectoral nerve and **two terminal branches**, the musculocutaneous nerve and lateral root of the median nerve.

The Lateral Pectoral Nerve

- This nerve pierces the **clavipectoral fascia** to supply the pectoralis major muscle.
- It also sends a branch to the medial pectoral nerve, which supplies the pectoralis minor muscle.
- This nerve is so named because **arises from the lateral cord; NOT** because it is lateral to the medial pectoral nerve.

The Musculocutaneous Nerve

- This nerve supplies the muscles of the anterior aspect of the arm.
- It enters deep to the surface of the coracobrachialis muscle and supplies it.
- It then continues to supply the biceps brachii and the brachialis muscles.
- Just proximal to the elbow joint, the musculocutaneous nerve pierces the deep fascia and becomes superficial.
- From here it is known as the **lateral antebrachial cutaneous nerve** (lateral cutaneous nerve of the forearm) and it supplies the skin on the lateral aspect of the forearm.

Lateral Root of the Median Nerve

- This is joined by the medial root of the median nerve, lateral to the axillary artery, to form the median nerve.

Branches of the Medial Cord

- The medial cord of the brachial plexus has **5 branches**; three side branches and two terminal branches.

The Medial Pectoral Nerve

- This enters the deep surface of the pectoralis minor muscle, and supplies it.
- It also supplies part of the pectoralis major muscle.
- This nerve is so named because it **arises from the medial cord**, NOT because it is medial to the lateral pectoral nerve.

Medial Brachial Cutaneous Nerve

- This nerve supplies the skin over the medial surface of the arm and the proximal part of the forearm.
- It usually communicates with the **intercostobrachial nerve**, which supplies the skin of the floor of the axilla and adjacent areas.

The Medial Antebrachial Cutaneous Nerve

- This nerve runs between the axillary artery and vein to supply the skin over the medial surface of the forearm.

The Ulnar Nerve

- The ulnar nerve passes through the arm into the forearm and hand, where it supplies one and one-half muscles in the forearm, most small muscles in the hand, and some skin.

The Medial Root of the Median Nerve

- This joins with the lateral root to form the median nerve.
- This supplies the flexor muscles of the forearm, except flexor carpi ulnaris, and part of the skin of the hand.

Branches of the Posterior Cord

- The posterior cord of the brachial plexus has **5 branches**; three side branches and two terminal branches.
- Generally, these nerves supply muscles that **extend the joints of the upper limb**.
- These branches also supply cutaneous nerves of the extensor surface of the upper limb.

The Upper Subscapular Nerve

- This small nerve supplies the subscapularis muscle.

The Thoracodorsal Nerve

- This nerve runs between the upper and lower subscapular nerves.
- It travels inferolaterally to supply the latissimusdorsi muscle.

The Lower Subscapular Nerve

- This nerve passes inferolaterally, deep to the subscapular artery and vein.
- It gives a branch to the subscapularis muscle and ends by supplying the teres major muscle.

The Axillary Nerve

- This nerve passes through the quadrangular space and then winds around the surgical neck of the humerus.
- It supplies the teres minor and deltoid muscles.

The Radial Nerve

- This nerve provides the major nerve supply to the extensor muscles of the upper limb.
- It also supplies cutaneous sensations to the skin to the extensor region, including the hand.

- As it leaves the axilla, the radial nerve runs posteriorly, inferiorly, and laterally between the long and medial heads of the triceps brachii muscle and enter the **radial groove** in the humerus.
- The radial nerve gives branches to the triceps brachii, anconeus and brachioradialis muscles, and to the extensor muscles of the forearm.

INJURIES OF BRACHIAL PLEXUS

- ★ **Injury to the superior roots** of the brachial plexus (C5–C6) may result from forceful pulling away of the head from the shoulder, as might occur from a heavy fall on the shoulder or excessive stretching of an infant's neck during childbirth.
 - The presentation of this injury is characterized by an upper limb in which the shoulder is adducted, the arm is medially rotated, the elbow is extended, the forearm is pronated, and the wrist is flexed. This condition is called **Erb-Duchenne palsy** or **waiter's tip position**. There is loss of sensation along the lateral side of the arm.
- ★ **Radial (and axillary) nerve injury** can be caused by improperly administered intramuscular injections into the deltoid muscle. The radial nerve may also be injured when a cast is applied too tightly around the mid-humerus.
 - Radial nerve injury is indicated by **wrist drop**, the inability to extend the wrist and fingers.
 - Sensory loss is minimal due to the overlap of sensory innervation by adjacent nerves.
- ★ **Median nerve injury** may result in **median nerve palsy**, which is indicated by numbness, tingling, and pain in the palm and fingers. Also inability to pronate the forearm and flex the proximal interphalangeal joints of all digits and the distal interphalangeal joints of the second and third digits. The wrist flexion is weak and is accompanied by adduction, and thumb movements are weak.
- ★ **Ulnar nerve injury** may result in **ulnar nerve palsy**, which is indicated by an inability to abduct or adduct the fingers, atrophy of the interosseous muscles of the hand, hyperextension of the metacarpophalangeal joints, and flexion of the interphalangeal joints, a condition called **clawhand**.
- ★ **Long thoracic nerve injury** results in paralysis of the serratus anterior muscle. The medial border of the scapula protrudes, giving it the appearance of a wing. When the arm is raised, the vertebral border and inferior angle of the scapula pull away from the thoracic wall and protrude outward, causing the medial border of the scapula to protrude; because the scapula looks like a wing, this condition is called **winged scapula**. The arm cannot be abducted beyond the horizontal position.

Axilla

- The **axilla** is the pyramidal space inferior to the glenohumeral joint and superior to the axillary fascia at the junction of the arm and thorax.

- The axilla provides a passageway, or “distribution center,” usually protected by the adducted upper limb, for the neurovascular structures that serve the upper limb.
- The axilla has an apex, a base, and four walls (three of which are muscular).
 - The **apex of axilla** is the **cervico-axillary canal**, the passageway between the neck and the axilla, bounded by the 1st rib, clavicle, and superior edge of the scapula. The arteries, veins, lymphatics, and nerves traverse to pass to or from the arm.
 - The **base** of the axilla or axillary fossa is bounded by the anterior and posterior axillary folds, the thoracic wall, and the medial aspect of the arm.
 - The **anterior wall of axilla** has two layers, formed by the pectoralis major and pectoralis minor and the pectoral and clavipectoral fascia associated with them.
 - The **anterior axillary fold** is the inferiormost part of the anterior wall that may be grasped; it is formed by the pectoralis major and the overlying integument.
 - The **posterior wall of axilla** is formed chiefly by the scapula and subscapularis on its anterior surface and inferiorly by the teres major and latissimusdorsi.
 - The **posterior axillary fold** is the inferiormost part of the posterior wall that may be grasped. It extends farther inferiorly than the anterior wall and is formed by latissimusdorsi, teresmajor, and overlying integument.

❖ *Contents of the Axilla*

- The axilla contains large nerves that are the branches of the brachial plexus.
- These nerves pass from the neck to the upper limb.
- The axilla also contains the **axillary vessels** (artery and branches, vein and tributaries, and lymph vessels).
- There are also several groups of **axillary lymph nodes**.

1) Axillary artery

❖ Compression of Axillary Artery

- The axillary artery can be palpated in the inferior part of the lateral wall of the axilla. Compression of the third part of this artery against the humerus may be necessary when profuse bleeding occurs (e.g., resulting from stab or bullet wound in the axilla).
- If compression is required at a more proximal site, the axillary artery can be compressed at its origin (as the subclavian artery crosses the 1st rib).

❖ Aneurysm of Axillary Artery

- The first part of the axillary artery may enlarge and compress the trunks of the brachial plexus, causing pain and anesthesia in the areas of the skin supplied by the affected nerves.
- Aneurysm of the axillary artery may occur in baseball pitchers because of their rapid and forceful arm movements.
- Injuries to Axillary Vein Wounds in the axilla often involve the axillary vein because of its large size and exposed position. When the arm is fully abducted, the axillary vein overlaps the axillary artery anteriorly. A wound in the proximal part of the axillary vein is particularly dangerous, not only because of profuse bleeding but

also because of the risk of air entering it and producing *air emboli* (air bubbles) in the blood.

- ❖ During **axillary node dissection**, two nerves are at risk of injury. During surgery, the long thoracic nerve to the serratus anterior is identified and maintained against the thoracic wall.
 - As mentioned earlier cutting the long thoracic nerve results in a winged scapula. If the thoraco dorsal nerve to the latissimusdorsi is cut, medial rotation and adduction of the arm are weakened, but deformity does not result.

2) Brachial plexus—discussed earlier

Cubital fossa

- The cubital fossa is an important area of transition between the arm and the forearm.
- this is a triangular **hollow area on the anterior surface of the elbow**.
- The three boundaries of the triangular cubital fossa are as follows :
 1. Superiorly, an imaginary line connecting the medial and lateral epicondyles.
 2. Medially, the mass of flexor muscles of the forearm arising from the common flexor attachment on the medial epicondyle; most specifically, the pronator teres.
 3. Laterally, the mass of extensor muscles of the forearm arising from the lateral epicondyle and supraepicondylar ridge; most specifically, the brachioradialis.
- ❖ The **floor of the cubital fossa** is formed by the brachialis and supinator muscles of the arm and forearm, respectively.
- ❖ The **roof of the cubital fossa** is formed by the continuity of brachial and ante brachial (deep) fascia reinforced by the bicipitalaponeurosis , subcutaneous tissue, and skin.

The contents of the cubital fossa are the

- This fossa is an important clinical area because it contains the biceps tendon, the brachial artery and its terminal branches (radial and ulnar arteries), the brachial veins, and part of the median and radial nerves.
- Superficially, in the subcutaneous tissue overlying the fossa are the **median cubital vein**, lying anterior to the brachial artery, and the medial and lateral antebrachial cutaneous nerves, related to the basilic and cephalic veins.

CUBITAL FOSSA--- CLINICAL

- ❖ BicipitalMyotatic Reflex
 - The biceps reflex is one of several deep-tendon reflexes that are routinely tested during physical examination. The relaxed limb is passively pronated and partially extended at the elbow. The examiner's thumb is firmly placed on the biceps tendon, and the reflex hammer is briskly tapped at the base of the nail bed of the examiner's thumb .
 - A normal (positive) response is an involuntary contraction of the biceps, felt as a momentarily tensed tendon, usually with a brief jerk-like flexion of the elbow. A positive

response confirms the integrity of the musculocutaneous nerve and the C5 and C6 spinal cord segments.

- Excessive, diminished, or prolonged (hung) responses may indicate central or peripheral nervous system disease or metabolic disorders (e.g., thyroid disease).

❖ Biceps Tendinitis

- The tendon of the long head of the biceps is enclosed by a synovial sheath and moves back and forth in the intertubercular sulcus of the humerus.
- Inflammation of the tendon (**biceps tendinitis**), usually the result of repetitive microtrauma, is common in sports involving throwing (e.g., baseball and cricket) and use of a racquet (e.g., tennis).
- It produce tenderness and **crepitus** (acrackling sound).

❖ Dislocation of Tendon of Long Head of Biceps Brachii

- Dislocated from the intertubercular sulcus in the humerus during traumatic separation of the proximal epiphysis of the humerus.

❖ Rupture of Tendon of Long Head of Biceps Brachii

- Rupture of the biceps tendon may result from forceful flexion of the arm against excessive resistance, as occurs in weight lifters. However, the tendon ruptures more often as the result of prolonged tendinitis that weakens it.
- The rupture results from repetitive overhead motions, such as occurs in swimmers and baseball pitchers.

❖ Venipuncture in Cubital Fossa

- The cubital fossa is the common site for sampling and transfusion of blood and intravenous injections because of the prominence and accessibility of veins.
- The median cubital vein is the most common. It crosses the bicipital aponeurosis, which separates it from the underlying brachial artery and median nerve and provides some protection to the latter.
- The median cubital vein is also a site for the introduction of cardiac catheters to secure blood samples from the great vessels and chambers of the heart.

Carpal tunnel

- The median nerve enters the hand through the **carpal tunnel**, between the tendons of flexor digitorum superficialis and flexor carpi radialis
- Clinical discussed in 1st part.

Triangle of Auscultation

- Near the inferior angle of the scapula is a small triangular gap in the musculature. The superior horizontal border of the latissimusdorsi, the medial border of the scapula, and the inferolateral border of the trapezius form a *triangle of auscultation*.
- This gap in the thick back musculature is a good place to examine posterior segments of the lungs with a stethoscope.

Lumbar puncture

- **LP** (spinal tap), the extraction or sampling of CSF from the lumbar cistern, is an important diagnostic tool for evaluating a variety of central nervous system (CNS) disorders.
 - **Meningitis and diseases of the CNS** may alter the cells in the CSF or change the concentration of its chemical constituents.
- Lumbar spinal puncture is performed with the **patient leaning forward or lying on the side with the back flexed**. Flexion of the vertebral column facilitates insertion of the needle by spreading the vertebral laminae and spinous processes apart, stretching the ligamentum flavum.
- Under **aseptic conditions**, the skin covering the lower lumbar vertebrae is anesthetized, and a **lumbar puncture needle** is inserted in the midline between the spinous processes of the **L3 and L4 (or L4 and L5)** vertebrae.
- Recall that a plane transecting the highest points of the iliac crests “the supracristal plane” usually passes through the L4 spinous process. After passing 4-6 cm in adults (more in obese persons), the needle punctures **the dura mater and arachnoid mater and enters the lumbar cistern**.
- CSF escapes at the rate of approximately one drop per second. If **subarachnoid pressure is high**, CSF flows out or escapes as a jet. So, **Lumbar puncture is not performed in the presence of increased intracranial pressure.**

Curvatures

- The vertebral column is positioned posteriorly in the body in a midline position. When viewed laterally, it has a number of **curvatures**
 - ❖ **Primary curvature** of the vertebral column is *concave* anteriorly.
 - ❖ **Secondary curvatures**, concave posteriorly, form in **the cervical and lumbar regions**, bring the center of gravity into a vertical line.

Clinical Considerations

- **Abnormal Curvatures of Vertebral Column**
 - Abnormal curvatures in some people result from developmental anomalies and in others from pathological processes such as **osteoporosis**.
- ★ **Vertebral body osteoporosis** occurs in all vertebrae but is most common in thoracic vertebrae and is an especially common finding in *postmenopausal women*.
 - The most prevalent metabolic disease of bone occurring in the elderly, especially in women, is osteoporosis.
 - It is characterized by net demineralization of the bones caused by a disruption of the normal balance of calcium deposition and resorption. As a result, the quality of bone is reduced and atrophy of skeletal tissue occurs.
 - Although osteoporosis affects the entire skeleton, the most affected areas are the neck of the femur (thigh bone), the vertebrae, the metacarpals (bones of the hand), and the radius (of the forearm). These bones become weakened and brittle, and are subject to fracture.
- ★ **Excess thoracic kyphosis** is characterized by an abnormal increase in the thoracic curvature; the vertebral column curves posteriorly. This abnormality can result from erosion of the anterior part of one or more vertebrae .

- **Dowager's hump** is a colloquial name for excess thoracic kyphosis in older women resulting from osteoporosis. However, excess kyphosis also occurs in elderly men .
- Osteoporosis especially affects the horizontal trabeculae of the trabecular (spongy)bone of the vertebral body .
- The remaining, unsupported vertical trabeculae are less able to resist compression and sustain compression fractures, resulting in short and wedge-shaped thoracic vertebrae. Progressive erosion and collapse of vertebrae also result in an overall loss of height, and the excess kyphosis leads to an increase in the AP diameter of the thorax.
- ★ **Excess lumbar lordosis** is characterized by an anterior rotation of the pelvis upper at the hip joints, producing an abnormal increase in the lumbar curvature; the vertebral column curves more anteriorly .
 - This abnormal extension deformity is often associated with weakened trunk musculature, especially the anterolateral abdominal muscles.
 - To compensate for alterations to their normal line of gravity, women develop a temporary excess lumbar lordosis during late pregnancy.
 - This lordotic curvature may cause low back pain, but the discomfort normally disappears soon after childbirth.
 - Obesity in both sexes can also cause excess lumbar lordosis and low back pain because of the increased weight of the abdominal contents anterior to the normal line of gravity.
 - Loss of weight and exercise of the anterolateral abdominal muscles facilitate correction of this type of excess lordosis.
- ★ **Scoliosis** (crooked or curved back) is characterized by an abnormal lateral curvature that is accompanied by rotation of the vertebrae .
 - The spinous processes turn toward the cavity of the abnormal curvature, and when the individual bends over, the ribs rotate posteriorly (protrude) on the side of the increased convexity.
 - Scoliosis is the most common deformity of the vertebral column in pubertal girls .
 - Asymmetrical weakness of the intrinsic back muscles (myopathic scoliosis), failure of half of a vertebra to develop (hemivertebra), and a difference in the length of the lower limbs are causes of scoliosis.
 - Habit scoliosis is supposedly caused by habitual standing or sitting in an improper position. When the scoliosis is entirely postural, it disappears during maximum flexion of the vertebral column.

Tension on vertebral disk

★ Aging of Intervertebral Discs

- With advancing age, the nuclei pulposi dehydrate and lose elastin and proteoglycans while gaining collagen. As a result, the IV discs lose their turgor, becoming stiffer and more resistant to deformation.
- As the nucleus dehydrates, the two parts of the disc appear to merge as the distinction between them becomes increasingly diminished.

- As these changes occur, the anulus fibrosis assumes an increasingly greater share of the vertical load and the stresses and strains that come with it.
- The lamellae of the anulus thicken and often develop fissures and cavities.
- Although the margins of adjacent vertebral bodies may approach more closely as the superior and inferior surfaces of the body become shallow concavities (the most probable reason for slight loss of height with aging), it has been shown that the intervertebral discs *increase* in size with age.
- Not only do they become increasing convex but, between the ages of 20 and 70, their anteroposterior (AP) diameter increases about 10% in females and 2% in males, while thickness (height) increases centrally about 10% in both sexes.
- **Overt or marked disc narrowing, especially when it is greater than that of more superiorly located discs, suggests pathology, not normal aging .**

★ Herniation of Nucleus Pulposus

- Herniation or protrusion of the gelatinous nucleus pulposus into or through the anulus fibrosus is a well-recognized cause of lower back pain (LBP) and lower limb pain .
- The water content of nuclei pulposi in young person is high (approaching 90%), giving them great turgor (fullness). However, violent hyperflexion of the vertebral column may rupture an IV disc and fracture the adjacent vertebral bodies. Flexion of the vertebral column produces compression anteriorly and stretching or tension posteriorly, squeezing the nucleus pulposus further posteriorly toward the thinnest part of the anulus fibrosus .
- If the anulus fibrosus has degenerated, the nucleus pulposus may herniate into the vertebral canal and compress the spinal cord or the nerve roots of the cauda equina .
- Herniations of the nucleus pulposus usually extend **posterolaterally**, where the anulus fibrosus is relatively thin and does not receive support from either the posterior or the anterior longitudinal ligaments.
- **Aposterolateral herniated IV disc is more likely to be symptomatic because of the proximity of the spinal nerve roots.**
- The *localized back pain* of a herniated disc, which is usually *acute pain*, results from pressure on the longitudinal ligaments and periphery of the anulus fibrosus and from local inflammation caused by chemical irritation by substances from the ruptured nucleus pulposus.
- *Chronic pain* resulting from compression of the spinal nerve roots by the herniated disc is usually **referred pain**, perceived as coming from the area (dermatome) supplied by that nerve.
- Because the IV discs are largest in the lumbar and lumbosacral regions, where movements are consequently greater, posterolateral herniations of the nucleus pulposus are most common here .

- **Sciatica**, pain in the lower back and hip radiating down the back of the thigh into the leg, is often caused by a herniated lumbar IV disc that compresses and compromises the L5 or S1 component of the sciatic nerve. The IV foramina in the lumbar region decrease in size and the lumbar nerves increase in size, which may explain why sciatica is so common.
- **IV discs may also be damaged** by violent rotation or flexing of the vertebral column. The general rule is that when an IV disc protrudes, it usually compresses the nerve root numbered one inferior to the disc; for example, the L5 nerve is compressed by an L4–L5 IV disc herniation.

★ Fracture of Dens

- The transverse ligament of the atlas is stronger than the dens of the C2 vertebra. Fractures of the dens make up about 40% of fractures of the axis. The most common dens fracture occurs at its base that is, at its junction with the body of the axis .
- Often these fractures are unstable because transverse ligament of the atlas becomes interposed between fragments and because the separated fragment (the dens) no longer has a blood supply, resulting in **avascular necrosis**
- Other dens fractures result from abnormal ossification patterns.

★ Lumbar Spinal Stenosis

- Lumbar spinal stenosis describes a stenotic (narrow) vertebral foramen in one or more lumbar vertebrae. This condition may be a hereditary anomaly that can make a person more vulnerable to age-related degenerative changes such as IV disc bulging.
- Lumbar spinal nerves increase in size as the vertebral column descends, but paradoxically, the IV foramina decrease in size. Narrowing is usually maximal at the level of the IV discs. However, stenosis of a lumbar vertebral foramen alone may cause compression of one or more of the spinal nerve roots occupying the inferior vertebral canal.

Tissue processing and microscopy

- **Tissue processing:** Tissues need to be processed in order to see them through the microscope. Because tissues are usually thick to let light pass through them, they need to be sectioned first. (this is called microtomy) (Thickness needed for light microscope= 5-10µm and electron microscope= 40-70µm)
- **Aims of tissue processing:** prevent the tissue from decomposition by bacteria, preserve its structures for microscopy etc.

So what are the processes required for tissue processing?

1. Fixation: make tissues soft to be cut and rigid to be sliced.
 - Formaldehyde is the ideal chemical fixating agent(although heat and freezing can be techniques too)
 - Immersion: for microscopic anatomy
 - Perfusion: for gross anatomy....for human body: **embalming**
- N.B** Most tissue components will be fixed as **macro molecules**.

2. Dehydration: remove any aqueous or liquid fluid using hydrophilic agent that dilute aqueous fluid. e.g.: Ethanol
3. Clearing: agents miscible with the used dehydrating agent- clearing agents e.g. xylene
4. Infiltration: infiltrating by embedding agent
5. Embedding: treat with agents that solidify in tissues, giving support and rigidity. E.g. Paraffin wax- a typical embedding agent.

Staining techniques: a. General stain: e.g. Hematoxylin and Eosin

b. Special stain: for specific components such as mucus, fiber...

Two types of dyes exist:-

- Basic dyes: react with acidic components.....e.g. Hematoxylin. Nucleus, RNA, Carboxyl group of proteins
- Acidic dyes: react with basic components.....e.g. eosin, mitochondria, collagen cytoplasmic filaments.
- Metachromasia: tissue components changing from blue to purple-red.

Microscopy:

1. Magnifying lens: convex lens
 2. Simple microscope: one or two lenses
 3. Compound microscope: combination of two lenses
 - Objective lens: forming the first image inside the microscope.
 - Also needed for resolution. Has $\times 4$, $\times 10$, $\times 40$ and $\times 100$ magn.
 - Eyepiece lens: forms a further enlarged image of the object.
 4. Electron microscope: electron beam instead of light.
- Magnification is the property of both ocular and objective lenses. Therefore total magnification is at least the product of the magnifying power of the two lenses.
 - Resolution: clarity and richness of details. is the property of only objective lens.

Organelle structure and interaction

- The protoplasm consists of: Cytoplasm, Organelles and Inclusions
- Organelles: metabolically active structures that can be membrane or non-membrane bounded.

- Plasma membrane	- Golgi apparatus
- Mitochondria	- Lysosomes
- Ribosomes	- Proteasomes
-Endoplasmic reticulum	- Peroxisomes, Microtubules

1) Plasma membrane:

- Structure: consists of **lipid bilayer**: In TEM, unilaminar appearance
- A. Outer leaflet facing the extracellular environment
 - B. Inner leaflet facing the cytoplasm

- Fluid mosaic model of plasma membrane: “protein icebergs in a lipid sea”

Lipid bilayer composed of:

- Phospholipids: amphipathic molecules, consisting of one polar (hydrophilic) head and two nonpolar (hydrophobic) fatty acid tails.
- Glycolipids: Polar carbohydrate residues of glycolipids extend from **the outer** leaflet into the extracellular space and form part of the **glycocalyx**.
- Cholesterol: maintain the structural integrity of the membrane.

Membrane proteins:

1. Integral proteins: are found complexed in the bi-lipid layer. May be one-pass or multiple-pass and are amphipathic. Most are glycoproteins.
2. Peripheral proteins: on cytoplasmic aspect of inner leaflet. most function as cytoskeletal components or intracellular secondary messenger system.

Functions of the cell membrane:

1. Selective permeability
-Passive diffusion, facilitated diffusion, active transport
2. Signal transduction
- Ion-channel, enzyme-linked, G-protein coupled receptors.
3. Endocytosis
4. Exocytosis
5. Compartmentalization
6. Storage, transport, secretion etc.

2) Mitochondria

- Membrane-enclosed organelles with an outer membrane and an inner one invaginating to form cristae, which greatly increase the surface area.
- Subdivided into an **intermembrane compartment** between the two membranes and an **inner matrix compartment**.
- Mitochondria is the site of aerobic respiration and except succinate dehydrogenase, they house all the necessary enzymes for it. ATP is synthesized via the kreb cycle. ATP also synthesized via a chemiosmotic coupling mechanism in the electron transport chain and ATP synthase in the cristae.
 - Mitochondrial deficiency diseases are most characterized by muscular dysfunction and that's because muscles have high-energy metabolism. These diseases are inherited along maternal line.

3) Ribosomes

- Small and large subunit composed of several types of rRNA and proteins. May be free or bound to membranes of RER.
- A cluster of ribosome along a single strand of mRNA and involved in protein synthesis—**polyribosome**

4) Endoplasmic reticulum

- Anastomosing network of intercommunicating channels or sacs in an enclosed-space called cisterna.

RER: rough because it is studded with ribosomes.

SER: lacks ribosomes hence the name smooth. It is less common than RER but found in abundance in cells synthesizing steroids, cholesterol etc.

Function: -Muscle contraction and relaxation by release of calcium in the SER of skeletal muscles- called **sarcoplasmic reticulum**.

- Drug detoxification in hepatocytes by proliferation of the SER following a drug-intake.
- Steroid hormone synthesis occurs in SER-rich cells.

5) Golgi apparatus

- Post translational modifications of proteins from RER. Smooth membranous **sacculles**.
- The entry or *cis-face* is where the RER buds off small transport vesicles.
- The exit or *trans-face* is where large condensing vacuoles accumulate and completed protein products are taken away. Vesicle formation is driven by assembly of various coat proteins – **clathrin**. The Golgi apparatus is important for glycosylation, sulfation, phosphorylation, and limited proteolysis of proteins. Furthermore, the Golgi apparatus initiates packing, concentration, and storage of secretory products.

6) Lysosomes:

- Are membrane-limited vesicles that contain about 40 different hydrolytic enzymes and are particularly abundant in cells with great phagocytic activity (eg, macrophages, neutrophils)
- Material taken from the cellular environment by endocytosis is digested when lysosomes fuse with the membrane of the phagosome or pinocytotic vesicle. The endocytosed material mixes with the hydrolytic enzymes. The composite structure is called **heterolysosome**. During the digestion of macromolecules, the indigestible material is retained within the vacuoles, now called residual bodies. They accumulate in some long-lived cells (neurons, heart muscles) and will be referred to as **lipofuscin**.

7) Proteasomes:

- Abundant cytoplasmic protein complexes not associated with membrane. They function to degrade denatured or otherwise nonfunctional polypeptides. Whereas lysosomes digest bulk material introduced into the cell, or whole organelles and vesicles, proteasomes deal primarily with proteins as individual molecules. This organelle recognizes proteins with ubiquitin attached to them. This will be unfolded by ATPases ready for destruction.

8) Peroxisomes:

- oxidize specific organic substrates by removing hydrogen atoms that are transferred to molecular oxygen (O_2), producing hydrogen peroxide (H_2O_2)- a harmful substance degraded by **catalase**. This set of reaction has many clinical implications: it oxidizes various potentially toxic molecules such as drugs. 50% of ingested ethyl alcohol is degraded to acetic aldehyde in peroxisomes of these cells. The oxidation of long-chain fatty acids to the synthesis of cholesterol is also carried out by several enzymes located in Peroxisomes.

CLINICAL CORRELATE

- The most common peroxisomal disorder is X-chromosome-linked **adrenoleukodystrophy**, caused by a defective integral membrane protein that participates in transporting very long-chain fatty acids into the peroxisome for β -oxidation.
- Deficiency in peroxisomal enzymes causes the fatal **Zellweger syndrome**, with severe muscular impairment, liver and kidney lesions and CNS impairment.

9) Microtubules:

- fine tubular structures in the cytoplasm; also found in the cilia and flagella. The protein subunit of a microtubule is a heterodimer composed of α and β -tubulin. Tubulin heterodimers polymerize to form microtubules. Microtubules have **microtubule-associated proteins** (MAPs), which stabilize them and bind them to other cytoskeletal components and organelles. They also are associated with **kinesin and cytoplasmic dynein**, two force-generating proteins, which serve as motors for vesicle or organelle movement.
- Kinesin- moves cargo towards the (+) end and Dynein towards the (-) end.

10) Centrioles:

- Each centriole has **nine microtubular triplets** and adjacent microtubules share some protofilaments.

Inclusions

- Accumulation of transient non-metabolically active(metabolic byproducts)
- They are storage forms of various nutrients
- **Glycogen granules:**
 - They are aggregates of carbohydrate which serves as a stored energy source degraded to glucose.
- **Lipid droplets:** Accumulations of lipid molecules that are prominent in adipocytes (fat cells), adrenal cortex cells, liver and other cells.
- **Lipofuscin granules:** residue of undigested material present in residual bodies. They increase with age and are found in stable non dividing cells such as neurons and cardiac muscle

COMPONENTS OF THE NUCLEUS

1. NUCLEAR ENVELOPE

- The nuclear envelope surrounds the nuclear material and consists of two parallel membranes separated from each other by a narrow perinuclear cisterna. These membranes fuse at intervals, forming openings called nuclear pores in the nuclear envelope.
- Outer nuclear membrane faces the cytoplasm and is continuous at certain sites with the rough endoplasmic reticulum. Intermediate filaments (vimentin) surround the outer nuclear membrane on its cytoplasmic aspect. Ribosomes stud the cytoplasmic surface of the outer nuclear membrane. These ribosomes synthesize proteins that enter the perinuclear cisterna.
- Inner nuclear membrane faces the nuclear material but is separated from it and is supported on its inner surface by the nuclear lamina. These intermediate filament proteins help organize the nuclear envelope and perinuclear chromatin and they are essential during the mitotic events.

Phosphorylation of lamins leads to disassembly, and dephosphorylation results in reassembly of the nuclear envelope.

- Nuclear pores are formed by fusion of the inner and outer nuclear membranes. They permit passage of certain molecules in either direction between the nucleus and the cytoplasm.
- Nuclear pore complex represents protein subunits surrounding the nuclear pore. Some of the proteins are arranged in eight fold symmetry around the margin of the pore. The nucleoplasmic side of the pore exhibits a nuclear basket, whereas the cytoplasmic side displays fibers extending into the cytoplasm. A transporter protein is located in the central core and is responsible for transporting proteins into and out of the nucleus via receptor-mediated transport. These proteins have clusters of certain amino acids known as **nuclear localization segments** that act as signals for transport. Transport mechanisms involve a group of proteins, exportins and importins. The function of these proteins is regulated by Ran, a group of guanosine triphosphate-binding proteins. The other group of proteins called nucleoporins facilitates the shuttling of cargo in both directions. Transport signals of this type are called nucleocytoplasmic shuttling signals.

Clinical correlate

- Certain mutations in the gene coding for laminA are associated with a subtype of the disorder progeria, which causes premature aging. In this and other rare “laminopathies,” the nuclear envelope is abnormal.

2. Nucleolus

- The nucleolus is a nuclear inclusion that is not surrounded by a membrane. It is observed in interphase cells that are actively synthesizing proteins. It contains mostly rRNA and protein along with a modest amount of DNA. It possesses nucleolar organizer regions, portions of the chromosomes where rRNA genes are located; these regions are involved in reconstituting the nucleolus during the G1 phase of the cell cycle. The nucleolus contains four distinct regions. 1. Fibrillar centers are composed of inactive DNA 2. The pars fibrosa contains transcriptionally active DNA 3. The pars granulosa is composed of maturing ribosomal precursor particles. 4. Nucleolar matrix is a fiber network participating in the organization of the nucleolus.
- The nucleolus is involved in the synthesis of rRNA and its assembly into ribosome precursors. The nucleolus also sequesters certain nucleolar proteins that function as cell cycle checkpoint signaling proteins.

3. CHROMATIN

- Chromatin consists of double-stranded DNA complexed with histones and acidic proteins. It resides within the nucleus as heterochromatin and euchromatin. The euchromatin/heterochromatin ratio is higher in malignant cells than in normal cells.
- Heterochromatin appears as coarse, electron-dense material in the electron microscope and as intensely basophilic clumps in the light microscope and is transcriptionally inactive. Heterochromatin corresponds to one of two X chromosomes and is therefore present in nearly all somatic cells of female mammals. During interphase, the inactive X chromosome is visible as a dark-staining body within the nucleus. This structure is called the **Barr body, or sex chromatin**.

- Euchromatin is the transcriptionally active form of chromatin visible as finely dispersed granular material in the electron microscope and as lightly stained basophilic areas in the light microscope.

Clinical correlate

- Barr bodies or gender chromatin permit gender to be determined microscopically in patients whose external sex organs do not permit that determination, as in hermaphroditism and pseudohermaphroditism. Sex chromatin analysis also helps reveal other anomalies involving the sex chromosomes

4. CHROMOSOME

- Chromosomes consist of chromatin extensively folded into loops; this conformation is maintained by DNA-binding proteins. Each chromosome contains a single DNA molecule and associated proteins, assembled into nucleosomes, the structural unit of chromatin packaging.
- Karyotype refers to the number and morphology of chromosomes and is characteristic for each species. Haploid number is the number of chromosomes in germ cells. Diploid number is the number of chromosomes in somatic cells.
- Genome, the total genetic complement of an individual, is stored in its chromosomes. In humans, the genome consists of 22 pairs of autosomes and 1 pair of sex chromosomes.

I. DNA

- DNA is a long double-stranded linear molecule composed of multiple nucleotide sequences. It acts as a template for the synthesis of RNA.
- Nucleotides are composed of a base (purine or pyrimidine), a deoxyribose sugar, and a phosphate group. The purines are adenine and guanine. The pyrimidines are cytosine and thymine.
- The DNA double helix consists of two complementary DNA strands held together by hydrogen bonds between the base pairs A-T and G-C.
- Exons are regions of the DNA molecule that code for specific RNAs.
- Introns are regions of the DNA molecule, between exons, that do not code for RNAs.
- A codon is a sequence of three bases in the DNA molecule that codes for a single amino acid.
- A gene is a segment of the DNA molecule that is responsible for the formation of a single RNA molecule.

Clinical correlate

- Oncogenes are the result of mutations of certain regulatory genes ,called proto-oncogenes, which normally stimulate or inhibit cell proliferation and development. oncogenes dominate the normal alleles (proto-oncogenes), causing deregulation of cell division, which leads to a cancerous state.

II. RNA

- RNA is a linear molecule similar to DNA; however, it is single stranded and contains ribose instead of deoxyribose sugar and uracil instead of thymine. RNA is

synthesized by transcription of DNA. Transcription is catalyzed by three RNA polymerases: I for rRNA, II for mRNA, and III for tRNA.

- mRNA carries the genetic code to the cytoplasm to direct protein synthesis mRNA contains codons that are complementary to the DNA codons from which it was transcribed, including one start codon (AUG) for initiating protein synthesis and one of three stop codons(UAA, UAG, or UGA) for terminating protein synthesis.
- tRNA possesses an anticodon which ensures that the tRNA transfers its activated amino acid molecule in the proper sequence to the growing polypeptide chain.
- Ribosomal RNA associates with many different proteins to form ribosomes.

Clinical correlate

- MicroRNAs are very small segments of RNA molecules that function to regulate gene expression. Mutations in miRNAs may be correlated with specific human cancers

EPITHELIAL TISSUE

SPECIALIZATIONS OF THE CELL SURFACE

- The **free or apical surface** of many types of epithelial cells contain specialized structures that increase the cell surface area or move substances or particles stuck to the epithelium.

1. APICAL SURFACE

i. MICROVILLI

- arefingerlike extensions found mainly on the free cell surface.
- Hundreds of microvilli are found in absorptive cells, such as the lining epithelium of the small intestine and the cells of the proximal renal tubule.
- In these absorptive cells the glycocalyx is thicker than it is in most other cells.
- The complex of microvilli and glycocalyx may be seen with the light microscope and is called the brush, or striated border.
- Within the microvilli are clusters of actin filaments that are cross-linked to each other and to the surrounding plasma membrane by several other proteins.

Clinical correlate

- Celiac disease, also called gluten-sensitive enteropathy or sprue, is a disorder of the small intestine in which one of the first pathologic changes is loss of the microvilli brush border of the absorptive cells. This is caused by an immune reaction against the wheat protein gluten during its digestion, which produces diffuse enteritis, changes to the epithelial cells leading to malabsorption, and eventually to pathologic changes in the intestinal wall. The malabsorption problems and structural changes are reversible when gluten is removed from the diet.

ii. STEREOCILIA

- long, non-motile extensions of cells of the epididymis and ductus deferens that are actually long and branched microvilli.
- Stereocilia increase the cell surface area, facilitating the movement of molecules into and out of the cell. More specialized stereocilia with a motion-detecting function are important components of inner ear sensory cells.

iii. **CILIA**

- Cilia are cylindrical, motile structures on the surface of some epithelial cells
- They are surrounded by the cell membrane and the nine doublets form an array around two central microtubules; the 9 + 2 assembly of microtubules is called an axoneme. The two microtubules of the peripheral pairs are joined to each other.
- Cilia are inserted into basal bodies, which are small cylindrical structures at the apical pole just below the cell membrane.
- In living organisms, cilia have a rapid back-and-forth movement.
- Ciliary movement is frequently coordinated to permit a current of fluid or particulate matter to be propelled in one direction over the epithelial surface.
- Adenosine triphosphate (ATP) is the source of energy for ciliary motion.

iv. **FLAGELLA**

- Flagella, present in the human body only in spermatozoa, are similar in structure to cilia but are much longer and are limited to one flagellum per cell.

Clinical correlate

- Several mutations have been described in the proteins of the cilia and flagella. They are responsible for the immotile cilia syndrome (Kartagener syndrome), whose symptoms are chronic respiratory infections caused by the lack of the cleansing action of cilia in the respiratory tract and immotile spermatozoa, causing male infertility.

2. LATERAL REGION

❖ INTERCELLULAR JUNCTIONS

- The lateral membranes of many epithelial cells often exhibit several types of membrane modifications called intercellular junctions.
- In several epithelia the various junctions are present in a definite order from the apex toward the base of the cell.
- Intercellular junctions: provide a mechanism for communication between adjacent cells. serve as sites of adhesion and as seals to prevent the flow of material through the space between epithelial cells.

i. TIGHT JUNCTIONS (zonulae or zonulaoccludens):

- These are the most apical of the junctions.
- "Zonula" refers to the fact that the junction forms a band completely encircling the cell, and "occludens" refers to the membrane fusions that close off the intercellular space.
- In properly stained thin sections viewed in the transmission electron microscope, the outer leaflets of membranes of neighbor cells are seen to fuse, giving rise to a pentalaminar sheet.
- One to several of these fusion sites may be observed, depending on the type of epithelium.

- When epithelial cells are submitted to cryofracture, it is possible to see anastomosing ridges and grooves that form a netlike structure corresponding to the fusion sites observed in conventional thin sections. The number of ridges and grooves, or fusion sites, has a high correlation with the leakiness of the epithelium.
- Epithelia with one or few fusion sites (eg, proximal renal tubules) are more permeable to water and solutes than are epithelia with numerous fusion sites (eg, urinary bladder).
- Thus, the principal function of the tight junction is to form a seal that prevents the flow of material between epithelial cells (called the paracellular pathway) in either direction (from apex to base or from base to apex. In this way the zonulaoccludens contributes to the formation of two functional compartments:
 1. Apical compartment that is composed of an organ cavity(such as the intestinal lumen) or the cavity of a secretory unit, and
 2. Basal compartment that begins at the junctions and encompasses the underlying tissues.

Clinical correlate

- Proteins of the zonulaoccludens provide the targets for certain common bacteria of medical importance. The enterotoxin secreted by *Clostridium perfringens*, which causes “food poisoning,” binds claudin molecules of intestinal cells, prevents insertion of these proteins during maintenance of tight junctions, and causes loss of tissue fluid into the intestinal lumen via the paracellular pathway. Similarly, *Helicobacter pylori*, which is important in the etiology of gastric ulcers, binds the extracellular domains of tight-junction proteins in cells of the stomach and inserts a protein into these cells and disrupts signaling from the junction.

ii. ZONULA ADHERENS:

- This junction encircles the cell and provides for the adhesion of one cell to its neighbor. Cell adhesion is mediated by cadherins
- A noteworthy feature of this junction is the insertion of numerous actin filaments into electron-dense plaques of material on the cytoplasmic surfaces of the junctional membranes.

iii. GAP OR COMMUNICATING JUNCTIONS:

- This junction can occur almost anywhere along the lateral membranes of epithelial cells.
- Actually, gap junctions are found in nearly all mammalian tissues, except in skeletal muscle.
- The individual unit of the gap junction is called a connexon.
- Each connexon is formed by six gap junction proteins called connexins, which join together leaving a hydrophilic pore in the center. Connexons of adjacent cells are aligned to form a hydrophilic channel between the two cells.
- Each gap junction is formed by tens or hundreds of aligned pairs of connexons.
- Connexins belong to a family of related proteins that is distributed differently and forms channels with differing physiological properties.
- Signaling molecules such as some hormones, cyclic AMP and GMP, and ions can move

through gap junctions, causing the cells in many tissues to act in a coordinated manner rather than as independent units. In heart muscle cells gap junctions are responsible for the heart's coordinated beat.

D. MACULA ADHERENSE or Desmosome (Gr. *desmos*, band, + *soma*, body)

- The desmosome is a complex disk-shaped structure at the surface of one cell that is matched with an identical structure at the surface of the adjacent cell.
- It is characterized by a dense plaque of intracellular attachment proteins, called desmoplakins, on the cytoplasmic surface of each opposing cell.
- In epithelial cells, groups of intermediate cyokeratin filaments are inserted into the attachment plaque or make hairpin turns and return to the cytoplasm. Because intermediate filaments of the cytoskeleton are very strong, desmosomes provide a firm adhesion among the cells. In nonepithelial cells, the intermediate filaments attached to desmosomes are made not of cyokeratin but of other proteins, such as desmin or vimentin.
- Proteins of the cadherin family participate in the adhesion provided by desmosomes.
- In the contact zone between certain epithelial cells and the basal lamina, hemidesmosomes can often be observed.
- These structures take the form of half a desmosome and bind the epithelial cell to the subjacent basal lamina. However, in desmosomes the attachment plaques contain mainly cadherins, whereas in hemidesmosomes the plaques are made of integrins.

Clinical correlate

- Various blistering (bullous) diseases, such as pemphigus vulgaris, involving the epidermis or stratified squamous epithelia of the oral mucosa, are due to abnormal desmosome function caused by autoimmune reactions against specific desmogleins that reduce cell-to-cell adhesion.

BASEMENT MEMBRANE

- All epithelial cells in contact with subjacent connective tissue have at their basal surfaces a specialized, feltlike sheet of extracellular material called basement membrane. With the TEM the basement membrane may be resolved into basal lamina and fibrous reticular lamina.
- The basal lamina is visible only with the electron microscope, where it appears as a dense layer consisting of a delicate network of very fine fibrils (lamina densa).
- In addition, basal laminae may have an electron-lucent layer on one or both sides of the lamina densa, this layer is called lamina rara or lamina lucida.
- The main components of basal laminae are: Collagen (type IV), Glycoproteins(laminin and entactin) and Proteoglycans. The precise molecular composition of these components varies between and within tissues.
- Basal laminae are attached to the underlying connective tissues (lamina propria) by anchoring fibrils formed by type VII collagen.
- All these components are secreted by epithelial, muscle, adipose, and Schwann cells.
- In some instances, reticular fibers (produced by connective tissues) are closely associated with the basal lamina, forming the reticular lamina.

- Basal laminae have many functions: They support the cells, form a barrier between connective tissue and cells of other tissues that limits or regulates the exchange of macromolecules, influence cell polarity, regulate cell proliferation and differentiation by binding with growth factors, influence cell metabolism, and serve as pathways for cell migration. The presence of the basal lamina around a muscle cell is necessary for the establishment of new neuromuscular junctions.

CONNECTIVE TISSUE: PROPER AND SPECIAL

- Classification is based on the proportion of cells to fibers and on the arrangement and type of fibers

A. Connective tissue proper

- Loose connective tissue (areolar tissue) possesses fewer fibers but more cells than dense connective tissue. This tissue is well vascularized, flexible, and not very resistant to stress. It is more abundant than dense connective tissue and is the connective tissue that fills in the spaces just deep to the skin.
- Dense connective tissue contains more fibers but fewer cells than loose connective tissue.
- It is classified by the orientation of its fiber bundles into two types: Dense, irregular connective tissue (most common), which contains fiber bundles that have no definite orientation. This tissue is characteristic of the dermis and capsules of many organs. Dense regular connective tissue, which contains fiber bundles and attenuated fibroblasts that are arranged in a uniform parallel fashion. It is present only in tendons and ligaments. This tissue may be collagenous or elastic.

B. Connective tissue special

- Cartilage is an avascular specialized fibrous connective tissue. It has a firm extracellular matrix that is less pliable than that of connective tissue proper, and it contains chondrocytes embedded in the matrix. Cartilage functions primarily to support soft tissues and assist in the development and growth of long bones. The three types of cartilage: hyaline cartilage, elastic cartilage, and fibrocartilage vary in certain matrix components.
- Bone is the primary constituent of the adult skeleton. It is a specialized type of connective tissue with a calcified extracellular matrix in which characteristic cells are embedded. Bone functions to protect vital organs, support fleshy structures, and provide a calcium reserve. It is a dynamic tissue that constantly undergoes changes in shape. Applied pressure results in bone resorption. Applied tension results in bone formation.
- Elastic tissue is composed of coarse, branching elastic fibers with a sparse network of collagen fibers and some fibroblasts filling the interstitial spaces. It is present in the dermis, lungs, elastic cartilage, and elastic ligaments and in large (conducting) blood vessels, where it forms fenestrated sheaths.
- Reticular tissue consists mostly of a network of branched reticular fibers (type III collagen). This tissue invests liver sinusoids, smooth muscle cells, and fat cells and forms the stroma of lymphatic organs, bone marrow, and endocrine glands. It also forms the reticular lamina of basement membranes.

- Adipose tissue is the primary site for storage of energy and has a rich neurovascular supply. White adipose tissue is composed of unilocular adipose cells. This tissue constitutes nearly all of the adult adipose tissue throughout the body. Brown adipose tissue is composed of multilocular adipose cells, which contain many large mitochondria. This tissue is found in infants and is much reduced in adults.

Clinical correlate

- Adipose tumors may be either benign or malignant. Lipomas are benign fatty tissue tumors. Liposarcomas are the most common malignant adipose tumors. Although liposarcomas are most frequently located in the leg and/or the retroperitoneal tissues, they are not restricted to these sites. Liposarcomas are difficult to diagnose since they resemble lipomas. Adipocytes of the malignant tissue resemble either unilocular or multilocular adipocytes. Presently, chromosome markers are used to differentially diagnose liposarcomas.

Connective tissue

- formed primarily of extracellular matrix, consisting of ground substance, and fibers, in which various connective tissue cells are embedded.

- supports organs and cells, acts as a medium for exchange of nutrients and wastes between the blood and tissues, protects against microorganisms, repairs damaged tissues, and stores fat.

Structure of Connective Tissue

Extracellular Matrix

- provides a medium for the transfer of nutrients and waste materials between connective tissue cells and the bloodstream.

A. **Ground substance-** is a colorless, transparent, gel-like material in which the cells and fibers of connective tissue are embedded.

1. It is a complex mixture of glycosaminoglycans(GAGs) , proteoglycans, and glycoproteins 2. Ground substance serves as a lubricant, helps prevent invasion of tissues by foreign agents, and resists forces of compression.

- **GAGs** - are long, unbranched polysaccharides composed of **repeating identical disaccharide units**.

1. An **amino sugar**, either *N*-acetylglucosamine or *N*-acetylgalactosamine, is always one of the repeating disaccharides.

2. Because GAGs are commonly **sulfated** and usually possess a **uronic acid sugar**, they are polyanions and hydrophilic.

3. GAGs are generally linked to a **core protein**.

4. strongly **resists compression**.

5. Their extended random coils occupy large volumes of space

6. GAGs may be classified into four main groups on the basis of their chemical structure

a. Hyaluronic acid is a very large unsulfated molecule that is not attached to a core protein.

- a major component of the synovial fluid, which fills joint cavities, and the vitreous body of the eye.

b. The other three GAG groups are **chondroitin sulfate** and **dermatan sulfate**, **heparin** and **heparan sulfate**, and **keratan sulfate**.

➤ **Proteoglycans** consist of a **core protein from which many GAGs extend**. These large molecules are shaped like a bottlebrush.

- may attach to hyaluronic acid via their core proteins to form large complex aggregates.

- are responsible for the highly viscous character of the ground substance.

- act as binding sites for **growth factors** (e.g., fibroblast growth factor) and **other signaling molecules**. They confer unique attributes to the extracellular matrix in certain locations (e.g., **selective permeability** in the filtration barrier of the glomerulus).

➤ **Glycoproteins** - compounds that contain a **protein moiety** to which **carbohydrates** are attached. promote adhesion between the cell and the matrix.

- **Fibronectin** - Interactions at these sites help to mediate normal cell adhesion and migration.
- **Laminin** - adhesion of epithelial cells to the basal lamina, a structure rich in laminin
- **Integrins** - The interactions that integrins mediate between the ECM and the cytoskeleton operate in both directions and play an important role in orienting both the cells and the ECM in tissues.

B. Fibers - are long and slender protein polymers.

1. **Collagen fibers**. Although there are 25 different types, the most common ones are type I and type III collagen, both consisting of many closely packed tropocollagen fibrils.

- are produced in a two-stage process, involving both intracellular and extracellular events.

- have great tensile strength, which imparts both flexibility and strength to tissues containing them.

- Bone, skin, cartilage, tendon, and many other structures of the body contain collagen fibers.

2. **Reticular fibers**

- are extremely thin

- composed primarily of type III collagen; they have higher carbohydrate content than other collagen fibers (thus stain black with silver salts)

3. **Elastic fibers**

- are coiled and may be stretched up to 150% of their resting length.

- They are composed of elastin and microfibrils of fibrillin.

- require special staining to be observed by light microscopy.

C. Connective Tissue Cells

A. Fixed Cells

- originate locally and remain in the connective tissue.

1. **Fibroblasts** - arise from mesenchymal cells and are the predominant cells in connective tissue proper. - they often possess an oval nucleus with two or more nucleoli.

- undergo mitosis only in **wound healing**.

- However, they may differentiate into other cell types under certain conditions.(may differentiate to adipose cells, or chondrocytes, or under pathological conditions, may differentiate to osteoblasts)

- Active fibroblasts are spindle shaped (fusiform) and contain well-developed RER and many Golgi complexes.

-Synthetically active, they produce procollagen and other components of the extracellular matrix.

2. **Pericytes**- are derived from embryonic mesenchymal cells and may retain a pluripotential role.

- contain actin, myosin, and tropomyosin and may function **as contractile cells** that modify **capillary** blood flow.

-During blood vessel formation and repair, they may differentiate into smooth muscle cells as well as endothelial cells.

3. **Adipose cells (adipocytes)**- arise from mesenchymal cells and perhaps from fibroblasts.

- fully differentiated cells so no cell division. However, they do increase in number in early neonatal life. - They are surrounded by a basal lamina and are responsible for the synthesis, storage, and release of fat.

a. Unilocular adipose cells (white adipose tissue)- contain a **single large** fat droplet which pushes the cytoplasm and nucleus to a thin rim.

- have receptors for insulin, growth hormone, norepinephrine, and glucocorticoids to control the uptake and release of free fatty acids and triglycerides.

- surrounded by a basal lamina and are responsible for the synthesis, storage, and release of fat.

b. Multilocular adipose cells (brown adipose tissue) - are smaller than unilocular adipose cells and the fat is stored as many small fat droplets, and thus the spherical nucleus is centrally located.

4. **Mast cells** - arise from myeloid stem cells in bone marrow and usually reside near small blood vessels.

- Although they share many structural and functional characteristics with basophils, they develop from different precursors and are not related.

- These cells are one of the largest cells of connective tissue proper.

- their cytoplasm is filled with coarse, deeply stained **metachromatic** granules; their contents (known as **primary mediators**).

- Two populations of mast cells exist.

- **Connective tissue mast cells** possess secretory granules containing **heparin**.
- the smaller **mucosal mast cells** whose secretory granules contain **chondroitin sulfate**, is located in the mucosa of the alimentary canal and of the respiratory tract.

- When mast cells become activated during a type I hypersensitivity reaction, phospholipids of their cell membranes can be converted into arachidonic acid by the enzyme phospholipase A2. Arachidonic acid is, in turn, converted into **secondary mediators** .

Table 1 Major mediators released by Mast cells

Substance	Intracellular Source	Action
Primary mediators		
Histamine	Granules	Vasodilator; increases vascular permeability; causes contraction of bronchial smooth muscle; increases mucus production
Heparin	Granules	Anticoagulant; inactivates histamine
Eosinophil chemotactic factor	Granules	Attractant for eosinophils to site of inflammation
Neutrophil chemotactic factor	Granules	Attractant for neutrophils to site of inflammation
Aryl sulfate	Granules	Inactivates leukotriene C ₄ , limiting inflammatory response
Chondroitin sulfate	Granules	Binds and inactivates histamine
Neutral proteases	Granules	Protein cleavage to activate complement; increases inflammatory response
Secondary mediators		
Prostaglandin D ₂	Membrane lipid	Causes contraction of bronchial smooth muscle; increases mucus secretion; vasoconstriction
Leukotrienes C ₄ , D ₄ , E ₄	Membrane lipid	Vasodilators; increases vascular permeability; contraction of bronchial smooth muscle
Bradykinins	Membrane lipid	Causes vascular permeability; responsible for pain sensation
Thromboxane A ₂	Membrane lipid	Causes platelet aggregation; vasoconstriction
Platelet-activating factor	Activated by phospholipase A ₂	Attracts neutrophils and eosinophils; causes vascular permeability; contraction of bronchial smooth muscle

B. Transient Cells

- originate elsewhere and remain only temporarily in connective tissue

1. Macrophages

- are the principal phagocytosing cells of connective tissue.

-are responsible for removing large particulate matter and assisting in the immune response.

- secrete substances that function in wound healing.

- originate in the bone marrow as monocytes, circulate in the blood- stream, then migrate into the connective tissue, where they mature into functional macrophages

- are members of the mononuclear phagocyte system. They display FcεRI receptors and receptors for complement.

- When activated, they display filopodia, an eccentric kidney-shaped nucleus, phago- cytic vacuoles, lysosomes, and residual bodies.

- When stimulated, they may fuse to form foreign body giant cells. These multinucleated cells surround and phagocytose large foreign bodies.

2. **Lymphoid cells** - arise from lymphoid stem cells during hemopoiesis .

-They are located throughout the body in the subepithelial connective tissue and accumulate in the respiratory system, GIT, and elsewhere in areas of chronic inflammation.

a. T lymphocytes(T cells) - initiate the cell-mediated immune response.

b. B lymphocytes (B cells)- following activation by an antigen, differentiate into plasma cells, which function in the humoral immune response.

c. Natural killer cells (NK cells) - lack the surface determinants characteristic of T and B lymphocytes but may display cytotoxic activity against tumor cells.

3. Plasma cells are antibody-manufacturing cells that arise from activated B lymphocytes and are responsible for humoral immunity.

- heterochromatin in eccentric nucleus, which appear to be arranged in a **wheel-spoke fashion**.
- Their cytoplasm is **deeply basophilic** because of an abundance of RER.
- They are most abundant at wound entry sites or in areas of chronic inflammation.

4. Granulocytes - are white blood cells that possess cytoplasmic granules and arise from myeloid stem cells during hemopoiesis.

- At sites of inflammation, they leave the bloodstream and enter the loose connective tissue, where they perform their specific functions

a. **Neutrophils** - phagocytose, kill, and digest bacteria at sites of acute inflammation. Pus is an accumulation of dead neutrophils, bacteria, extracellular fluid, and additional debris at an inflammatory site.

b. **Eosinophils** - bind to antigen-antibody complexes on the surface of parasites (e.g., helminths) and then release cytotoxins that damage the parasites.

(1) They are most prevalent at sites of chronic or allergic inflammation.

(2) Eosinophils are attracted by eosinophil chemotactic factor (ECF), which is secreted by mast cells and basophils, to sites of allergic inflammation. There, eosinophils release enzymes that cleave histamine and leukotriene C, thus moderating the allergic reaction.

(3) These cells also phagocytose antibody-antigen complexes.

c. **Basophils** - are similar to mast cells in that they possess

- FcεRI receptors;
- their granules house the same primary mediators;
- and the same secondary mediators are manufactured de novo from the phospholipids of their plasmalemma.

- They differ, however, in that they circulate via the bloodstream, whereas mast cells do not.

Types of Connective Tissue

Classification is based on the proportion of cells to fibers and on the arrangement and type of fibers

A. Embryonic connective tissue

1. **Mucous tissue (Wharton jelly)** - is a loose connective tissue that is the main constituent of the umbilical cord.

- It consists of a jellylike matrix with some collagen fibers in which large stellate fibroblasts are embedded.

2. **Mesenchymal tissue** - is found only in embryos.

- It consists of a gel-like amorphous matrix containing only a few scattered reticular fibers, in which **star-shaped**, pale-staining **mesenchymal cells** (which are pluripotent) are embedded.

B. Connective tissue proper

1. **Loose connective tissue (areolar tissue)** - possesses fewer fibers but more cells than dense connective tissue.

- This tissue is well vascularized, flexible, and not very resistant to stress.

- It is more abundant than dense connective tissue and is the connective tissue that fills in the spaces just deep to the skin.

2. **Dense connective tissue** - contains more fibers but fewer cells than loose connective tissue.

a. **Dense, irregular connective tissue** (most common),

- fiber bundles that have no definite orientation.

- This tissue is characteristic of the dermis and capsules of many organs.

b. **Dense regular connective tissue**,

- are arranged in a uniform parallel fashion.

- It is present only in tendons and ligaments and may be collagenous or elastic.

3. **Elastic tissue** is composed of coarse, branching elastic fibers with a sparse network of collagen fibers and some fibroblasts filling the interstitial spaces.

- It is present in the dermis, lungs, elastic cartilage, and elastic ligaments and in large (conducting) blood vessels, where it forms fenestrated sheaths.

4. **Reticular tissue** - consists mostly of a network of branched reticular fibers (type III collagen).

- This tissue invests liver sinusoids, smooth muscle cells, and fat cells and forms the stroma of lymphatic organs, bone marrow, and endocrine glands.

- It also forms the reticular lamina of basement membranes.

5. **Adipose tissue** - is the primary site for storage of energy (in the form of triglycerides) and has a rich neurovascular supply.

a. **White adipose tissue** is composed of unilocular adipose cells.

- This tissue constitutes nearly all of the adult adipose tissue throughout the body.

- It stores and releases lipids

- also secretes leptin, a hormone that stimulates cells in the hypothalamus, to regulate appetite and to increase energy consumption.

b. **Brown adipose tissue** - is composed of multilocular adipose cells, which contain many large mitochondria.

- This tissue is capable of generating

- This tissue is found in infants (also in hibernating animals) and is much reduced in adults.

C. Specialized connective tissue

- Cartilage and bone.; blood

1. Cartilage

- avascular** specialized **fibrous connective tissue**

- support soft tissues and assist in the development and growth of long bones.

Types of Cartilage

A. Hyaline cartilage - is the most abundant cartilage in the body

- it also serves as a temporary skeleton in the fetus until it is replaced by bone.

- **type II collagen**

- **Perichondrium** is a layer of dense, irregular connective tissue that surrounds hyaline cartilage except at articular surfaces.

(1) It consists of an **outer fibrous layer** containing **type I collagen**, fibroblasts, and blood vessels and an **inner cellular layer** containing chondrogenic cells and chondroblasts.

(2) It **provides the nearest blood supply** to the avascular cartilaginous tissue.

- **Chondroblasts** manufacture the cartilage matrix through which nutrients and waste materials pass
- **Chondrocytes** are mature cartilage cells that are embedded within **lacunae** in the matrix.

(1) They arise by differentiation of mesenchymal **chondrogenic cells** and from chondrogenic cells within the inner layer of the perichondrium into chondroblasts, which are the earliest cells to produce cartilage matrix. Once these cells become totally enveloped by matrix, they are referred to as **chondrocytes**

(2) Chondrocytes located **superficially** are ovoid and positioned with their longitudinal axis parallel to the cartilage surface. Those located **deeper** are more nearly spherical and may occur in groups of four to eight cells (**isogenous groups**).

2. Histogenesis of hyaline cartilage is similar to that of elastic cartilage and fibrocartilage

a. Interstitial growth results from cell **division of preexisting chondrocytes**.

-This type of growth occurs only during the early stages of cartilage formation and in articular cartilage and the epiphyseal plates of long bones.

b. Appositional growth - results from differentiation of chondrogenic cells in the perichondrium.

- This type of growth results in the formation of chondroblasts and/or new chondrocytes, which elaborate a new layer of cartilage matrix at the periphery.

3. Degeneration of hyaline cartilage occurs when chondrocytes undergo hypertrophy and die and the matrix becomes calcified, a process that becomes more frequent with age.

- Degeneration of hyaline cartilage is a normal part of endochondral bone formation

B. Elastic cartilage. - possesses a perichondrium and is nearly identical to hyaline cartilage except for a network of elastic fibers, which impart a **yellowish** color.

- Although it contains type II collagen, it is less prone to degeneration than hyaline cartilage and is located in areas where **flexible support** is required.

C. Fibrocartilage - **lacks** an identifiable perichondrium.

-It is characterized by alternating rows of chondrocytes surrounded by scant matrix and thick parallel bundles of **type I collagen** fibers.

- located in areas where **support and tensile strength** are required.

table 7.1 Cartilage Types, Characteristics, and Locations

Type of Cartilage	Identifying Characteristics	Perichondrium	Location
Hyaline	Type II collagen, basophilic matrix, chondrocytes usually arranged in groups (isogenous groups)	Perichondrium usually present except on articular surfaces	Articular ends of long bones, nose, larynx, trachea, bronchi, ventral ends of ribs, template for endochondral bone formation
Elastic	Type II collagen elastic fibers	Perichondrium present	Pinna of ear, auditory canal and tube, epiglottis, some laryngeal cartilages
Fibrocartilage	Type I collagen, acidophilic matrix, chondrocytes arranged in parallel rows between bundles of collagen, always associated with dense collagenous connective tissue and/or hyaline cartilage	Perichondrium absent	Intervertebral discs, articular discs, pubic symphysis, insertion of tendons, meniscus of knee

BONE

- **calcified** extracellular matrix

- protect vital organs, support fleshy structures, and provide a calcium reserve (bone contains about 99 % of the body's calcium)

.- **Applied pressure results in bone resorption. Applied tension results in bone formation.**

B. Structure

1. Bone matrix

a. The **inorganic (calcified) portion of the bone matrix** (about 65% of the dry weight)- is composed of minerals primarily of **hydroxyapatite crystals** $\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$.

b. The **organic portion of the bone matrix (osteoid)** (about 35% of the dry weight) consists primarily of **type I collagen** (95%).

- It has a ground substance that contains **chondroitin sulfate** and **keratan sulfate**.

2. The **periosteum** is a layer of **noncalcified** connective tissue covering bone on its **external** surfaces, except at synovial articulations and muscle attachments.

a. It is composed of an outer dense **fibrous** collagenous layer and an inner cellular **osteoprogenitor (osteogenic)** layer.

b. **Sharpey fibers** (type I collagen) attach the periosteum to the bone surface.

c. The periosteum functions to distribute blood vessels to bone.

3. The **endosteum** is a thin specialized connective tissue that lines the **marrow cavities** and supplies **osteoprogenitor cells** and **osteoblasts** for bone growth and repair.

C. Bone cells

1. Osteoprogenitor cells

- located in the **periosteum** and the **endosteum**.

- capable of differentiating into osteoblasts. However, at **low oxygen tensions**, may change into **chondrogenic cells**.

2. Osteoblasts

- They possess cytoplasmic processes with which they contact the processes of other osteoblasts and osteocytes and form **gap junctions**.

- These cells become entrapped in **lacunae** but maintain contact with other cells via their cytoplasmic processes. Entrapped osteoblasts are known as osteocytes.

3. Osteocytes

- **mature bone cells** housed in their own lacunae.

- have narrow cytoplasmic processes that extend through **canaliculi** in the calcified matrix

- maintain communication with each other via **gap junctions** between their processes.

- are nourished and maintained by nutrients, metabolites, and signal molecules carried by the extracellular fluid that flows through the lacunae and canaliculi.

4. Osteoclasts are large, motile, multinucleated cells (up to 50 nuclei) that resorb bone (osteolysis).

- They are derived from cells of the **mononuclear-phagocyte system**.

- They form and reside in depressions known as **Howship's lacunae**, which represent areas of bone resorption.

- acidophilic cytoplasm

- The **ruffled border** is the site of active bone resorption.

d. Bone resorption - involves the following events:

(1) Osteoclasts secrete **acid**, which decalcifies the surface layer of bone.

(2) Acid hydrolases, collagenases, and other proteolytic enzymes secreted by osteoclasts then **degrade the organic portion**

(3) Osteoclasts resorb the organic and inorganic residues of the bone matrix and release them into connective tissue capillaries.

D. Classification of bone

1. Gross observation - cross-sections of bone reveals two types.

a. Spongy (cancellous) bone - , which is composed of interconnected trabeculae.

- Bony trabeculae surround cavities filled with bone marrow.

- Spongy bone is always surrounded by compact bone.

b. Compact (dense) bone has no trabeculae or bone marrow cavities.

2. Microscopic observation of bone reveals two types:

a. Primary bone, also known as **immature** or **woven** bone

- osteocytes and large, irregularly arranged type I collagen bundles

- It has a low mineral content.

- It is the first compact bone produced during fetal development and bone repair.

- It is **remodeled and replaced by secondary bone** except in a few places (e.g., tooth sockets, near suture lines in skull bones, and at insertion sites of tendons).

b. Secondary bone, is also known as **mature** or **lamellar** bone

- Secondary bone is the compact bone of adults.

- It has a **calcified matrix arranged in regular layers, or lamellae**.

- It contains osteocytes in lacunae between, and occasionally within, lamellae.

E. Organization of lamellae in compact bone (e.g., diaphysis of long bones) is characteristic and consists of the following elements

1. Haversian systems (osteons) - are long cylindrical structures that run approximately parallel

to the long axis of the diaphysis.

- are composed of 4 to 20 lamellae surrounding a central haversian canal, which contains blood vessels, nerves, and loose connective tissue.

- They are interconnected by **Volkman canals**, which also connect to the periosteum and endosteum and carry the **neurovascular supply**.

2. Interstitial lamellae are irregularly shaped lamellae between haversian systems.

- They are remnants of remodeled haversian systems.

3. Outer and inner circumferential lamellae are located at the external and internal surfaces of the diaphysis, respectively.

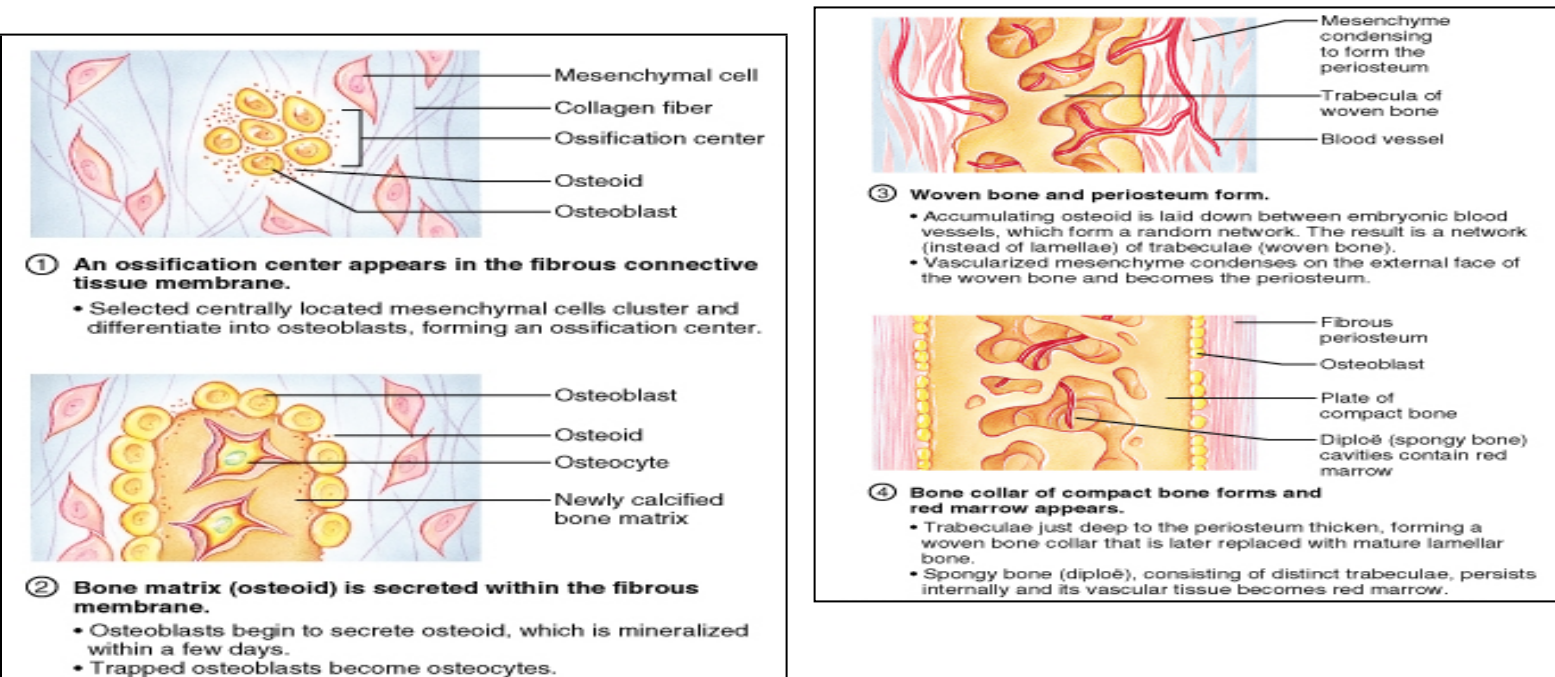
F. Histogenesis of bone

- two processes and both produce bones physiologically identical.

- combination of bone formation and resorption- remodeling- which occurs throughout life.

1. Intramembraneous bone formation

- for formation of most flat bones (Eg. Skull, mandible, maxilla)



2. Endochondral Ossification

- Most bones formed this way
- Cartilage model replaced by bone
- Replacement begins in diaphysis (primary ossification centers)
- Replacement follows in epiphyses (secondary ossification centers)

• Zones of epiphyseal plates

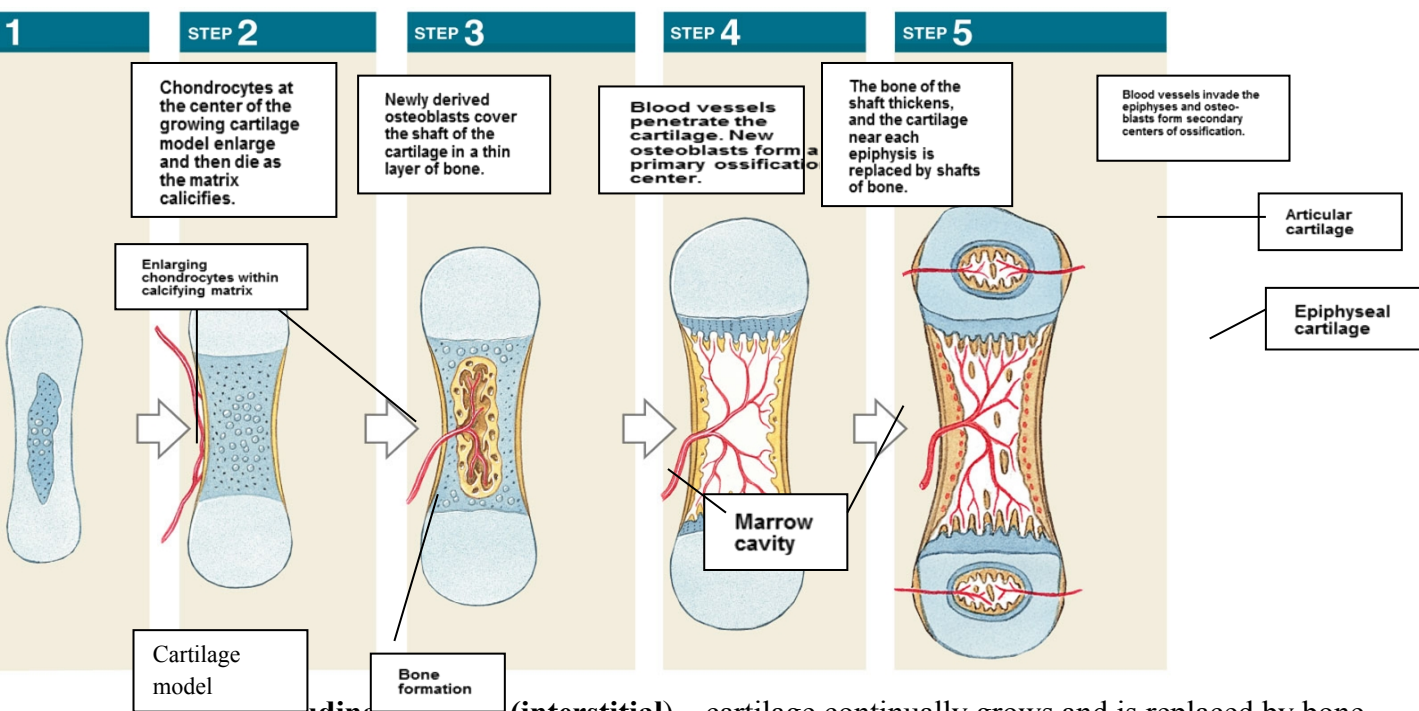
a. Zone of reserve is located at the epiphyseal side of the plate and possesses small, randomly arranged, inactive chondrocytes.

b. Zone of proliferation is a region of rapid mitotic divisions, giving rise to rows of isogenous cells.

c. **Zone of cell hypertrophy and maturation** is the region where the chondrocytes are greatly enlarged.

d. **Zone of calcification** is the region where remnants of cartilage matrices become calcified and chondrocytes die.

e. **Zone of ossification** is where bone is elaborated upon the calcified cartilage, followed by the resorption of the calcified bone-calcified cartilage complex.



• **Longitudinal Growth (interstitial)** – cartilage continually grows and is replaced by bone

- Bones lengthen entirely by growth of the **epiphyseal plates**
- **Appositional growth** – growth of a bone by addition of bone tissue to its surface.
- Growing bones widen as they lengthen

Role of vitamins in bone formation

1. Vitamin D

- is necessary for **absorption of calcium** from the small intestine.
- Vitamin D deficiency causes **rickets** in children and **osteomalacia** in adults
- is also necessary for **bone formation** (ossification).
- excess causes bone resorption.

2. Vitamin A

- deficiency inhibits proper bone formation and growth, while an excess accelerates ossification of the epiphyseal plates.
- deficiency or excess results in **small stature**.

3. Vitamin C

- is necessary for **collagen formation**.
- deficiency results in **scurvy**,

Role of hormones in bone formation

1. Parathyroid hormone

- activates osteoclasts to **resorb bone**, thus **elevating blood calcium levels**.

Calcitonin

- eliminates the ruffled border of osteoclasts.
- **inhibits bone-matrix resorption**, thus preventing the release of calcium.

3. Pituitary growth hormone

- stimulates overall growth, especially that of epiphyseal plates.
- excess during growing years causes **pituitary gigantism** and in adults causes **acromegaly**.
- deficiency during growing years causes **dwarfism**.

Nervous tissue

- The nervous system, by far the most complex system in the body, is formed by a network of many billion nerve cells (neurons), all assisted by many more supporting cells called glial cells.
- Nerve tissue is distributed throughout the body as an integrated communications network, anatomically organized in to two major divisions:
 - Central nervous system (CNS): consisting of the *brain and spinal cord*, developed from neural tube
 - Peripheral nervous system (PNS): composed of the cranial, spinal, and peripheral nerves and *ganglia*, developed from neural crest cells

Note: both central and peripheral nerve tissue is composed of neurons and neuroglial cells

- Neurons have ion gradient (membrane potential) across their cell membrane (neurolemma) that can be changed rapidly in response to stimuli (muscle cells and some gland cells can do the same). Thus neurons are called *excitable or irritable cells* (responding to stimuli). The rapid change or reversal of ion gradient is known as membrane depolarization. Membrane depolarization is called *action potential (depolarization wave or nerve impulse)* when it spreads and propagates across the neurons entire membrane.

The neuron

The functional unit nervous tissue in both the CNS and PNS, composed of

- **Cell body(soma, priakaryon):**
 - contains spherical, unusually large, euchromatic (pale staining) nucleus and most organelles in the neuroplasm
 - serve as synthetic or trophic centre of the neuron

- receive nerve endings conveying excitatory or inhibitory stimuli generated in other nerve cells
- the neuroplasm contains nissl substance (nissl body, chromatophilic substance)- concentrated RER and polyribosome which appear basophilic histologically
- Golgi apparatus is found only in the soma, but mitochondrial is also found in axon terminal abundantly
- the soma also contain intermediate filaments (neurofilaments) and microtubules

➤ **Dendrites:**

- Short and branched processes form the cell body and numerous in most neurons
- The principal signal reception and processing site of a neuron
- Usually each dendrite has many synaptic connection sites called dendritic spines
- Dendritic spines increase signal receptive area and enable a neuron connect with a great number of axon terminals from other neurons
 - ✓ Purkinje cells of cerebellum can make up to 200,000 synaptic connections
- Dendritic spines are important in bringing about synaptic plasticity underlying adaptation, learning and memory
- Synaptic plasticity- activity dependent change in synaptic strength

➤ **Axon :**

- Single and longest process of the cell body that could be myelinated or unmyelinated
- Has uniform diameter unlike dendrites
- originate from a pyramid-shaped region of the perikaryon called the axon hillock
- Plasma membrane of the axon is often called the axolemma and its contents are known as axoplasm.
- Action potential initiate from the initial segment(unmyelinated area just beyond the axon hillock)-the site where an inhibitory and excitatory signals impinging on the neuron are sum up to decide 'to produce or not to produce' action potential
- To lesser extent axon can give collateral branches(such as in, interneurons and some motor neurons)
- Axoplasm contains mitochondria, microtubules, neurofilaments, and smooth ER, but polyribosomes or RER,
- Axon ends in *axon terminal* that make synaptic connection with dendrites of other neurons
- Have bidirectional transport system
 - ✓ **Anterograde transport**- from perikaryon to axonterminal (transport of organelles and macromolecules)
 - ✓ **Retrograde transport**- from axon-terminal to perikaryon (transport of viruses and toxins)

❖ Neurons communicate through synapse(specialized junctions between neurons or between neuron and effector organ). Synapse composed of presynaptic neuron, synaptic cleft and postsynaptic neuron or effector organ. Morphologically there are three synapses:

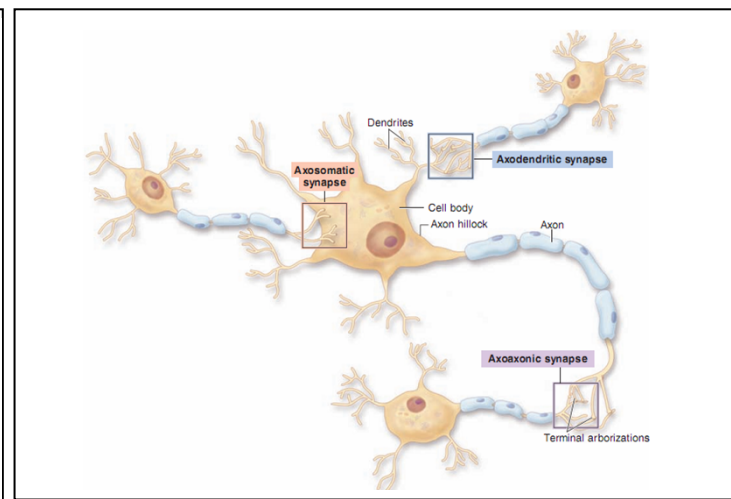
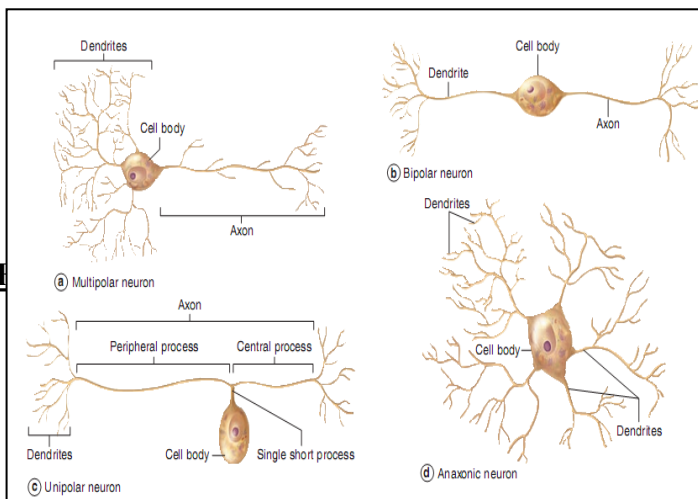
- **Axodendritic** - between axons and dendrites;
- **Axosomatic** - between axons and the cell body
- **Axoaxonic** - between axons and axons

Based on mechanism of conduction of the nerve impulses and the way the action potential is generated in the target cells. Thus, synapses may also be classified as the following.

- **Chemical synapses:** impulse is conducted across synapse by the release of chemical substances (neurotransmitters) from the presynaptic neuron
 - ✓ Main synapse in mammals
 - ✓ Between neurons, neuron and effector organs
- **Electrical synapses:** contain gap junctions that permit movement of ions (direct spread of electrical current) between cells
 - ✓ Common in invertebrates
 - ✓ mammalian equivalents include the gap junctions in smooth muscles and cardiac muscles

Difference among neurons lie in the

- Size of cell body- as large as pyramidal cells of cerebral cortex or as small as granular cells of cerebellar cortex
 - Length of axons- as long as motor neurons or as small as some interneurons
 - Number of processes - can be classified into four groups
- ❖ **Multipolar neurons-**
 - ✓ Has One axon and two or more dendrites
 - ✓ The most common neurons, most of interneurons
 - ❖ **Bipolar neurons**
 - ✓ One dendrite and one axon
 - ✓ Found in special sensory organs- retina, olfactory mucosa, taste buds, inner ear (cochlear and vestibular ganglia)
 - ❖ **Unipolar or pseudounipolar neurons**
 - ✓ Has single process that bifurcates close to the perikaryon with
 - ✓ longer branch extending to a peripheral ending and the other toward the CNS
 - ✓ found in spinal ganglia (dorsal root ganglia) and most cranial ganglia (such as trigeminal ganglia)- are first degree sensory neurons
 - ❖ **Anaxonic neurons**
 - ✓ Has many dendrites but no true axon
 - ✓ Do not produce action potential (b/c no axon) but regulate electrical changes of adjacent neurons
 - ✓ Mostly are interneurons (note that interneurons could also be multipolar)



- ❖ **Sensory neurons(afferent neurons)**- receive stimuli from receptor to CNS
 - ✓ Somatic afferent neurons- convey sensations of pain, temperature, touch, and pressure from the body surface, and pain and proprioception (nonconscious sensation) from organs within the body (muscles, tendons, and joints)
 - ✓ Visceral(autonomic) afferent neurons- transmit mainly pain impulses from internal organs, mucous membranes, glands, and blood vessels
- ❖ **Motor neurons(efferent neurons)**- send impulse from CNS or ganglia to effector organs (muscle fibers and glands)
 - ✓ Somatic motor neurons- are under voluntary control innervate skeletal muscles,
 - ✓ Autonomic motor neurons – control involuntary activities of glands, cardiac muscle and most smooth muscles
- ❖ **Interneurons(intercalated neurons)** – form complex functional networks or circuits and compose about 99% of neurons in the CNS

Clinical---Alzheimer disease,

- common type of dementia in the elderly, affects both neuronal perikarya and synapses within the cerebrum
- Functional defects are due to
 - ✓ **neurofibrillary tangles**, which are accumulations of tau protein associated with microtubules of the neuronal perikaryon and axon hillock regions,
 - ✓ **neuritic plaques**, which are dense aggregates of β -amyloid protein that form around the outside of these neuronal regions

Myelination

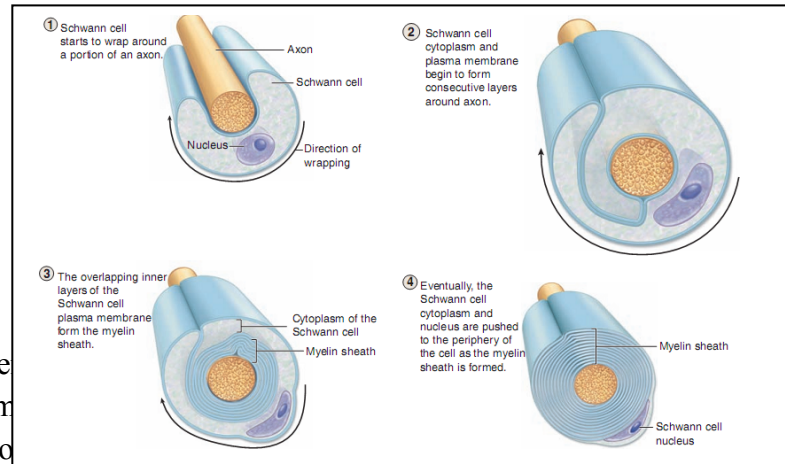
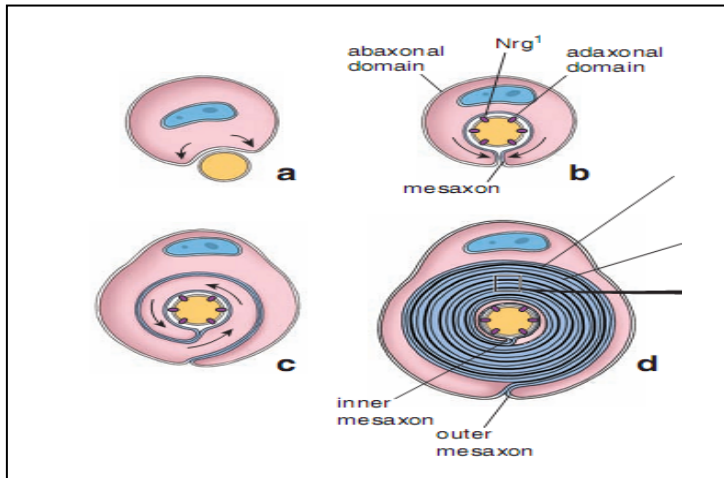
Myelin sheath

- lipid-rich layer(composed of about 80% lipid) that surrounds the axons
 - ✓ myelin is large lipoprotein complex
- Isolates (insulate) the axon from the surrounding extracellular compartment of endoneurium.
- Ensures the rapid conduction of nerve impulses.
- The axon hillock and the axon terminal where the axon synapses with its target cells are not covered by myelin
- Produced by the Schwann cells in PNS and oligodendritic cells in the CNS

Myelination in PNS

- begins when a Schwann cell surrounds the axon and its cell membrane becomes polarized
- the schwann membrane forming myelin sheath have three domains
 - ✓ **abaxonal domain** : part of the Schwann cell membrane that is exposed to the external environment or endoneurium,
 - ✓ **adaxonal or periaxonal domain**: plasma membrane, which is in direct contact with the axon
 - ✓ **mesaxon**: the third domain which is a double membrane that connects the abaxonal and adaxonal membranes and encloses the narrow extracellular space

- outer mesaxon -mesaxon of the last layer
- inner mesaxon -
- The myelin sheath develops from compacted layers of Schwann cell mesaxon wrapped concentrically around the axon



- **inner collar of Schwann cell cytoplasm**
 - ✓ Internal to the concentric layers of the developing myelin sheath
 - ✓ surrounded by the adaxonal plasma membrane
- **Node of Ranvier:**
 - ✓ Represents the junction between two adjacent Schwann cells.
 - ✓ Devoid of myelin sheath
 - ✓ At these nodes the
 - ★ axolemma is exposed to ions in the interstitial fluid and
 - ★ has a much higher concentration of voltage-gated Na⁺ channels,
 - ★ which renew the action potential and produce saltatory conduction (L. saltare, to jump) of nerve impulses,
 - ★ saltatory conduction- causes the rapid conduction of impulses in myelinated than in unmyelinated
- **internodal segment:**
 - ✓ The myelin between two sequential nodes of Ranvier
 - ✓ A schwann cell form only single internodal segment
- cytoplasm of the schwan cell is found at several location in the concentric layers
 - ✓ **The inner collar of Schwann cell cytoplasm** - between the axon and the myelin;
 - ✓ **Schmidt-Lanterman clefts(myelin clefts)**- small islands within successive lamellae of the myelin;
 - ✓ **perinodal cytoplasm** - at the node of Ranvier;
 - ✓ **the outer collar of perinuclear cytoplasm**- around the myelin
- Unmyelinated axons in the peripheral nervous system are enveloped by Schwann cells and their external lamina
 - ✓ Single axon(somatic) or group of axons(autonomic) fit into grooves in the surface of the cell
 - ✓ Voltage –gated ion channels are evenly distributed along the axon
 - ✓ no nodes of Ranvier and thus no sultatory conduction resulting in slow impulses

Myelination in CNS

- formed by concentric layers of oligodendrocyte plasma membrane
- multiple processes of a single oligodendrocyte may myelinate one axon or several nearby axons (producing multiple internodal segments)

Difference in CNS and PNS myelination

- myelin in CNS
 - ✓ exhibits fewer Schmidt-Lanterman clefts
 - ✓ little or no cytoplasm may be present in the outermost layer of the myelin sheath
 - ✓ myelin of adjacent axons may come into contact(may share an internodal line)
 - ✓ the nodes of Ranvier in the CNS are larger than those in the PNS
 - ✓ unmyelinated neurons in the CNS are often found to be bare (they are not embedded in glial cell processes)

Supporting cells and neuroglia

Glial cells

- Supporting cells of the nervous tissue which develop from neural tube as neurons of CNS
- ten times more abundant in the mammalian brain than the neurons
- substitute the connective tissue found in other tissues(no connective tissue in brain except around larger vessels)
- support, protect and nourish neurons

Central neuroglia

- **Oligodendrocytes**
 - ✓ predominant glial cells in CNS white matter,
 - ✓ produce the myelin sheaths around axons that provide the electrical insulation for neurons in the CNS
- **Astrocytes**
 - ✓ have a large number of radiating processes
 - ✓ the most numerous glial cells of the CNS and
 - ✓ structurally grouped in to
 - ★ fibrous astrocytes- with relatively few, long processes and are typical in white matter;
 - ★ protoplasmic astrocytes - with many shorter, branched processes which are predominate in the gray matter
 - ✓ Functions associated with various astrocytes
 - ★ Regulating the extracellular ionic concentrations around neurons, with particular importance in buffering extracellular K⁺ levels
 - ★ Cover capillaries of the BBB by their expanded foot processes
 - ★ Regulating vasodilation within the CNS and moving nutrients, wastes, and other metabolites between neurons and capillaries

- ★ Extending processes that associate with or cover synapses in the CNS, affecting the formation, function, and plasticity of synapses
- ★ Forming a barrier layer of expanded processes, called the glial limiting membrane just beneath the pia matter of the meninges
- ✓ Most brain tumors are astrocytomas derived from these glial cells
- **Ependymal Cells**
 - ✓ columnar or cuboidal cells that line the ventricles of the brain and central canal of the spinal cord
 - ✓ have cilia at their apical ends, which facilitate the movement of cerebrospinal fluid (CSF)
 - ✓ joined apically by junctional complexes similar to those of epithelial cells
 - ✓ but unlike a true epithelium there is no basal lamina
- **Microglia**
 - ✓ small cells with short irregular processes evenly distributed throughout gray and white matter
 - ✓ Less numerous than oligodendrocytes or astrocytes but nearly as common as neurons
 - ✓ Unlike other glial cells, microglia are migratory, scanning the tissue for damaged cells and invading microorganisms
 - ✓ Are major immune defence cells in the CNS
 - ✓ Do not originate from neural progenitor cells like other glia, but from circulating blood monocytes
 - ✓ When activated by damage or invaders, microglia retract their processes, proliferate, and assume the morphologic characteristics and functions of antigen presenting cells

Clinical---Multiple sclerosis (MS)

- Autoimmune disorder where myelin sheaths are damaged.
- T lymphocytes and microglia, which phagocytise and degrade myelin debris, play major roles in progression of this disease

Peripheral neuroglia

- **Schwann Cells**
 - ✓ found only in the PNS and differentiate from precursors in the neural crest
 - ✓ are trophic and produce myelination of peripheral neurons
 - ✓ forms only one intermodal segment around a segment of one axon
- **Satellite Cells of Ganglia**
 - ✓ Small cells derived from the embryonic neural crest,
 - ✓ Form an intimate covering layer over the large neuronal cell bodies in the ganglia of the PNS
 - ✓ Exert a trophic or supportive effect on these neurons, insulating, nourishing, and regulating their microenvironments.

Peripheral nervous system

The main components of the peripheral nervous system (PNS) are the

- Nerves – bundle of nerve fibers(axons)
- ganglia,
- nerve endings

Nerve Organization

- Nerve- is bundle of nerve fibers
- Nerve fibers are analogous to tracts in the CNS that contain axons enclosed within sheaths of glial cells
- Nerve is organized that it contains
 - ✓ **Endoneurium**- contain axons and Schwann cells are enclosed within layers of connective tissue(reticular fibers, fibroblasts, capillaries)
 - ✓ **Perineurium**- encloses bundles of multiple endoneurim containing axons and schwann cells – called fascicles
 - These connective tissue form blood-nerve barrier
 - ✓ **Epineurium**- external fibrous connective tissue that cover the nerve and send deep extensions to fill the gap between fascicles
- Most peripheral nerves contain both afferent and efferent fibers- creat communication between CNS and effector organs
- Nerves could be
 - ✓ Sensory nerves-containing only afferent fibers, cranial nerves(I, II,VIII)
 - ✓ Motor nerves-containing only effector fibers- cranial nerves(IV, VI,XI,XII)
 - ✓ Mixed nerves-have both afferent and efferent fibers (the most common nerves- all spinal nerves, and cranial nerves-III, V, VII, IX, and X)

Ganglia

- ovoid structures containing neuronal cell bodies and their surrounding glial satellite cells
- supported by delicate connective tissue and surrounded by a denser capsule
- serve as relay stations to transmit nerve impulses,
- two types
 - ✓ **sensory ganglia**-
 - relay afferent impulse to the CNS
 - have pseudounipolar neurons that relay information from the ganglion to spinal cord
 - cranial sensory ganglion- associated with cranial nerves,
 - dorsal root ganglion of spinal nerves
 - ✓ **autonomic ganglia**-
 - small bulbous dilatations in autonomic nerves
 - relay efferent impulse to effector organs

- composed of multipolar neurons
- contains two neurons circuit (preganglionic and postganglionic fibers)
 - **sympathetic ganglion**
 - cell body of preganglionic fiber in thoracic and lumbar spinal cord
 - cell body of postganglionic fiber(the ganglia) located along the vertebral column
 - **parasympathetic ganglion**
 - cell body of preganglionic fiber is in medulla, midbrain and sacral part of spinal cord
 - cell body of postganglionic fibers(the ganglion) located within or near the effector organ, e.g walls of the stomach and intestine
 - sometimes called intramural ganglia
 - lack distinct capsules

Congenital anomalies: Types and Causes (Genetic Factors and Environmental Factors and conditions that affect them)

- A birth defect is a structural abnormality of any type that is present at birth.
- Certain stages of embryonic development are more vulnerable to disruption than others
 - first 2 weeks - teratogenic agents usually kill the embryo or have no effects (spontaneous abortion)
 - **organogenetic period (weeks 3 and 8)** - disrupt development and may cause major congenital anomalies
 - fetal period - produce morphologic and functional abnormalities, particularly of the brain and eyes

Factors determining the capacity of an agent to produce birth defects. Susceptibility to teratogenesis depends on

1. the **genotype** of the conceptus and the *maternal genome*
2. the **developmental stage** at the time of exposure (*3rd–8th weeks*)
3. the **dose and duration of exposure** to a teratogen
4. specific **ways (mechanisms) in which teratogens act** on developing cells and tissues

There are four clinically significant **types of birth defect**: malformation, disruption, deformation, and dysplasia

Malformation:

- A morphologic defect of an organ that results from an **intrinsically abnormal developmental process**
- The primordium of the **organ is abnormal from the beginning**, such as a chromosomal abnormality of a gamete.
- Associated with a disorder of tissue development **often occur in the first trimester** (during formation of structures)

Disruption:

- A morphologic defect of an organ that results from the **extrinsic breakdown of an originally normal developmental process**
- exposure to **teratogens**
- A disruption **cannot be inherited**, but inherited factors can predispose to a disruption

Deformation:

- Due to **mechanical forces that mold a part of fetus over a prolonged period of time**
- Intrauterine compression that may result from **oligohydramnios, leiomyoma or multiple pregnancies**
- often occur in the **2nd or 3rd trimesters (weeks 35-38)**
- such as **meningomyelocele and clubfoot**
- involve the musculoskeletal system and **may be reversible postnatally**

Dysplasia:

- An **abnormal organization of cells in tissue(s)** and its morphologic result(s)
- It is the process and the consequence of **dyshistogenesis (abnormal tissue formation)**
- causally nonspecific and often affects several organs because of the nature of the underlying cellular disturbances.

The **causes of birth defects** are divided into:

- **Genetic factors** such as chromosome abnormalities
- **Environmental factors** such as drugs and viruses
- **Multifactorial inheritance** (genetic and environmental factors acting together)

GENETIC FACTORS

- are the most important causes of birth defects (approximately one third of all defects)
- Two kinds of changes occur in chromosome complements: numerical and structural

Numerical Chromosomal Abnormalities

- usually result from **nondisjunction** as a result, the embryo's cells may be **hypodiploid** (45, X0 (monosomy X) as in Turner syndrome) **or hyperdiploid** (47, as in trisomy 21 or Down syndrome)
 - **Turner syndrome** (45, X0): phenotype is **female**. **Features**: short stature, webbed neck, absence of sexual maturation, broad chest with widely spaced nipples, lymphedema of the extremities and mental retardation
 - **Down syndrome** (trisomy 21): **features**: single transverse palmar *flexion crease* (simian crease) and the *clinodactyly* (incurving) of the fifth digit, up slanting palpebral fissures, epicanthal folds, and flat nasal bridge

- **Edwards syndrome** (Trisomy 18): **features**: growth retardation, clenched fists with characteristic positioning of the fingers (second and fifth ones overlapping the third and fourth ones), short sternum, and low-set malformed ears
- **Patau syndrome** (Trisomy 13): **features**: severe central nervous system malformations, sloping forehead, malformed ears, microphthalmia, bilateral cleft lip and/or palate, polydactyly
- **Klinefelter syndrome** (XXY trisomy): male phenotype, **features**: *gynecomastia*, small testes with aspermatogenesis, often tall with disproportionately long lower limbs
- Trisomy increases frequency as maternal age increases
- **Mosaicism**: two or more cell types containing different numbers of chromosomes, leads to a less severe phenotype
- **Tetrasomy and pentasomy**: extra sex chromosomes do not accentuate sexual characteristics. the greater the number of sex chromosomes present, the **greater the severity of mental deficiency and physical impairment**

Structural Chromosomal Abnormalities

- result from **chromosome breakage**, followed by **reconstitution in an abnormal combination**
- only two aberrations of chromosome structure that are transmitted from a parent to an embryo are inversion and translocation
- **Translocation**: transfer of a piece of one chromosome to a nonhomologous chromosome
- **Deletion**: When a chromosome breaks, part of it may be lost
 - **Cri du chat syndrome** - partial terminal deletion from the short arm of **chromosome 5**. **Features**: weak cat-like cry, microcephaly, severe mental deficiency and congenital heart disease
- **Microdeletion and microduplication**: very small interstitial and terminal deletions
 - **Prader-Willi syndrome**: deletion arises in the **paternal chromosome 15**. **Features**: short stature, mild mental deficiency, obesity, hyperphagia, and hypogonadism.
 - **Angelman syndrome**: deletion arises in the **maternal chromosome 15**. **Features**: severe mental deficiency, microcephaly, brachycephaly (shortness of head), seizures, and ataxic (jerky) movements of the limbs and trunk

Mutant Genes

- **Achondroplasia**: mutation in the **fibroblast growth factor receptor**. **Features**: short stature, short limbs and fingers, normal length of trunk, bowed legs, a relatively large head, prominent forehead, and depressed nasal bridge
- **Fragile X syndrome**: X-linked. **Features**: mental deficiency, long face and prominent ears

ENVIRONMENTAL FACTORS

- By teratogens
- Are not effective in causing defects until cellular differentiation has begun

- exact mechanisms by which environmental factors disrupt embryonic development still remain obscure
- **Alcohol:** Fetal alcohol syndrome, intrauterine growth restriction, mental deficiency, microcephaly, ocular anomalies, joint abnormalities, short palpebral fissures
- **Cigarette smoking:** intrauterine growth restriction, premature delivery, low birth weight, ...

Multifactorial factors

- **isregulation of gene expression by an environmental factor.**
- **Epigenetic regulation:** changes in phenotype (appearance) or gene expression caused by mechanisms other than changes in the underlying DNA sequence.
 - by modifying transcriptional factors, DNA methylation, or histone modification.
- Several birth defects such as stress or altered nutrition rather than due to changes in DNA sequences
 - The sex of the transmitting parent will influence expression or non-expression of certain genes in the offspring

Regulation of development

- Normal embryogenesis is regulated by several complex signaling cascades
- **Transcriptional factors** act by influencing the pattern of gene expression of own cells or adjacent cells.
- **Morphogens**, elicit many responses depending on their level of expression with other cells.
- Many inductive interactions occur between epithelial and mesenchymal cells and are called **epithelial–mesenchymal interactions**
- By **paracrine interactions** or **juxtacrine interactions**.
- **Juxtacrine interactions**
 - three ways: (1) analogous to paracrine signaling. **Notch pathway.**
 - (2) Ligands in the extracellular matrix secreted by one cell interact with their receptors on neighboring cell
 - (3) direct transmission of signals from one cell to another by **gap junctions**
- **Paracrine Signaling Factors:** whereby proteins synthesized by one cell diffuse over short distances to interact with other cells.
 - The diffusible proteins are called **paracrine factors** or **growth and differentiation factors (GDFs)**.
 - Fibroblast Growth Factors
 - Hedgehog Proteins
 - WNT Proteins
 - **The TGF- β Superfamily**

Gametogenesis

- Process of formation and development of gametes, chromosome number is reduced by half (**haploid number**)

- The **first meiotic division is a reduction division** because the chromosome number is reduced from diploid to haploid
- The X and Y chromosomes are not homologs, but they have homologous segments at the tips of their short arms
- By the end of the first meiotic division, each new cell formed (secondary spermatocyte or secondary oocyte) has the haploid chromosome number.
- The *second meiotic division* follows the first division **without a normal interphase** (i.e., without a step of DNA replication)
- the haploid number of chromosomes (23) is retained and each daughter cell (now a single-chromatid chromosome)
- **meiosis**: Allows **random assortment**, **crossing over** of chromosome segments “shuffles”

SPERMATOGENESIS

- takes approximately 2 months
- **Spermatogonia** are dormant in the seminiferous tubules of the testes during the fetal and postnatal periods
- Spermatogonia are transformed into **primary spermatocytes**, the **largest** germ cells in the seminiferous tubules
- primary spermatocyte undergoes a **reduction division**, the first meiotic division, to form two haploid **secondary spermatocytes**
- secondary spermatocytes undergo a **second meiotic division** to form four haploid **spermatids**
- The spermatids are gradually transformed **into four mature sperms** by a process known as **spermiogenesis**
 - loss of cytoplasm, development of the tail, and formation of the acrosome (anterior 2/3 of the head)
 - The **acrosome is derived from the Golgi region** of the spermatid, contains **enzymes** that assist the sperm in penetrating the corona radiata and zona pellucida surrounding the secondary oocyte
- sperms enter the seminiferous tubules then passively to the epididymis where they are stored and become functionally mature

OOGENESIS

- maturation of the oocytes begins before birth and is completed after puberty
- **Oogonia** (primordial female sex cells) enlarge to form **primary oocytes before birth**
- As the primary oocytes form, connective tissue cells surround them and form a **single layer of flattened, follicular cells - primordial follicle**
- As the primary oocyte enlarges during puberty, the follicular epithelial cells become **cuboidal** in shape and then columnar - **primary follicle**
- The primary oocyte becomes surrounded by a covering of amorphous acellular glycoprotein material - **the zona pellucida**

- Primary oocytes begin the first meiotic divisions before birth, but completion of **prophase** does not occur until adolescence
 - The follicular cells secrete a substance, **oocyte maturation inhibitor**
- **No primary oocytes form after birth**, in contrast to the continuous production of primary spermatocytes
- As a follicle matures, the primary oocyte increases in size and **shortly before ovulation**, completes the first meiotic division to give rise to a **secondary oocyte** and the first polar body
- The secondary oocyte receives almost all the cytoplasm, and the *first polar body* receives very little.
- At ovulation, the nucleus of the secondary oocyte **begins the second meiotic division, but progresses only to metaphase**
- If fertilization, the second meiotic division is completed, and most cytoplasm is again retained by one cell, the fertilized oocyte

OVULATION

- Around midcycle, the ovarian follicle, under the influence of FSH and LH, undergoes a sudden growth spurt, producing a cystic swelling or bulge on the surface of the ovary. A small avascular spot, the **stigma**, soon appears on this swelling. Before ovulation, the **secondary oocyte** and some cells of the **cumulus oophorus** detach from the interior of the distended follicle .
- Ovulation is triggered by a surge of LH production. Ovulation usually follows the LH peak by 12 to 24 hours. The LH surge, elicited by the high estrogen level in the blood, appears to cause the stigma to balloon out, forming a vesicle. The stigma soon ruptures, **expelling the secondary oocyte with the follicular fluid**. Expulsion of the oocyte is the result of **intrafollicular pressure** and possibly **contraction of smooth muscle in the theca externa owing to stimulation by prostaglandins**.
- The expelled secondary oocyte is surrounded by the zona pellucida and one or more layers of follicular cells, which are radially arranged as the corona radiata forming the oocyte-cumulus complex. The LH surge also seems to induce resumption of the first meiotic division of the primary oocyte. Hence, mature ovarian follicles contain secondary oocytes.

Clinical

- **ANOVULATION**--Some women do not ovulate (cessation of ovulation— anovulation) because of an **inadequate release of gonadotropins**. In some of these women, ovulation can be induced by the administration of gonadotropins or an ovulatory agent such as clomiphene citrate. This drug stimulates the release of pituitary gonadotropins (FSH and

LH), resulting in maturation of several ovarian follicles and multiple ovulations. The incidence of multiple pregnancy increases when ovulation is induced.

OVARIAN CYCLE

- FSH and LH produce cyclic changes in the ovaries resulting in the development of follicles, ovulation, and corpus luteum formation. During each cycle, FSH promotes growth of several primordial follicles into 5 to 12 primary follicles; however, only one primary follicle usually develops into a mature follicle and ruptures through the surface of the ovary, expelling its oocyte.
- Follicular Development of an ovarian follicle is characterized by:
 - Growth and differentiation of primary oocyte
 - Proliferation of follicular cells
 - Formation of zona pellucida
 - Development of the theca folliculi
- As the primary follicle increases in size, the adjacent connective tissue organizes into a capsule, the **theca folliculi**. The theca soon differentiates into two layers, an internal vascular and glandular layer, the **theca interna**, and a capsule-like layer, the **theca externa**. Thecal cells are thought to produce an angiogenesis factor that promotes growth of blood vessels in the theca interna, which provide nutritive support for follicular development.
- The follicular cells divide actively, producing a **stratified layer** around the oocyte. The ovarian follicle soon becomes oval and the oocyte eccentric in position. Subsequently, **fluid-filled spaces** appear around the follicular cells, which coalesce to form a single large cavity, the **antrum**, which contains follicular fluid. After the antrum forms, the ovarian follicle is called a **vesicular or secondary follicle**. The primary oocyte is pushed to one side of the follicle, where it is surrounded by a mound of follicular cells, the **cumulus oophorus**, that projects into the antrum.
- The follicle continues to enlarge until it reaches maturity and produces a swelling on the surface of the ovary. The **early development** of ovarian follicles is induced by **FSH**, but **final stages** of maturation require **LH** as well.
- Growing follicles produce **estrogen**. The vascular theca interna produces follicular fluid and some estrogen. Its cells also secrete androgens that pass to the follicular cells, which, in turn, convert them into estrogen. Some estrogen is also produced by widely scattered groups of stromal secretory cells, known collectively as the interstitial gland of the ovary.

MENSTRUAL CYCLE

- The menstrual cycle is the time during which the oocyte matures, is ovulated, and enters the uterine tube. The hormones produced by the **ovarian follicles and corpus luteum** (estrogen and progesterone) produce cyclic changes in the endometrium. These monthly changes in the **internal layer of the uterus** constitute the **endometrial cycle**, commonly referred to as the menstrual cycle or period.
- The endometrium is a “**mirror**” of the ovarian cycle because it responds in a consistent manner to the fluctuating concentrations of gonadotropic and ovarian hormones. The average menstrual cycle is **28 days**, with day 1 of the cycle designated as the day on which menstrual flow begins. Menstrual cycles vary in . In 90% of women, ranges between 23 and 35 days. Almost all these **variations** result from **alterations in the duration of the proliferative phase** of the menstrual cycle

Process of the menstrual cycle

- Changes in the estrogen and progesterone levels cause cyclic changes in the structure of the female reproductive tract, notably the endometrium.
1. **Menstrual phase**--The functional layer of the uterine wall is sloughed off and discarded with the menstrual flow, called menses (monthly bleeding), which usually lasts 4 to 5 days.
 2. **Proliferative phase**-- lasts approximately 9 days, coincides with growth of ovarian follicles and is controlled by **estrogen** secreted by these follicles.
 - Surface epithelium reforms and covers the endometrium
 - 2-3X increase **in the thickness of the endometrium**.
 - **glands increase in number and length and the spiral arteries elongate**.
 3. **Luteal phase**--lasts approximately 13 days, coincides with the formation, functioning, and growth of the corpus luteum.
 - The **progesterone** produced by the corpus luteum stimulates the glandular epithelium to secrete a **glycogen-rich material**.
 - Glands become wide, tortuous, and saccular, and the endometrium thickens because of the influence of progesterone and estrogen from the corpus luteum, and because of **increased fluid in the connective tissue**.
 - **Spiral arteries** grow into the superficial compact layer, they become **increasingly coiled**. The venous network becomes complex and large **lacunae** (venous spaces) develop. **Direct arteriovenous anastomoses** are prominent features of this stage.
- ❖ **If fertilization does not occur:**
- Corpus luteum degenerates.

- Estrogen and progesterone levels fall and the secretory endometrium enters an ischemic phase.
- Menstruation occurs.

★ **Ischemic phase**--occurs when the oocyte is not fertilized. Ischemia occurs as the spiral arteries constrict, giving the endometrium a pale appearance. This constriction results from the decreasing secretion of hormones, primarily progesterone, by the degenerating corpus luteum.

- In addition the hormone withdrawal results in the stoppage of glandular secretion, a **loss of interstitial fluid, and a marked shrinking of the endometrium.**
- Toward end of the ischemic phase, the **spiral arteries become constricted** for longer periods. This results in venous stasis and patchy ischemic necrosis (death) in the superficial tissues. Eventually, rupture of damaged vessel walls follows and blood seeps into the surrounding connective tissue. The torn ends of the arteries bleed into the cavity, resulting in a loss of 20 to 80 ml of blood. Eventually, over 3 to 5 days, the entire compact layer and most of the spongy layer of the endometrium are discarded in the menses.
- Remnants of the **spongy and basal layers** remain to undergo regeneration during the subsequent proliferative phase of the endometrium.

If fertilization occurs:-

- **Cleavage** of the zygote and blastogenesis (formation of blastocyst) begin.
- The **blastocyst implants** in the endometrium on approximately the sixth day of the luteal phase (day 20 of a 28-day cycle).
- **Human chorionic gonadotropin**, a hormone produced by the syncytiotrophoblast, keeps the corpus luteum secreting estrogens and progesterone.
- The luteal phase continues and menstruation does not occur.
- The menstrual cycles cease and the endometrium passes into a **pregnancy phase**.
- With the termination of pregnancy, the **ovarian and menstrual cycles resume** after a variable period (usually 6 to 10 weeks if the woman is not breast-feeding her baby).

FERTILIZATION

- The usual site of fertilization is in the ampulla of the uterine tube. If the oocyte is not fertilized there, it slowly passes along the tube to the body of the uterus, where it degenerates and is resorbed. Although fertilization may occur in other parts of the tube, it does not occur in the body of the uterus.

- Chemical signals (attractants), secreted by the oocyte and surrounding follicular cells, guide the capacitated sperms (**sperm chemotaxis**) to the oocyte. .
- The fertilization process takes approximately **24 hours**.
- Carbohydrate binding molecules and gamete-specific proteins on the surface of the sperms are involved in **sperm-egg recognition** and their union.

Phases of Fertilization--is a sequence of coordinated events, defects at any stage in the sequence of these events might cause the zygote to die

- 1) Passage of a sperm through the corona radiata---dispersal of corona radiata result from
 - action of the enzyme hyaluronidase released from the acrosome of the sperm.
 - Tubal mucosal enzymes also appear to assist the dispersal.
 - Movements of the tail of the sperm are also important.
- 2) Penetration of the zona pellucida---Passage of a sperm through the zona pellucida is the important phase in the initiation of fertilization. Formation of a pathway also results from the
 - Action of enzymes released from the acrosome. The enzymes esterases, acrosin, and neuraminidase appear to cause lysis of the zona pellucida, thereby forming a path for the sperm to follow to the oocyte.
 - The most important of these enzymes is acrosin, a proteolytic enzyme. Once the sperm penetrates the zona pellucida, a **zona reaction**—a change in the properties of the zona pellucida—occurs that makes it **impermeable** to other sperms.
- 3) Fusion of cell membranes of the oocyte and sperm---The plasma or cell membranes of the oocyte and sperm fuse and break down at the area of fusion.
 - The head and tail of the sperm enter the cytoplasm of the oocyte, but the sperm's cell membrane (plasma membrane) and mitochondria remains behind.
- 4) Completion of the second meiotic division of oocyte and formation of female pronucleus---Penetration of the oocyte by a sperm activates the oocyte into completing the second meiotic division and forming a mature oocyte and a second polar body.
 - Following decondensation of the maternal chromosomes, the nucleus of the mature oocyte becomes the female pronucleus.
- 5) Formation of the male pronucleus---Within the cytoplasm of the oocyte, the nucleus of the sperm enlarges to form the male pronucleus and the tail of the sperm degenerates.
 - Morphologically, the male and female pronuclei are indistinguishable.

- Pronuclei--DNA-1n (haploid), 2c (two chromatids), the oocyte containing the two haploid pronuclei is called an **ootid**.
- 6) As the pronuclei fuse into a single diploid aggregation of chromosomes, the ootid becomes a **zygote**.
- The zygote is genetically unique because half of its chromosomes came from the mother and half from the father. The zygote contains a new combination of chromosomes that is different from that in the cells of either of the parents.
 - basis of biparental inheritance and variation of the human species.
 - Meiosis allows **independent assortment** of maternal and paternal chromosomes among the germ cells. **Crossing over** of chromosomes, by relocating segments of the maternal and paternal chromosomes, “**shuffles**” the genes, thereby producing a **recombination** of genetic material.
 - The embryo’s chromosomal sex is determined at fertilization by the kind of sperm (X or Y) that fertilizes the oocyte.
 - X-bearing sperm → 46, XX zygote → female
 - Y-bearing sperm → 46, XY zygote → male.

Fertilization

- Stimulates oocyte to complete the second meiotic division.
- Restores the normal diploid number of chromosomes (46) in the zygote.
- Results in variation of the human
- Determines the chromosomal sex of the embryo.
- Causes metabolic activation of the ootid (nearly mature oocyte) and initiates cleavage of the zygote.

THE PROCESS OF IMPLANTATION

- Implantation of the blastocyst in the uterine endometrium begins at the end of the first week and is completed by the end of the second week (6 to 10 days after ovulation). As the blastocyst implants, more trophoblast contacts the endometrium and differentiates into two layers:
 - **Inner layer**, the cytotrophoblast that is mitotically active and forms new cells that migrate into the increasing mass of syncytiotrophoblast, where they fuse and lose their cell membranes.

- Outer layer, the syncytiotrophoblast, a rapidly expanding, multinucleated mass in which no cell boundaries are discernible.

❖ **Implantation may be summarized as follows:**

1. The zona pellucida degenerates (**day 5**).
 1. Its disappearance results from **enlargement of the blastocyst** and degeneration caused by **enzymatic lysis**. The lytic enzymes are released from the acrosomes of the sperms that surround and partially penetrate the zona pellucida.
2. The blastocyst adheres to the endometrial epithelium (**day 6**).
3. The trophoblast differentiates into two layers: the syncytiotrophoblast and cytotrophoblast (**day 7**).
4. The syncytiotrophoblast erodes endometrial tissues and the blastocyst begins to embed in the endometrium (**day 8**).
5. Blood-filled lacunae appear in the syncytiotrophoblast (**day 9**).
6. The blastocyst sinks beneath the endometrial epithelium and the defect is filled by a closing plug (**day 10**).
7. Lacunar networks form by fusion of adjacent lacunae (**days 10 and 11**).
8. The syncytiotrophoblast erodes endometrial blood vessels, allowing maternal blood to seep in and out of lacunar networks, thereby establishing a uteroplacental circulation (**days 11 and 12**).

Clinical

❖ **Ectopic tubal pregnancy (ETP)**

- ETP occurs when the blastocyst implants within the uterine tube due to delayed transport.
- The ampulla of the uterine tube is the most common site of an ectopic pregnancy. The rectouterine pouch (pouch of Douglas) is a common site for an ectopic abdominal pregnancy.
- ETP is most commonly seen in women with endometriosis or pelvic inflammatory disease.
- ETP leads to uterine tube rupture and hemorrhage if surgical intervention is not performed.
- ETP presents with abnormal uterine bleeding, unilateral pelvic pain, increased levels of human chorionic gonadotropin (hCG) (but lower than originally expected with uterine implantation pregnancy), and a massive first-trimester bleed.
- ETP must be differentially diagnosed from appendicitis, an aborting intrauterine pregnancy, or a bleeding corpus luteum of a normal intrauterine pregnancy.

Gastrulation and its regulation

- Gastrulation is the process that forms all three germ layers of the growing embryo.
- It is the main event that takes place in the **third week** of gestation.
- It begins with the formation of the **primitive streak** on the surface of the epiblast.
 - Cells of the epiblast migrate to the primitive streak and upon arrival they detach from it and slip beneath into the primitive pit. This process is called **invagination**.
- Once the cells have invaginated, some displace the hypoblast, creating the embryonic **endoderm**, and others come to lie between the epiblast and newly created endoderm to form **mesoderm**. Cells remaining in the epiblast then form **ectoderm**.
- Thus, the epiblast, through the process of gastrulation, is the source of all of the germ layers, and cells in these layers will give rise to all of the tissues and organs in the embryo.
- By the end of the third week, three basic **germ layers**, consisting of **ectoderm**, **mesoderm**, and **endoderm (trilaminar disk)**, are established in the head region, and the process continues to produce these germ layers for more caudal areas of the embryo until the end of the 4th week. Tissue and organ differentiation has begun, and it occurs in a cephalocaudal direction as gastrulation continues.

Expression of genes responsible in regulation of gastrulation

- Before the primitive streak is formed, cephalic and caudal ends of the embryo are established. Thus, cells in the hypoblast (endoderm) at the cephalic margin of the disc form the **anterior visceral endoderm (AVE)** that expresses head-forming genes and the transcription factors **OTX2**, **LIM1**, and **HESX1** and the secreted factor **cerberus**. Then **Nodal**, a member of the **TGF- β** family of genes, is activated and initiates and maintains the integrity of the node and streak.
- **BMP-4 (Bone Morphogenetic Protein-4)**, another **TGF- β** family), in the presence of **FGF (fibroblast growth factor)**, ventralizes mesoderm during gastrulation so that it forms intermediate and lateral plate mesoderm.
- **Chordin**, **noggin**, and **folistatin**, activated by *Goosecoid*, antagonize BMP-4 activity and dorsalize mesoderm to form the notochord and somitomeres in the head region. *Goosecoid* is normally expressed in the node and is a major regulator of head development.
- **HNF-3 β** maintains the node and later induces regional specificity in the forebrain and midbrain areas. Without *HNF-3 β* , embryos fail to gastrulate properly and lack forebrain and midbrain structures.
- Formation of **dorsal mesoderm** in more caudal regions is regulated by the **Brachyury (T)** gene. Gene product absence causes caudal dysgenesis.
- **Left-right asymmetry** is regulated by a cascade of genes; first, **FGF-8**, secreted by cells in the node and streak, induces **Nodal** and **Lefty-2** expression on the left side. These genes upregulate **PITX2**, a transcription factor responsible for left sidedness. **Sonic hedgehog (SHH)** may also function in this role as well as serving as a repressor for left sided gene expression on the right. In addition, the neurotransmitter **serotonin (5HT)** plays an important role in L/R axis determination.

Teratogenesis involved with gastrulation

- **Caudal dysgenesis (Sirenomelia)** – insufficient mesoderm in caudal region. Abnormalities in the areas associated with the mesoderm, i.e fusion of lower limbs, imperforate anus, renal agenesis, genital anomalies. It is associated with maternal diabetes and genetic anomaly.
- **Situs inversus** – organ reversal in the thorax and abdomen.
- Laterality sequences – **abnormal sidedness**. Bilaterally left sidedness or right sidedness.
- Tumors
 - **Sacroccygeal teratomas** – persistence of remnants in the primitive streak in the sacroccygeal region. It normally consists of tissues from all three germ layers. These tumors may become malignant and are most common in females.

Embryonic folding and The body form

- A significant event in the establishment of body form is folding of the flat trilaminar embryonic disc into a somewhat cylindrical embryo.
- Folding occurs in both the **median and horizontal planes** and results from rapid growth of the embryo. The growth rate at the sides of the embryonic disc fails to keep pace with the rate of growth in the long axis as the embryo increases rapidly in length.
- Folding at the cranial and caudal ends and sides of the embryo occurs simultaneously. Concurrently, there is relative constriction at the junction of the embryo and umbilical vesicle (yolk sac).

Folding of the Embryo in the Median Plane

- Folding of the ends of the embryo ventrally produces head and tail folds that result in the cranial and caudal regions moving ventrally as the embryo elongates cranially and caudally

1. Head Fold

- By the beginning of the fourth week, the neural folds in the cranial region have thickened to form the primordium of the brain.
- Initially, the developing brain projects dorsally into the amniotic cavity. Later, the developing forebrain grows cranially beyond the oropharyngeal membrane and overhangs the developing heart.
 - Concomitantly, the **septum transversum** (transverse septum), primordial heart, pericardial coelom, and oropharyngeal membrane move onto the ventral surface of the embryo.
 - During folding, part of the endoderm of the umbilical vesicle is incorporated into the embryo as the **foregut**. The foregut lies between the brain and heart, and the **oropharyngeal membrane** separates the foregut from the **stomodeum**.
 - After folding, the **septum transversum** lies caudal to the heart where it subsequently develops into the **central tendon of the diaphragm**.
- The head fold also affects the arrangement of the embryonic coelom (primordium of body cavities).
 - Before folding, the coelom consists of a flattened, horseshoe-shaped cavity.
 - After folding, the pericardial coelom lies ventral to the heart and cranial to the septum transversum. At this stage, the intraembryonic coelom communicates widely on each side with the extraembryonic coelom.

2. Tail Fold

- Folding of the caudal end of the embryo results primarily from growth of the distal part of the neural tube-the primordium of the spinal cord.
- As the embryo grows, the **caudal eminence** (tail region) projects over the **cloacal membrane** (future site of anus).
- During folding, part of the endodermal germ layer is incorporated into the embryo as the **hindgut** (primordium of descending colon). The terminal part of the hindgut soon dilates slightly to form the **cloaca** (primordium of urinary bladder and rectum).
- Before folding, the **primitive streak** lies cranial to the cloacal membrane; after folding, it lies caudal to it .
- The **connecting stalk** (primordium of umbilical cord) is now attached to the ventral surface of the embryo, and the **allantois**-a diverticulum of the umbilical vesicle-is partially incorporated into the embryo.

Folding of the Embryo in the Horizontal Plane

- Folding of the sides of the embryo produces right and left **lateral folds**.
- Lateral folding is produced by the rapidly growing spinal cord and somites.
- The primordia of the **ventrolateral wall** fold toward the median plane, rolling the edges of the embryonic disc ventrally and forming a roughly **cylindrical embryo**.
- As the abdominal walls form, part of the endoderm germ layer is incorporated into the embryo as the **midgut** (primordium of small intestine, etc.).
- Initially, there is a wide connection between the midgut and umbilical vesicle, however; after lateral folding, the connection is reduced to an **omphaloenteric duct**.
 - The region of attachment of the amnion to the ventral surface of the embryo is also reduced to a relatively narrow umbilical region.
- As the umbilical cord forms from the connecting stalk, ventral fusion of the lateral folds reduces the region of communication between the intraembryonic and extraembryonic coelomic cavities to a narrow communication.
 - As the amniotic cavity expands and obliterates most of the extraembryonic coelom, the amnion forms the epithelial covering of the umbilical cord.

Extra-embryonic membranes and the placenta

Yolk Sac (Umbilical vesicle)

- It is a membranous sac attached to an embryo, formed by the cells of the hypoblast adjacent to the embryo disk. It's found between the amnion and the chorion. It usually disappears by the 20th week and onwards. Significances:
 - Transfer of nutrients in the 2nd and 3rd week of gestation
 - Helps in identification of early pregnancy

- During the fourth week, its **endoderm** is incorporated into the embryo as the **primordial gut**. Its endoderm, derived from epiblast, gives rise to the epithelium of the trachea, bronchi, lungs, and digestive tract.
- Primordial germ cells appear in the endodermal lining of the wall of the yolk sac in the third week and subsequently migrate to the developing gonads. They differentiate into spermatogonia in males and oogonia in females.

The Allantois

- The allantois (Greek, sausage) appears on approximately day 16 as a small, sausage-shaped diverticulum (out pouching) from the caudal wall of the umbilical vesicle(yolk sac) that extends into the connecting stalk
- In humans, the allantoic sac remains very small, but allantoic mesoderm expands beneath the chorion and **forms blood vessels that will serve the placenta.**
- The proximal part of the original allantoic diverticulum persists throughout much of development as a stalk called the **urachus**, which extends from the bladder to the umbilical region.
- The urachus is represented in adults by the **median umbilical ligament.**
- The blood vessels of the allantoic stalk become the umbilical arteries.
- It is involved with early blood formation and associated with urinary bladder development.

The Placenta

- The placenta is the primary site of nutrient and gas exchange between the mother and fetus. The placenta is a fetomaternal organ that has two components:
 - A fetal part that develops from the chorionic sac
 - A maternal part that is derived from the endometrium
- ❖ **Development:** The placenta develops as the embryo invades the endometrium of the uterus and as the trophoblast forms the villous chorion. The formation of the villous chorion proceeds through three stages: **primary chorionic villi**, **secondary chorionic villi**, and **tertiary chorionic villi**.
 - Primary villi consists of syncytiotrophoblast and cytotrophoblast protrusion (day 13)
 - Secondary villi have a core of extra embryonic mesoderm (day 16)
 - Tertiary villi have blood vessels in the mesoderm (day 21)
 - **Cytotrophoblast:** act as supportive frame work and divide to produce Syncytiotrophoblast cells; compact layer of trophoblast
 - **Syncytiotrophoblast:** produce hormones such as, hCG, progesterone, estrogen, somatomammotropin; loose layer of trophoblast
- The villi start to extend to the chorionic plate to the decidua basalis. Then Cytotrophoblast cells erode maternal sinusoids and intervillous spaces (lacunar spaces) are filled with maternal blood.

- By the beginning of the 4th month cytotrophoblastic cells and some CT of the villi disappear which leaves the syncytium and the endothelial from the fetal capillaries as the only layers that separate the mother and fetal circulation.
- Villi at the embryonic pole continue to grow as pregnancy progresses. This will produce the **villous chorion**. Villi at the abembryonic pole degenerate and the chorionic sac will look smooth, thus giving the **smooth chorion**.
- **Placental Barrier (Placental membrane)** - prevents the mixing of the maternal and the fetal blood. Formed by:
 - Endothelium of fetal capillaries and their BM
 - Surrounding mesoderm
 - Cytotrophoblast and its BM
 - Syncytiotrophoblast

The Decidua and Placental components

- The Decidua: the functional layer of the endometrium. Sheds during parturition. Contains 3 regions after implantation:
 - **Decidua basalis**: attached to the villous chorion. It is the maternal part of the placenta
 - **Decidua capsularis**: it's the layer over the smooth chorion at the abembryonic pole. Eventually degenerates with the growth of the chorionic sac.
 - **Decidua parietalis**: the decidua overlying the rest of the uterus which is not in contact with the chorionic sac.
 - After the D.capsularis disintegrates and the fetus enlarges it comes into contact with the smooth chorion and fuses with it. Thus the uterus is completely covered.

N.B:- The amniotic sac enlarges faster than the chorionic sac. As a result, the amnion and smooth chorion fuse to form the **amniochorionic membrane**. This composite membrane fuses with the **decidua capsularis** and, after disappearance of D.capsularis, adheres to the **decidua parietalis**.

- It is the amniochorionic membrane that ruptures during labor and results in the escaping of the amniotic fluid. Premature labor = preterm rupture of this membrane.
- ❖ Placenta (by 4th month) = villous chorion (fetal) + decidua basalis (maternal)
- Septa from the **decidua** grow into the intervillous spaces dividing the placenta into **cotyledons** which have “cobblestone” like appearance. Each cotyledon has a number of anchoring villi and its branches.
- At the end of pregnancy the placenta is 15- 25 cm wide, weighs about 600 grams, and 3cm thick. The placenta along with the deciduas is shed after parturition.

Circulation of the placenta

- Spiral arteries (maternal) → Intervillous space → Villi → Umbilical Vein → Fetal Circulation → Umbilical Arteries → Villi → Endometrial Veins

N.B:- **Umbilical arteries** carry deoxygenated blood from the fetus to the placenta. They are two in number. **Umbilical vein** carries oxygenated blood from the placenta to the fetus. It is one in number(originally two but the right one disappears).

Functions of Placenta:-has three main functions

- **Metabolism** – synthesizes glycogen, fatty acids, and cholesterol to support the fetus and to support the placenta itself in its other functions.
- **Transport** - transport of substances in both directions between the fetal and maternal blood is facilitated by the great surface area of the placental membrane. Many substances cross the placenta and some don't. (see the table on page 59 of BRS embryology 6th edition).
- **Endocrine Secretion** – production of hormones such as, hCG, hPL, progesterone, estrogen

Body fluid compartments:

- Two main compartments of body fluids exist:
 - **Extracellular:-** 20% of body wt, 1/3 of TBW (14L)
 - **Interstitial:-** 75% (11L)
 - **Plasma** volume:- 25% (3L)
 - **transcellular fluid:-**A small compartment of fluid, that consists of synovial, CSF, peritoneal, pericardial and intraocular fluid.
 - **Intracellular:-** 40% (28L)
- So in 70 kg person, total body water/**TBW**/ is **60%** of the wt(=42L). TBW is highest in newborns and adult males but low in adult female due to high fat composition.
- Membranes can serve as barriers, The 2 important membranes are the cell membrane is a relative barrier for Na⁺ b/n ECF and ICF, while the capillary membrane is a barrier for plasma proteins b/n ISF and plasma.

How to measure body fluid compartments:

- A known amount of substance is given whose volume of distribution is the body fluid compartment of given interest.
- The commonly used are indicator or dye dilution.
 - e.g. D₂O and HTO for total body water
Insulin, mannitol...for ECF (these substances are confined to ECF)
RISA and Evans blue -- marker for plasma volume.

Volume of distribution = Amount of the substance given/concentration in plasma

Distribution of cations and anions in the body:

- Body fluid has equal amount of **cations** and **anions** because positive charges = Negative charges
 - ECF: Major cation of the ECF is Na⁺.
(Na⁺ in ECF 142-145mEq/L and in ICF 4mEq/L)
Major anions of the ECF are Cl⁻ and HCO₃⁻.
- ICF: Major cations of the ECF are K⁺ and Mg²⁺.

(K⁺ in plasma: 4.2mEq/L Mg²⁺= 0.8mEq/l)

Major anions are proteins and organic phosphates.

Regulation of Body fluids

- Two types of body fluid regulations can be sited:
 1. Osmotic regulation
 2. Volume regulation
- 1. Any change in the osmolality of the ECF will be detected by the osmoreceptors of the hypothalamus especially an increase from the normal osmolality range (280-300meq) and upon which they start firing an AP. Any increase in the osmolality of ECF in any part of the body will have the same result in the hypothalamus where the osmoreceptors are located. Their action of firing of AP will be sent to the hypothalamus to secrete ADH to reabsorb water and this way water balance is maintained. In the same way if the ECF osmolality is less, ADH secretion is low and excess water will be removed to maintain water balance.====**OSMOTIC REGULATION**
- 2. When the ECF volume decreases, BP drops and the RAAS will be activated. Renin helps in the conversion of angiotensinogen into angiotensin 1, which will later go on to the lungs to be converted to angiotensin II by ACE. This molecule then helps in aldosterone release which will let Na⁺ reabsorption take place. Following sodium, water will enter in and that will result in ECF volume adjustment.====**VOLUME REGULATION**

Clinical correlates:- Edema: disturbance of fluid balance. This mainly results in fluid accumulation in the interstitial compartment.

- Edema in general occurs when the increase in tissue fluid volume is 30% or more of the basal volume. Two types are there:
 1. Pitting edema: aka extracellular edema
 2. Non-pitting edema: aka cellular edema
- **Pitting edema:-** causes
 1. Increased hydrostatic pressure which is secondary to heart failure and this can be due to:
 - a. excessive kidney retention of salt and water,
 - b. High venous pressure; localized increased venous pressure→↑pressure in venules, capillaries and arterioles→↑hydrostatic pressure →↑filtration.
 - c. decreased arteriolar resistance
 2. Decreased osmotic pressure due to low plasma proteins secondary to conditions like nephrotic syndrome, burns, wounds, failure to produce proteins due to liver disease, chronic malnutrition, etc.
 3. Increased capillary permeability due to immune reactions, toxins, infections, vit. deficiency, etc
 4. Lymphatic obstruction secondary to infection, surgery, etc.
- **Cellular/non-pitting:-**edema may be caused by:

1. Depression of cellular metabolism or lack of adequate nutrition → $\downarrow \text{Na}^+ - \text{K}^+$ pump → $\uparrow \text{Na}^+$ influx into cell → Water enters following Na^+ → **Cellular edema**
 2. Inflammation of tissue areas → $\uparrow$ membrane permeability to sodium and other ions that diffuse into cell with water → **cellular edema**.
- Another condition is **water intoxication**--accumulation of water inside cells. Occurs when excessive volumes of water are absorbed too quickly → results in nausea, vomiting and shock-conditions caused by undue drop in plasma osmolality before adequate inhibition of ADH secretion occurs.

Homeostasis, feedback mechanisms and allostasis

Homeostasis: it is the ability of the body to maintain internal constant environment despite a changing surrounding.

- Any alteration in the body is detected by a **receptor**. Environmental alteration is transmitted to an **integrating centre** through **afferent nerves or hormones** for assessment and discrimination using a **reference or set value**. The **discrepancy** that occurs is sent to **effector** organs as an **error signal**. Then a **response** is made by the effector organ in accordance with the need for **adjustment**.
- Regulatory process involves:-
 - Intrinsic regulation—occur in cells, tissue, organ
 - Adjust themselves or adjusted by ERM
 - Extrinsic regulation—involves nervous and endocrine element
 - Long distance communication
- Higher level physiological adjustment in the body:
 - **Nervous system**: Serves as a means of rapid adjustment to changes
 - **Endocrine System**: This is used for slower chronic metabolic adjustment to changes.
 - **Neuro-endocrine**: This is integrated in many instances to provide optimal environment for the cell
- Responses to stimuli occur in loops.
 - **open loop system**: Input is not affected by output
 - Input → detector → amplifier → effector → controlled output
 - **Control is heavily dependent on precise knowledge of disturbances.**
 - **Mechanism is based on compensation for a given effect emanating from a source of disturbance.**
 - **Determine effects of disturbances in a quantitative manner.**
 - **Adjusted as per measured magnitudes of disturbances.**
 - **closed loop system**: communication b/n biological component receptor, coordinating center, effector...

- Input→detector→amplifier→effectors→regulated output→feedback
- Input affected by output

Feedback mechanisms

i. Negative Feedback

- Output returns to the normal input, i.e. response is in the opposite direction as the stimulus.
- The system is restored back to its original state by resisting or opposing any attempts to change the system, it favor stability
- Regulation of **arterial blood pressure** is one example: when there is a change in the blood pressure, stretch in the aortic arch and carotid sinus will be detected and by the sensory nerve taken to the vasomotor center and the autonomic nervous system will, via the sympathetic or parasympathetic nervous system will make adjustment that is to counterbalance the effect.
- The RAAS system, regulation of PaCO₂, temp regulation, plasma volume regulation, hormonal secretion regulation is some other example of negative feedback.

ii. Positive Feedback

- Output not returned to the normal input, i.e. response is in the same direction as the stimulus.
- May be part of a negative feedback mechanism.
- Rare in biology because in many cases it causes instability leading to death, i.e. it favors instability.
- Eg:-1. Action potential generation 2. LH secretory surge in mid-cycle
3. Uterine contraction during labor 4. Severe shock
5. Viral infection: virus invades cell→it replicates in cell→becomes many viruses→invasion of other cells→further replication.

▪ Feedback factor-reflect the strength of regulation

- Eg, Control mechanism for body temp is high while it is low for ABP
- **FF**=uncontrolled dce of disturbance/controlled dce of disturbance
- $FF < 1$ indicate +ve feedback and $FF > 1$ indicate -ve feedback

▪ Feedback has got three disadvantages:

1. Compensation is generally incomplete
2. Responses may be slow
3. Too much feedback causes instability.

❖ **Allostasis** :-means maintaining stability (homeostasis) through acute change, i.e. attaining viability through change.

- The body adapts to potentially diverse and dangerous situations through the activation of neural, hormonal, or immunological mechanisms.
- **Stress mediators** such as catecholamines and cortisol are used.
- Strong connection between **allostasis and happiness** because with happiness there are alterations in various physiological parameters like heart rate, blood pressure, etc.
- **Allostatic load**:- refers to the cost of adaptation to a stressful environment eliciting repeated and sometimes prolonged adaptive responses.
 - Manifestations of allostatic load include: decreased cognitive function during aging, abdominal obesity, increased risk for hypertension and cardiovascular disease, insulin-dependent diabetes, and decreased immune responses etc
- **Types of allostatic overload**
 1. **Over stimulation** by frequent stress, resulting in excessive stress hormone exposure;
 2. **Failure to inhibit allostatic responses** when they are not needed or an inability to habituate to the same stressor, both of which result in overexposure to stress hormones; and
 3. **Inability to stimulate allostatic responses** when needed, in which case other systems (e.g., inflammatory cytokines) become hyperactive and produce other types of wear and tear.

Transport across cell membrane:

- Mainly two types of transport mechanisms exist in cells:
1. **Passive transport**
 - i. **Simple Diffusion**: Free movement of a substance from one part of a solution to another occurring in both directions. Net movement occurs from an area of high concentration to another area of low concentration.

Rate of diffusion (Flux) = $D \cdot A \cdot P \cdot T \div L \eta$, where D= diffusion coefficient, A=surface area, P= partial pressure difference, T= temperature of fluid, L=membrane thickness, η =viscosity of fluid.
 - ii. **Facilitated diffusion** includes :
 - a) **carrier-mediated transport**: Mechanism built into membrane structure
 - Molecules bind to specific proteins on membrane surface →movement through membrane,
 - e.g. Glucose transport
 - b) **carrier-mediated exchange**:this mechanism mediates only a solute exchange.
 - E.g. Cl-HCO₃ exchanger of RBC, also known as “band 3” protein.
 - c) **ion channels and channels for other small molecules**:
 - Open

- Gated, i.e. have gates that open or close
 - voltage-gated (gated by alterations in membrane potential)
 - Ligand-gated —opened or closed when they bind a ligand (**external** – a neurotransmitter or a hormone; **internal**– Ca^{++} , cAMP, G-proteins).
 - Depending on the number of substances that carriers transport, carrier proteins may be referred to as
 - **Uniport**---carrying only one substance
 - **Symport**---carrying more than one substance in one direction
 - **Antiport**---facilitating exchange of substances
- iii. **osmosis** needs no explanation:
2. **Active transport:**
- i. **Na^+ - K^+ ATPase pump:**
- This pump uses ATPase enzyme, a transmembrane protein responsible for controlling volume of cells. Without this function, cells swell until they burst.
 - Moves 3Na^+ out and 2K^+ into the cell leading to movement with each revolution of pump leading to a net of one positive charge into the exterior → positivity outside the cell, i.e. the Na^+ - K^+ pump is **electrogenic** because it creates an electrical potential across the cell membrane.
- ii. **Electrogenic co-transport (secondary active transport)**
- Binding of more than one molecule in one direction – symport, e.g. Na^+ - glucose
 - All co-transports have Na^+ as one of the transported solutes; exception is yeast.
 - Na^+ moves down its conc. gradient but co-transported solutes (e.g. glucose, a.a) are pumped up against gradient-- **secondary active transport of co-transport solutes**. This co-transport mechanism uses diffusion energy liberated by the downhill movement of sodium, i.e. metabolic energy is not used. The transport carrier protein **has two binding sites** on its exterior side.
3. **Vesicular transport:** - transport for macromolecules includes endocytosis, phagocytosis and pinocytosis.

Physiology of ANS

Two major categories of motor neurons: - somatic and autonomic.

- **Somatic motor neurons:** - have their cell bodies within the CNS and send axons to skeletal muscles, which are usually under voluntary control.
- **Autonomic motor nerves:** - innervate organs whose functions are not usually under voluntary control. The effectors that respond to autonomic regulation include cardiac muscle, smooth muscles, and glands. These effectors are part of the visceral organs and of blood vessels. The

involuntary effects of autonomic innervation contrast with the voluntary control of skeletal muscles (by way of somatic motor neurons.)

Table 9.1 | Comparison of the Somatic Motor System and the Autonomic Motor System

Feature	Somatic Motor	Autonomic Motor
Effector organs	Skeletal muscles	Cardiac muscle, smooth muscle, and glands
Presence of ganglia	No ganglia	Cell bodies of postganglionic autonomic fibers located in paravertebral, prevertebral (collateral), and terminal ganglia
Number of neurons from CNS to effector	One	Two
Type of neuromuscular junction	Specialized motor end plate	No specialization of postsynaptic membrane; all areas of smooth muscle cells contain receptor proteins for neurotransmitters
Effect of nerve impulse on muscle	Excitatory only	Either excitatory or inhibitory
Type of nerve fibers	Fast-conducting, thick (9–13 μ m), and myelinated	Slow-conducting; preganglionic fibers lightly myelinated but thin (3 μ m); postganglionic fibers unmyelinated and very thin (about 1.0 μ m)
Effect of denervation	Flaccid paralysis and atrophy	Muscle tone and function persist; target cells show denervation hypersensitivity

- From CNS to effectors, autonomic system has 2 neurons:--preganglionic and postganglionic
 - **Preganglionic neurons**:- of ANS originate in the midbrain and hindbrain and in the upper thoracic to the fourth sacral levels of the spinal cord.(CNS)
 - **Autonomic ganglia** are located in the head, neck, and abdomen; chains of autonomic ganglia also parallel the right and left sides of the spinal cord.
 - **postganglionic neurons** :-have cell bodies within autonomic ganglia.
- ❖ The sympathetic and parasympathetic divisions of ANS differ in the particular locations of their preganglionic neuron cell bodies within the CNS, and in the location of their ganglia.

Clinical note

- The involuntary effectors are somewhat independent of their innervation. Smooth muscles maintain a resting tone (tension) in the absence of nerve stimulation, for example. In fact, damage to an autonomic nerve makes its target tissue more sensitive than normal to stimulating agents. This phenomenon is **called denervation hypersensitivity**. Such compensatory changes can explain why, for example, the ability of the stomach mucosa to secrete acid may be restored after its neural supply from the vagus nerve has been severed. (This procedure is called vagotomy, and is sometimes performed as a treatment for ulcers.)

ANS



I. Sympathetic Division

- the thoracolumbar division of ANS

- preganglionic fibers exit the spinal cord, in the ventral roots of spinal nerves, from T1 to L2 levels and synapse with postganglionic neurons within a double row of sympathetic ganglia, called paravertebral ganglia, located on either side of the spinal cord .
- Ganglia within each row are interconnected, forming a sympathetic chain of ganglia that parallels the spinal cord on each lateral side.
- The myelinated preganglionic sympathetic axons exit the spinal cord in the ventral roots of spinal nerves, but they soon diverge from the spinal nerves within short pathways called **white rami communicantes**. The axons within each ramus enter the sympathetic chain of ganglia, where they can travel to ganglia at different levels and synapse with postganglionic sympathetic neurons. The axons of the postganglionic sympathetic neurons are unmyelinated and form the **gray rami communicantes** as they return to the spinal nerves and travel as part of the spinal nerves to their effector organs. Because sympathetic axons form a component of spinal nerves, they are widely distributed to the skeletal muscles and skin of the body, where they innervate blood vessels and other involuntary effectors.
- The divergence of impulses from the spinal cord to the ganglia and the convergence of impulses within the ganglia results in the mass activation of almost all of the postganglionic sympathetic neurons. This explains why the sympathetic system is usually activated as a unit, affecting all of its effector organs at the same time. However, it does appear that the sympathetic division can increase its stimulation of a particular organ, such as the heart, in some cases.
- **Collateral Ganglia**
 - Many preganglionic fibers that exit the spinal cord below the level of the diaphragm pass through the sympathetic chain of ganglia without synapsing. Beyond the sympathetic chain, these preganglionic fibers form splanchnic nerves. Preganglionic fibers in the splanchnic nerves synapse in collateral, or prevertebral, ganglia. These include the celiac, superior mesenteric, and inferior mesenteric ganglia. Postganglionic fibers that arise from the collateral ganglia innervate organs of the digestive, urinary, and reproductive systems.
 - Like a sympathetic ganglion, the cells of the adrenal medulla are innervated by preganglionic sympathetic fibers only. The adrenal medulla secretes epinephrine into the blood in response to this neural stimulation.

II. Parasympathetic Division

- The craniosacral division of the autonomic system.
 - This is because its preganglionic fibers originate in the brain (specifically, in the midbrain, pons, and medulla oblongata) and in the S2-S4 levels of the spinal column. These preganglionic parasympathetic fibers synapse in ganglia that are located next to—or actually within—the organs innervated.
 - These **parasympathetic ganglia, called terminal ganglia**, supply the postganglionic fibers that synapse with the effector cells.
 - Most parasympathetic fibers do not travel within spinal nerves, as do sympathetic fibers.
- ❖ As a result, cutaneous effectors (blood vessels, sweat glands, and arrector pili muscles) and blood vessels in skeletal muscles receive sympathetic but not parasympathetic innervation.
- Four of the 12 pairs of cranial nerves contain preganglionic parasympathetic fibers. These are the oculomotor (III), facial (VII), glossopharyngeal (IX), and vagus (X) nerves.

- Parasympathetic fibers within the first three of these cranial nerves synapse in ganglia located in the head; fibers in the vagus nerve synapse in terminal ganglia located in widespread regions of the body.
- Visceral sensory information (from blood pressure receptors in certain arteries, for example) evokes autonomic reflex motor responses (of heart rate, for example).

Table 9.3 | The Parasympathetic Division

Nerve	Origin of Preganglionic Fibers	Location of Terminal Ganglia	Effector Organs
Oculomotor (third cranial) nerve	Midbrain (cranial)	Ciliary ganglion	Eye (smooth muscle in iris and ciliary body)
Facial (seventh cranial)	Pons (cranial)	Pterygopalatine and submandibular ganglia	Lacrimal, mucous, and salivary glands
Glossopharyngeal (ninth cranial) nerve	Medulla oblongata (cranial)	Otic ganglion	Parotid gland
Vagus (tenth cranial) nerve	Medulla oblongata (cranial)	Terminal ganglia in or near organ	Heart, lungs, gastrointestinal tract, liver, pancreas
Pelvic spinal nerves	S2 to S4 (sacral)	Terminal ganglia near organs	Lower half of large intestine, rectum, urinary bladder, and reproductive organs

Function of ANS

- Difference in function between sympathetic and parasympathetic division is due to the different neurotransmitters released by post ganglionic fibers--- Where the axons of postganglionic autonomic neurons enter into their target organs, they have numerous swellings, called varicosities, that contain the neurotransmitter molecules. Neurotransmitters can thereby be released along a length of axon, rather than just at the axon terminal. Thus, autonomic neurons are said to form **synapses en passant** (“**synapses in passing**”) with their target cells.

Table 9.4 | Effects of Autonomic Nerve Stimulation on Various Effector Organs

Effector Organ	Sympathetic Effect	Parasympathetic Effect
Eye		
Iris (radial muscle)	Dilation of pupil	—
Iris (sphincter muscle)	—	Constriction of pupil
Ciliary muscle	Relaxation (for far vision)	Contraction (for near vision)
Glands		
Lacrimal (tear)	—	Stimulation of secretion
Sweat	Stimulation of secretion	—
Salivary	Saliva becomes thick	Increased secretion; saliva becomes thin
Stomach	—	Stimulation of secretion
Intestine	—	Stimulation of secretion
Adrenal medulla	Stimulation of hormone secretion	—
Heart		
Rate	Increased	Decreased
Conduction	Increased rate	Decreased rate
Strength	Increased	—
Blood Vessels	Mostly constriction; affects all organs	Dilation in a few organs (e.g., penis)
Lungs		
Bronchioles (tubes)	Dilation	Constriction
Mucous glands	Inhibition of secretion	Stimulation of secretion
Gastrointestinal Tract		
Motility	Inhibition of movement	Stimulation of movement
Sphincters	Closing stimulated	Closing inhibited
Liver	Stimulation of glycogen hydrolysis	—
Adipose (Fat) Cells	Stimulation of fat hydrolysis	—
Pancreas	Inhibition of exocrine secretions	Stimulation of exocrine secretions
Spleen	Contraction	—
Urinary Bladder	Muscle tone aided	Contraction
Arrector Pili Muscles	Erection of hair and goose bumps	—
Uterus	If pregnant: contraction; if not pregnant: relaxation	—
Penis	Ejaculation	Erection (due to vasodilation)

Adrenergic and Cholinergic Synaptic Transmission

- Acetylcholine (ACh) is the neurotransmitter of all preganglionic fibers (both sympathetic and parasympathetic). Acetylcholine is also the transmitter released by most parasympathetic postganglionic fibers at their synapses with effector cells. Transmission at these synapses is thus said to be cholinergic.
- The neurotransmitter released by most postganglionic sympathetic nerve fibers is norepinephrine (noradrenaline). Transmission at these synapses is thus said to be adrenergic.

Exceptions

- Some sympathetic fibers that innervate blood vessels in skeletal muscles, as well as sympathetic fibers to sweat glands, release ACh (are cholinergic).

Receptors of effector organs

I. Adrenergic receptors

- The two major classes of these receptor proteins are designated alpha- (α) and beta- (β) adrenergic receptors.
- each class of adrenergic receptor has two major subtypes: α_1 and α_2 ; β_1 and β_2 .
- All adrenergic receptors act via G-proteins. the binding of epinephrine and norepinephrine to their receptors causes the group of three G-proteins (designated α , β , and γ) to dissociate into an α subunit and a $\beta\gamma$ complex. In different cases, either the α subunit or the $\beta\gamma$ complex causes the opening or closing of an ion channel in the plasma membrane, or the activation of an enzyme in the membrane. This begins the sequence of events that culminates in the effects of epinephrine and norepinephrine on the target cells.

Table 9.5 | Selected Adrenergic Effects in Different Organs

Organ	Adrenergic Effects of Sympathoadrenal System	Adrenergic Receptor
Eye	Contraction of radial fibers of the iris dilates the pupils	α_1
Heart	Increase in heart rate and contraction strength	β_1 , primarily
Skin and visceral vessels	Arterioles constrict due to smooth muscle contraction	α_1
Skeletal muscle vessels	Arterioles constrict due to sympathetic nerve activity	α_1
	Arterioles dilate due to hormone epinephrine	β_2
Lungs	Bronchioles (airways) dilate due to smooth muscle relaxation	β_2
Stomach and intestine	Contraction of sphincters slows passage of food	α_1
Liver	Glycogenolysis and secretion of glucose	α_1 , β_2

- All subtypes of beta receptors stimulate the production of cyclic AMP within the target cells.
- α_1 receptors effect is mediated by a different second messenger system—a rise in the cytoplasmic concentration of Ca^{2+} .
- α_2 -adrenergic receptors, located on presynaptic axon terminals, when stimulated, cause a decreased release of norepinephrine. This may represent a form of negative feedback control.
- vascular smooth muscle cells also have α_2 -adrenergic receptors on the postsynaptic membrane, where they can be activated to produce vasoconstriction. This action would cause a rise in blood pressure. However, drugs that activate α_2 -adrenergic receptors are used to lower blood pressure. This is because they stimulate presynaptic α_2 -adrenergic receptors in the brain, and this somehow reduces the activity of the entire sympathetic nervous system.
- The stimulation of α_1 -adrenergic receptors consistently causes contraction of smooth muscles. We can thus state that the vasoconstrictor effect of sympathetic nerves always results from the activation of alpha- adrenergic receptors.
- The effects of beta-adrenergic activation are more diverse; stimulation of beta-adrenergic receptors promotes the relaxation of smooth muscles (in the digestive tract, bronchioles, and uterus, for example) but increases the force of contraction of cardiac muscle and promotes an increase in cardiac rate. Adrenergic stimulation wrought by activation of the sympathetic division produces an increase in cardiac pumping (a β_1 effect), vasoconstriction and thus reduced blood flow to the visceral organs (an α_1 effect), dilation of pulmonary bronchioles (a β_2 effect), and so on, preparing the body for physical exertion .

Table 9.5 | Selected Adrenergic Effects in Different Organs

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Eye	Contraction of radial fibers of the iris dilates the pupils	α_1
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	Arterioles dilate due to hormone epinephrine	β_2
Lungs	Bronchioles (airways) dilate due to smooth muscle relaxation	β_2
Stomach and intestine	Contraction of sphincters slows passage of food	α_1
Liver	Glycogenolysis and secretion of glucose	α_1, β_2

II. Cholinergic receptors

- All somatic motor neurons, all preganglionic neurons (sympathetic and parasympathetic), and most postganglionic parasympathetic neurons are cholinergic.
- ACh released by somatic motor neurons and by preganglionic autonomic neurons are always excitatory.
- The effects of ACh released by postganglionic parasympathetic axons are usually excitatory, but in some cases they are inhibitory. For example, the cholinergic effect of the postganglionic parasympathetic axons innervating the heart (a part of the vagus nerve) slows the heart rate.
- there are two types of cholinergic receptors— nicotinic and muscarinic.
- Nicotinic R---- stimulated by ACh released by somatic motor neurons and by preganglionic autonomic neurons.
 - blocked by curare (tubocurarine)
 - are ligand-gated ion channels(Na^+ diffuses inward when Ach binds to Receptor
→depolarization)
 - effect is always stimulatory
- Muscarinic R----stimulated by ACh released by postganglionic parasympathetic axons to produce the parasympathetic effects.(on viscera)
 - blocked by atropine (or belladonna)
 - coupled to G-proteins
 - effects can be either excitatory or inhibitory

Table 9.6 | Cholinergic Receptors and Responses to Acetylcholine

Receptor	Tissue	Response	Mechanisms
Nicotinic	Skeletal muscle	Depolarization, producing action potentials and muscle contraction	ACh opens cation channel in receptor
Nicotinic	Autonomic ganglia	Depolarization, causing activation of postganglionic neurons	ACh opens cation channel in receptor
Muscarinic (M_1, M_2)	Smooth muscle, glands	Depolarization and contraction of smooth muscle, secretion of glands	ACh activates G-protein coupled receptor, opening Ca^{2+} channels and increasing cytosolic Ca^{2+}
Muscarinic (M_2)	Heart	Hyperpolarization, slowing rate of spontaneous depolarization	ACh activates G-protein coupled receptor, opening channels for K^+

Other Autonomic Neurotransmitters

- ❖ “nonadrenergic, noncholinergic fibers.”
- ❖ Proposed neurotransmitters for these axons include ATP, a polypeptide called vasoactive intestinal peptide (VIP), and nitric oxide (NO).
 - E.g blood vessels of the penis and cerebral arteries

=>**Dual innervation** (SNS and PNS)-----in most organs, usually antagonism

=> **Complementary action** (same effect of sns and pns)

- e.g salivary gland-> watery saliva is by pns,
->thicker, more viscous saliva is by sns
- In both cases saliva is secreted.

=>**cooperative action** (when pns and sns produce different effects that work together to promote a single action)

- e.g on reproductive and urinary system
 - penis -----erection(pns), ejaculation and orgasm(sns)
 - micturition (urination) reflex.
 - Although the contraction of the urinary bladder is largely independent of nerve stimulation, it is promoted in part by the action of parasympathetic nerves. This is exploited clinically in helping people with incontinence (involuntary urination) caused by overactive bladder. In this condition, contractions of the detrusor muscle are stimulated by ACh released by parasympathetic axons.
 - drugs (darifenacin and solifenacin) are available to block the specific muscarinic receptor subtypes (primarily M₃) that mediate the parasympathetic stimulation of bladder contractions.
- ❖ The micturition reflex is also enhanced by sympathetic nerve activity, which increases the tone of the detrusor (bladder) muscle; Emotional states that are accompanied by high sympathetic nerve activity (such as extreme fear) may thus result in reflex urination at bladder volumes that are normally too low to trigger this reflex.

Organs Without Dual Innervation

1. the adrenal medulla;
 2. the arrector pili muscles in the skin;
 3. the sweat glands in the skin; and
 4. most blood vessels.
- Regulation is achieved by increases or decreases in the tone (firing rate) of the sympathetic fibers.
 - The sympathoadrenal system is required for **nonshivering thermogenesis**: animals deprived of their sympathetic system and adrenals cannot tolerate cold stress. The sympathetic system itself is required for proper thermoregulatory responses to heat. In a hot room, for example, decreased sympathetic stimulation produces dilation of the blood vessels in the skin, which increases cutaneous blood flow and provides better heat radiation. During exercise, by contrast, sympathetic

activity increases, causing constriction of the blood vessels in the skin of the limbs and stimulation of sweat glands in the trunk.

- The sweat glands in the trunk secrete a watery fluid in response to cholinergic sympathetic stimulation. Evaporation of this dilute sweat helps to cool the body. The sweat glands also secrete a chemical called bradykinin in response to sympathetic stimulation. Bradykinin stimulates dilation of the surface blood vessels near the sweat glands, helping to radiate some heat despite the fact that other cutaneous blood vessels are constricted. At the conclusion of exercise, sympathetic stimulation is reduced, causing cutaneous blood vessels to dilate. This increases blood flow to the skin, which helps to eliminate metabolic heat. Notice that all of these thermoregulatory responses are achieved without the direct involvement of the parasympathetic system.

❖ Hypothalamus is the major regulator of ANS

Table 9.7 | Adrenergic and Cholinergic Effects of Sympathetic and Parasympathetic Nerves

Organ	Effect of			
	Sympathetic		Parasympathetic	
	Action	Receptor ^c	Action	Receptor ^d
<i>Eye</i>				
<i>Iris</i>				
Radial muscle	Contracts	α_1	—	—
Circular muscle	—	—	Contracts	M
<i>Heart</i>				
Sinoatrial node	Accelerates	β_1	Decelerates	M
Contractility	Increases	β_1	Decreases (atria)	M
<i>Vascular Smooth Muscle</i>				
Skin, splanchnic vessels	Contracts	α_1, β	—	—
Skeletal muscle vessels	Relaxes	β_2	—	—
	Relaxes	M**	—	—
Bronchiolar Smooth Muscle	Relaxes	β_2	Contracts	M
<i>Gastrointestinal Tract</i>				
<i>Smooth muscle</i>				
Walls	Relaxes	β_2	Contracts	M
Sphincters	Constricts	α_1	Relaxes	M
Secretion	Decreases	α_1	Increases	M
Myenteric plexus	Inhibits	α_1	—	—
<i>Genitourinary Smooth Muscle</i>				
Bladder wall	Relaxes	β_2	Contracts	M
Urethral sphincter	Constricts	α_1	Relaxes	M
Uterus, pregnant	Relaxes	β_2	—	—
	Contracts	α_1	—	—
Penis	Ejaculation	α_1	Erection	M
<i>Skin</i>				
Pilomotor smooth muscle	Contracts	α_1	—	—
<i>Sweat glands</i>				
Thermoregulatory	Increases	M	—	—
Apocrine (stress)	Increases	α_1	—	—

Heterostasis

- In case of homeostasis internal environment's constancy is kept at a set point or within a narrow range so if fluctuation occurs from this set point the homeostatic system will intervene via various mechanisms such as -ve feedback mechanism to go back to the normal equilibrium but, this is not always the case.
- Regulatory levels are not rigidly fixed, but may be actively altered in the face of exogenous challenges or pathogens. **heterostasis simply is maintenance of physiologic stability in circumstances of change, whether predictable or unpredictable, through "setting new regulated point" or "it is remaining stable by being variable."**
- External world → problems to be solved → structural evolutionary changes to solve problems of existence → adaptation- a bilateral process in which:
 - Organism is adapted ↔ environment is modified
- **example of heterostatic regulation is that of fever** in this case, illness induced fever reflects adoption of new higher set point. this is vital in mobilizing energy resources to combat infection and other adaptive challenges.

E.g. . Grazing animals vs. herbs

Trees vs. soil

- NB: Birds are more adaptable to high altitudes because of high cardiac output and more efficient respiratory system. Mammals are rarely found at about 6000 metres above sea level but some birds nest at 6500m and others migrate above 9000m. The physiological adjustment in these birds is in P50 and not in Hct.

❖ Long- term adaptation to cold

I. Physiological Adaptation

- Tribes adapted to freezing cold temp.
 - e.g Aborigines of Australia
- In cold climate:- 25—50% loss in BMR → cooling of peripheral and deep lying tissues → ↓ average skin temperature up to 15-12°C → ↓ gradient of heat flow
 - Thermal conductivity drops from 8 kcal/m²/hr/°C to 5 kcal → ↓ temp. Gradient → reduced heat loss.

II. Cultural Adaptation

e.g. The Eskimos: Live in the Arctic Zone

- a. **Housing:** Low and nearly round → prevention of heat loss + perfect ventilation; hot air escapes through apertures in the domed roof and can be stopped at will. Cold air enters through from below through a tunnel
- b. **Clothing:** skins and fur consisting of two layers → imprisoning of heat so effectively → Temperature of tropical summer
- c. **Food:** Walrus soup which is mainly fat

❖ Long term Adaptation to heat

i. Genetic

- Body shape---Long limbs, Protruding organs
- ii. **Physiological**
E.g. Activity of sweat glands
 - Hot climate→all sweat glands become active→increased ability of sweating
 - Temperate climate→some sweat glands are inactive→reduced ability of sweating
- iii. **Cultural**
E.g. **Clothing** of Arabian Bedouins (white, loose and long garments)→ Evaporation of heat
 - insulating protection
 - circulation of air

❖ **Adaptation to high altitude**

- **Homeostatic:** e.g. CVS, Resp. System
 - **Adaptive:-** Hyperhaemoglobinaemia,
- Hypervascularization,
- Hyperactivity of cell enzymes →↑mitochondrial enzymes→↑O₂ utilization

Composition of biomolecules

1. Carbohydrates

- Composed of carbon (carbo-), hydrogen and oxygen (-hydrate). The simplest CHO are the monosaccharides- have general formula (CH₂O)_n where N is 3 or more.
- A monosaccharide or simple sugar, consists of a carbon chain with a number of hydroxyl (OH) groups and either one aldehyde group (often written as -CHO) or one ketone group(C=O).
- A sugar that bears an aldehyde group is called an aldose
- a sugar with a ketone group is a ketose.
- The smallest carbohydrates, for which n is 3, are called trioses.
- E.g glyceraldehyde is a triose that has an aldehyde group -can also be called an aldotriose.
- Similarly, dihydroxyacetone is a triose with ketone group-also called ketotriose.

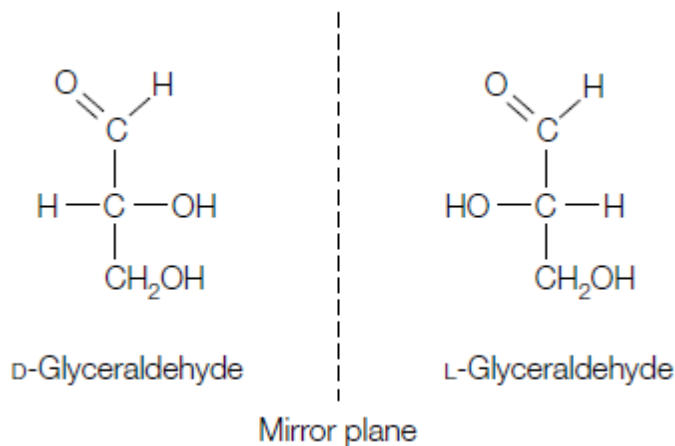
❖ **Reducing sugars:**

- Sugars that contain a free aldehyde or ketone group in the open-chain configuration.
- can reduce cupric ions (Cu²⁺) to cuprous ions (Cu⁺)----This is the basis of the Fehling's and Benedict's tests for reducing sugars. The reducing end of such a sugar chain is thus the end that bears the aldehyde or ketone group.

❖ **Structural forms:**

- **Stereoisomers: compounds having same molecular formula but different orientation of groups (atoms)**
 - Note that glyceraldehyde and dihydroxyacetone have the same chemical composition, C₃H₆O₃, but differ in structure (i.e. they are structural isomers).
- **Enantiomers (mirror images), optic isomers, D & L forms**

- Glyceraldehyde has a single asymmetric carbon atom (the central one) and so two Enantiomers that is two forms of glyceraldehyde, denoted as D- and L-glyceraldehyde, Which are mirror images of each other.



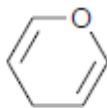
- **Generally, D-monosaccharides are metabolizable in cells but not L-forms**
- **Stereoisomers also exist for amino acids.**

❖ Ring structures

- The aldehyde or ketone group can react with a hydroxyl group to form a covalent bond.
- the rxn between an aldehyde and the hydroxyl group of a sugar (an alcohol) creates a hemiacetal
- ketone group reacts with a hydroxyl group of the sugar(alcohol) to form a hemiketal
 - For tetroses and larger sugars, the rxn can take place within the same molecule so that the straight chain form of the sugar cyclizes.

For example, if the C-5 OH group of glucose reacts with the aldehyde group, a six-membered ring is formed whereas if the C-4 OH reacts with the aldehyde group, a five-membered ring is formed.

Because of their similarity to the ring compound called pyran, six-membered ring structures are called pyranoses.



Pyran

- cyclization of the ketohexose, fructose, gives a five membered ring.
- Because of their similarity to the ring compound called furan, five-membered ring structures are called furanoses.



Furan

- In general, aldoses and ketoses with five or more carbons can form either pyranose or furanose rings. so that in solution a mixture of these exist.
- Which is the more stable ring form, and hence predominant, depends on the chemical structure of the monosaccharide, including the nature of the substituent groups.
- Usually aldohexoses such as glucose exist mainly in the pyranose ring form.

❖ Anomeric forms: α - β -anomers

- during cyclization of glucose (aldohexose) a new asymmetric center is formed, at C-1, the carbon that carried the carbonyl moiety. (in straight structure asymmetric center was at C-2)

Thus two isomers of D-glucose exist,

- α -D-glucose (in which the OH group at C-1 lies below the plane of the ring)
- β -D-glucose (in which the OH group at C-1 lies above the plane of the ring).
- The carbonyl carbon (C-1 in this case) is called the anomeric carbon atom and so the α and β forms are called anomers.
- In aqueous solution, the α - and β -forms rapidly interconvert via the open-chain structure, to give an equilibrium mixture. This process is called mutarotation.-mediated by mutarotase enzyme.
- In the case of the ketose, fructose, the anomeric carbon atom (that carried the carbonyl moiety) is C-2 and hence two isomers (anomers) exist which differ in their configuration about that carbon atom, i.e. the α/β designation refers to the configuration about C-2 not C-1

❖ Epimers: stereoisomers which differ in orientation of groups at single asymmetric center

❖ glucose & galactose epimers at C-4

❖ glucose & mannose epimers at C-2

Disaccharides

- The aldehyde or ketone group on the anomeric carbon atom of one monosaccharide can react with the hydroxyl group of a second monosaccharide to form a disaccharide. The covalent bond formed is called a glycosidic bond.
 - Lactose $\Rightarrow$ C-1 of D-galactose and C-4 of D-glucose. $\beta(1 \rightarrow 4)$ linkage
 - Maltose $\Rightarrow$ C-1 and C-4 positions of two glucose units. $\alpha(1 \rightarrow 4)$ linkage.
- For lactose and maltose, one of the anomeric carbons has been used to form the bond, leaving the second anomeric carbon free. Thus both lactose and maltose have a reducing end.
 - sucrose $\Rightarrow$ anomeric C-1 of glucose and the anomeric C-2 of fructose so that sucrose lacks a free reducing group., lipids, Amino acids, peptides and proteins, Nucleotides and nucleic acids

Sugar derivatives

- By replacing hydroxyl groups of simple sugar with other groups.
 - ✓ For example, phosphorylated sugars such as glucose 6-phosphate are important metabolites in glycolysis.
 - ✓ In amino sugars, one or more hydroxyl groups are each replaced by an amino group (which is often acetylated), for example acetyl -D-N acetyl glucosamine. This and other sugar derivatives are common components of many glycoproteins.
 - ✓ Nucleotides, such as ATP, each consist of a sugar in which the anomeric carbon atom has formed a covalent bond with a nitrogenous base. Since the bond is between the anomeric carbon of the sugar and a nitrogen atom of the base, it is called an N-glycosidic bond.
 - ✓ Other modifications include oxidation of one of the carbons to form a carboxylate group, so generating a uronic acid. For example, oxidation of carbon-6 of D-glucose in this way yields D-glucuronic acid. Uronic acids are important components of many polysaccharides.
 - ✓ Aldoses and ketoses can also be reduced to yield polyhydroxy alcohols called alditols such as sorbitol (D-glucitol), D-mannitol and D-xylitol, which are sweet-tasting and are used to flavor sugarless gum and mints, and glycerol which is an important lipid component.

Polysaccharides

- Long chains of monosaccharides joined together, The major storage polysaccharides are glycogen (in animals), starch (in plants) and dextran (in yeast and bacteria). Cellulose is a structural polysaccharide found in plant cell walls.
- **Glycogen** is a branched-chain polysaccharide containing glucose residues linked by α 1–4 bonds with α 1–6 branchpoints (every 10 units or so).
 - The branched nature of glycogen makes it more accessible to glycogen phosphorylase during degradation, since this enzyme degrades the molecule by sequential removal of glucose residues from the nonreducing ends.
 - The α 1–6 branches themselves are removed by a debranching enzyme
- **Starch** is a mixture of unbranched amylose (glucose residues joined by α 1–4 bonds) and branched amylopectin (glucose residues joined α 1–4 but with some α 1–6 branch points).
 - exists in plants as insoluble starch granules in chloroplasts
- **Dextran** consists of glucose residues linked mainly by α 1–6 bonds but with occasional branch points that may be formed by α 1–2, α 1–3 or α 1–4 bonds.
- **Cellulose** is a straight-chain polymer of glucose units linked by β 1–4 bonds. The polysaccharide chains are aligned to form fibrils that have great tensile strength. Cellulases, enzymes that degrade cellulose, are absent in mammals but are produced by some bacteria, fungi and protozoa.
 - Ruminant animals such as cattle have cellulose producing bacteria in their digestive tracts and so can digest cellulose

Oligosaccharides

- are short chains of monosaccharides linked together by glycosidic bonds. In the case of oligosaccharides linked to proteins (glycoproteins) or lipids (glycolipids), the oligosaccharide

is not a repeating unit but consists of a range of different monosaccharides joined by a variety of types of bonds.

In glycoproteins, two main types of oligosaccharide linkages exist:

- O-linked oligosaccharides attached to the protein via O-glycosidic bonds, to the OH groups of serine or threonine side-chains
- N-linked oligosaccharides attached to the protein via N-glycosidic bonds, to the NH₂ groups of asparagine side-chains
- All N-linked oligosaccharides have a common pentasaccharide core. High-mannose N-linked oligosaccharides have additional mannose residues linked to the core whilst the complex type N-linked oligosaccharides have branches comprising combinations of GlcNAc, Gal, sialic acid and L-fucose.

Amino acids and proteins

1) Aminoacids

- All proteins are made up from the same set of 20 standard amino acids.
- A typical amino acid has **Components:**
 - ◆ central carbon
 - ◆ amino group
 - ◆ carboxyl group (acid)
 - ◆ side chain (R group)
 - variable group, attached to α carbon
 - confers unique chemical properties

a) **Enantiomer**

- All of the 20 standard amino acids, except for glycine, have four different groups arranged tetrahedrally around the C α atom and thus can exist in either the D or L configuration. These two enantiomers are Non superimposable mirror images. Only the L isomer is found in proteins.
- The 20 standard amino acids have difference only on the side-chains or R groups and display different physicochemical properties (polarity, acidity, basicity, aromaticity, bulkiness, conformational inflexibility, ability to form hydrogen bonds, ability to cross-link and chemical reactivity).

❖ **functions** of aminoacids include:-

- ▶ Cys and met are sources of S in the body
- ▶ Carbon skeleton and N of AAs used for: synthesis of; Purine & pyrimidine

- ▶ Gly and Met help in the detoxification mechanisms
- ▶ Met can act as a methyl group donor in methylation reactions
- ▶ Glycine is a precursor for
 - ▶ Heme
 - ▶ Creatine
- ▶ Tyrosine is the precursor for number of hormones:
 - ▶ Thyroxine & Triiodothyronine
 - ▶ Epinephrine&nor-epinephrine
 - ▶ Skin pigment melanine
- ▶ Tryptophan can give rise to
 - ▶ Niacin
 - ▶ Serotonin
- ▶ Histidine can be converted to Histamine

❖ Classification of AA by side chain character

- Glycine is the only AA whose side chain is hydrogen.
 - Non-polar, aliphatic R groups:
 - glycine
 - valine
 - alanine
 - leucine, isoleucine
 - proline
 - methionine
 - Aromatic R groups:
 - phenylalanine
 - tyrosine
 - tryptophan
 - Polar uncharged R groups:
 - serine
 - threonine
 - cysteine
 - asparagine
 - glutamine
 - Negatively charged R group:
 - aspartate
 - glutamate
 - Positively charged (H⁺ acceptor):
 - lysine
 - arginine
 - histidine

❖ Classification based on nutritive value

- **Indispensable or essential AAs**
Not synthesizable in the body in adequate amounts

- **Dispensable or non-essential AAs**
Synthesizable in the body and there is no diet dependency for them
- **Ten AAs are essential:**
Val, Ile, Thr, Trp, Arg, Leu, Lys, Met, Phe and His.
‘VITTAL LyMPH’

▶ **Arg & His are semi-essential:**

- ☐ Healthy adult human body synthesizes just enough amount
- ☐ But not enough in conditions like childhood growth period, sickness and during pregnancy

▶ **Some AAs become essential under certain conditions:**

- ☐ Tyr also turns essential upon dietary deficiency of Phe
- ☐ Cys turns essential upon dietary deficiency of Met

❖ **Classification based on metabolic fate**

▶ **Ketogenic**

- ▶ Only two AAs are purely ketogenic: Leu and Lys
- ▶ Catabolically give intermediates convertible into Acetyl-CoA or Acetoacetyl-CoA

▶ **Mixed type**

- ▶ C-skeleton of which is catabolized to produce glycolytic intermediates or acetyl CoA derivatives

Four in no.: -Ile, Tyr, Trp, Phe

▶ **Glucogenic**

- ▶ The rest 14 of them give rise to intermediates of glycolysis or Krebs's cycle

❖ **Uncommon AAs found in protein:-**

- **Hydroxylysine and hydroxyproline**
 - Mainly in collagen
 - Found in **connective tissues**
- **Tyroxine and Triiodotyronine**
 - Produced from degradation of **thyroglobulin**
 - Act as hormones for growth & development
- **N-methylarginine and N-acetyllysine**
 - found in histone proteins
- **Methylhistidine, -N-methyllysine, & -N,N, N-trimethyllysine**
 - Methylated amino acids in **myosin**

- **γ -Carboxyglutamic acid**
 - In blood clotting proteins & Ca^{2+} containing proteins
E.g. Prothrombin
- **Desmosine**
 - Derivative of four Lys residues
 - Found in fibrous protein elastin
- **Selenocysteine**
 - Derived from serine
 - **Introduced during protein synthesis**
 - Has antioxidant property
- e.g. **Gluthione peroxidase active site**
- **Pyrolysine**
 - Found in methyl-transferase enzyme

❖ **AAs not found in protein**

- **GABA** (decarboxylation of glutamic acid)
 - a potent neurotransmitter
- **Histamine** (decarboxylation of histidine) & **Serotonin** (from tryptophan)
 - function in smooth muscle contraction, as neurotransmitter, vascular permeability, etc
- **β -Alanine**
 - a component of pantothenic acid (vitamin B), CoA & Acyl carrier protein
- **Ornithine & Citrulline**
 - in urea cycle, Arg synthesis
- **Epinephrine**
- **Dopa (3,4-dihydroxyphenylalanine): Precursor of melanin**
- **S-adenosyl methionine (SAM): a methyl donor in transmethylation rxn**

❖ **Acid-base character of AAs**

- At a physiological pH(7.4), the amino group on AAs carries a positive charge, and the carboxylic acid group is negatively charged. The pK_a of the primary carboxylic acid groups for all of the amino acids is around 2. At pH values much lower than the pK_a (higher hydrogen ion concentrations), all of the carboxylic acid groups are protonated. At the pK_a , 50% of the molecules are dissociated into carboxylate anions and protons, and at a pH of 7.4, over 99% of the molecules are dissociated. at the lower pH of 7.4, most of the amino groups are fully protonated and carry a positive charge.
- The form of an amino acid that has both a positive and a negative charge is called a **zwitterion—is neutral**. Because these charged chemical groups can form hydrogen bonds with water molecules, all of these amino acids are water soluble at physiological pH. (Ammonium ion acts as an acid, the carboxylate as a base)
- The histidine side chain is an imidazole ring with a pK_a of about 6 that changes from a predominantly protonated positively charged ring to an uncharged ring at this pH. The amino

group on the α -carbon titrates at a much higher pH (between 9 and 10), and the charge changes from positive to zero as the pH rises.

- The pH at which the net charge on the molecules in solution is zero is called the **isoelectric point (pI)**. At this pH, the molecules will not migrate in an electric field toward either a positive pole (cathode) or a negative pole (anode) because the number of negative charges on each molecule is equal to the number of positive charges. at PI zwitteron concentration is maximum.

PI depends on side chain.

- ✓ Electrophoresis, a technique used to separate proteins on the basis of charge, has been extremely useful in medicine to identify proteins with different amino acid composition
- ✓ The PI can be calculated as the average of pK_{a1} and pK_{a2}
- ✓ PI value is <6 for acidic AA, is approximately 6 for neutral AA and >6 for basic AA
- ✓ Relation of pK_a value with PI can be depicted on titration curve.

❖ **Peptide bond**

- ✓ Condensation rxn
- ✓ Linkage between carboxylic grp of one AA and α -amino grp of the other
- ✓ Formation of dipeptide, tripeptides, etc....
- ✓ Forms N-terminal and C-terminal
- ✓ Adding -yl for the residues except the last residue
- ✓ Acid-base behavior of a peptide can be predicted from its free -amino and -carboxyl groups as well as the nature and number of its ionizable R groups.
- ✓ Have characteristic titration curves and a characteristic isoelectric pH (pI)
- ✓ pK_a value for an ionizable R group can change somewhat when an amino acid becomes a residue in a peptide.

❖ **Planar peptide groups in a polypeptide chain**

- Amide nitrogens are non-basic because their unshared electron pair is delocalized by interaction with the carbonyl group.
- Rotation around C-N bond is restricted due to the double bond nature of the resonance hybrid form
- Peptide groups are therefore planar.

➤ **Peptide Bond Steric Restrictions**

- ✓ The angles of rotations, **phi (ϕ)** & **psi (ψ)** angles can be used to predict and define some higher order protein structures.
- ✓ The **six atoms** composing the peptide bond group tend to be coplanar forming **amide plane**

- ✓ Only **two degrees of freedom** per residue for the peptide chain
- ✓ $\phi=180^\circ$ and $\psi=0^\circ$ are not allowed: forbidden overlap of H of amide nitrogen
- ✓ $\phi = 0^\circ$ and $\psi = 180^\circ$ are forbidden; unfavorable overlap between carbonyl oxygen

Examples of Biologically Active Peptides

- ✓ Tyrotrophine relasing factor. angiotensin, ADH, methionine, enkephaline, little gastrin, glucagon, bradykinin (vasodilator)
- ✓ **Glutathione**
Formed from γ -glutamic acid, Cysteine & Glycine
protects the cell membrane from damage,
e.g., prevents hemolysis of erythrocytes
- **Slightly larger peptides/ Oligopeptides**
 - ▶ **Insulin**
 - which contains two polypeptide chains,
 - ▶ **Corticotropin**
- Commercial peptides:-Aspartame/ L-Aspartyl-L-phenylalanine methyl ester/Neurasweet
Artificial sweetener

Protein

- Structure determines function
 - Native, folded structure of the protein depends on several factors
 - (1) interactions with solvent molecules
 - (2) the pH and ionic composition of the solvent
 - (3) the sequence of the protein
- 1) Primary structure:----**Describes sequence and number of amino acids in chain**
 - Peptide bond Determined by gene (DNA sequence)
 - Contain all information necessary for folding into its “native” structure
 - AA sequence determines protein function for e.g oxytocin and vasopressin differ by a single AA(Leu→Arg), hence very different function.
 - sickle cell anemia arise from a single 6th AA sequence disturbance in β -hemoglobin(Glu→Val)
 - 2) secondry structure:-----local folding
 - Describes about folding pattern along short sections of polypeptide
 - The Development of regular patterns of hydrogen bonding result in distinct folding patterns

❖ α -helix

- Peptide carbonyl H-bonded to peptide N-H group four residues farther
- One turn of the helix represents 3.6 amino acid residues (13 atoms from the O to the H of the H bond - 3.6₁₃ helix)
- All of the H bonds lie parallel to the helix axis
 - α -helix is formed if $\phi \sim -60^\circ$ and $\psi = -45$ to -50° .

. e.g. **Myoglobin**--eight stretches of α -helix forming a **box** to contain the heme prosthetic grp.

❖ Other forms of Helix

- ³₁₀ helix
- - 3.0 residues per turn (with 10 atoms in the ring formed by making the hydrogen bond three residues up the chain)
- ₂⁷ ribbon

❖ β -Pleated Sheet

- Each strip of paper as a single peptide strand
- Peptide backbone makes a zigzag pattern along the strip
- α -carbons lying at the folds of the pleats.
- Formation of inter strand H-bonds
- Side chains oriented perpendicular or normal to the plane of the sheet,

◆ Two types

- **Parallel pleated sheet**
 - Typically large structures
 - Usually > five strands
 - Tend to be more regular
- **Antiparallel pleated sheet**
 - Usually having hydrophobic side chain on one side, so it requires alternating hydrophobic & hydrophilic residues arrangement in primary structure

★ β -turn/bend/tight turn

- **Abundant** in globular structures
- Proline and glycine dominate in the structure

★ There are also Super-secondary structures commonly found in some DNA-binding proteins

- Like Zn finger and Leu zipper.

3) Tertiary structure:-Shows the whole molecule folding pattern

- Describes about overall 3-D structure of a polypeptide
- Determined by interactions between R groups
- Bonds that determine 3D--structure are:
 - Hydrogen bond
 - Disulfide bridges
 - Ionic bonds
 - Hydrophobic interactions

-Van der Waals Force

e.g of domains in tertiary structure include:-bundle, saddle and β -barrel

4) Quaternary structure:- Formed from more than one polypeptide chain

- Organized by the same types of bonds as tertiary structure
- Most intracellular enzymes are oligomers

★ Advantage:-

- **Stability**
 - Favorable reduction of the protein's surface-to-volume ratio
 - Interactions stabilizes the protein energetically
 - Shield hydrophobic residues from solvent water
- **Genetic Economy and Efficiency**
 - Less DNA is required to code for a monomer
- **Bringing Catalytic Sites Together**
 - Monomer may not constitute a complete enzyme active site
 - May also carry out different but related reactions on different subunits
- **Cooperativity**
 - Regulate catalytic activity by subunit interactions
 - Ligand binding at one site changes affinity of the other site

e.g. Hb for O₂

- Some proteins are inactive when oligomers

e.g. Insulin hexamer formation in solution

Some important proteins

➤ **Regulatory Proteins**

- **Regulating other proteins**

Eg. Insulin

- **Regulation of gene expression**

e.g. Transcription activators

➤ **Transport Proteins**

e.g. Hb:

Serum albumin: Fatty acid transport

Membrane transporters (channels): e.g. AA & Glucose transporters

➤ **Storage Proteins:**

e.g. Casein: Major source of P in mammalian infants

Ferritin: for Hb synthesis

➤ **Contractile and Motile Proteins**

- Cell motility, cell division, muscle contraction

e.g. Actin & Myosin:

Tubulin:

Dynein and Kinesin (Motor proteins): drive the movement of vesicles, granules, and organelles along microtubules serving as established cytoskeletal “tracks.”

➤ **Structural Proteins**

- Provide strength & protection to cells & tissues

e.g. α -keratin: hair, horns, and fingernails

Collagen: bone, connective tissue, tendons, cartilage,

Elastin: an important component of ligaments

Collagen and proteoglycans: in ECF that cushion and lubricate

➤ **Scaffold Proteins (Adapter Proteins)**

eg. Can act as a scaffold onto which a set of different proteins assembled

Anchoring (or targeting) proteins: bind other proteins, causing them to associate with other structures in the cell

➤ **Protective proteins**

- active role in cell defense & protection

- Eg. **Immunoglobulins** or antibodies by lymphocytes

- thrombin and fibrinogen:

▶ **Proteins maintaining Osmotic Potential:**

- plasma proteins are responsible for most effective osmotic pressure of the blood this osmotic pressure plays a central role in many processes, e.g., urine formation

Classification

★ **Based on Shape:**

1. Fibrous proteins:- long, string-like, insoluble, high tensile strength

- Triple helix

e.g α -Keratins Collagen,elastin,fibrinogen,myosin

- Serve structural role in cells (e.g. Hair, wool, skin,

e.g nail,tendon,bone,cartilage,teeth,blood vessels)

- **Most common collagen types**

Type I - bones, tendons, and skin – two $\alpha 1$ (I) & one $\alpha 2$ (I)

Type II – Cartilage

Type III - Blood vessels

➤ Tropocollagen: Basic structural unit of collagen , Three intertwined polypeptide chains, Mainly composed of Glycine (one in three) & proline

- Sequence are repeats of a Gly-x-y motif, where

➤ x is frequently Pro

➤ y is frequently Pro or Hyp

- hydroxylation is involved in the process and it requires ascorbic acid.
- 2. Globular proteins: spherical, Usually very soluble in aqueous solution
 - Most soluble cellular proteins
 - All contain substantial amounts of the α -helices and β -sheets
 - e.g. Insulin, Hb, Enzymes, Abs, Carrier proteins
 - can be further classified, Based on type and arrangement of secondary structure
 - Antiparallel α -helix
 - Parallel or mixed β -sheet
 - Antiparallel β -sheet
 - Small metal- and disulfide-rich proteins
- **Membrane proteins:** Hydrophobic side chain towards the membrane
 - Hence, insoluble in aqueous but solubilized by detergents
 - Have fewer hydrophilic amino acids than cytosolic proteins

★ **Based on Nutritional Value**

- Complete or high quality proteins:
 - contain all the necessary essential amino acids; eggs, meat, etc.
- Incomplete or low-quality proteins:
 - deficient in one or more essential amino acids
 - Eg. corn deficient with Lys, Met, Try

★ **Based on composition:** Simple Vs Conjugated

- Simple proteins: only amino acids
- **Conjugated proteins:** amino acids + something else
 - e.g:- glycoprotein
 - lipoprotein(transport of lipid)
- **nucleoprotein(:** for storage & transmission of genetic information)
 - e.g:- Ribosomes, Chromosomes
- **Phosphoproteins:** Esterified with phosphates by ser, thr or tyr
- e.g. **Casein:** bring phosphorous to growing infant
- **Metalloproteins:** - metal storage proteins (e.g. ferritin)
 - metal containing enzymes e.g SOD
- **Hemoproteins**
 - Subclass of metalloproteins
 - Having heme prosthetic grp
- **Flavoproteins** - Containing flavin (vitamin B2 derivative)
 - E.g. oxidoreductases

★ **Protein folding**

- Is formation of a defined spatial structure characteristic for the native protein.
- Native structure represents thermodynamically most favored status
- Several motifs pack together to form **domains**

- Folding can make several conformations/ 3-D structure
 - Usually only one conformation is biologically active
 - Misfolding can result in many diseases such as Alzheimer's and various cancers
- Folding is a very fast process within seconds/milliseconds.
- Catalyzed by specific enzymes such as
 - **Protein disulfide isomerase (PDI)**
 - **Peptide prolyl cis-trans isomerase (PPI)**
- Folding error increases with increase in protein size
- Intermediates of folding process may expose hydrophobic residue and results in protein aggregation
- Incorrect types of bonding (eg. disulfide) may also cause aggregation
- Nascent polypeptides do not contain complete information necessary for folding.
- Crowding of cytosolic components may cause exposed hydrophobic groups to misfold and aggregate
- Cytosolic proteins should fold fast
- Proteins for another compartment should be delayed in unfolded state for translocation through membrane.
- Spontaneous unfolding and loss of native form may occur due to random motion and thermal variation

➤ chaperones

- Guide proper folding
- Also known as heat shock proteins (**Hsps**)
- Their family name is based on molecular weight e.g hsp 40
 - Keep the new protein segregated from cytoplasmic influences
- Minimize heat and stress damage to proteins
- Facilitate correct folding of proteins by minimizing aggregation and other misfolding
- Bind to nascent polypeptides to prevent premature folding
 - Enable refolding into the active 3D structure
- Facilitate membrane translocation/import by preventing folding prior to membrane translocation
- Facilitate assembly/disassembly of multiprotein complexes

❖ Denaturation

- Is a phenomenon that involves transformation of a native form of a protein to an unfolded state under non-physiological conditions.
- Occurs when secondary, tertiary, or quaternary structure is disrupted.
- Leading to changes in its physical and chemical characteristics and loss of biological activity
- Results in non-functional form
- Greater tendency to aggregate

➤ **Chemical denaturants**

- Extreme pH
- Solutes: urea, guanidine hydrochloride or isothiocyanate,
- Detergents: e.g., sodium dodecyl sulfate.
- Organic solvents & detergents act primarily on hydrophobic
- Alkaloidal reagents (trichloroacetic, picric, & phosphotungstic acids)
- Heavy metals, formaldehyde, formamide and glyoxal

➤ **Physical denaturants**

- Vigorous shaking
- Ionizing irradiations
- Heat
 - Heat breaks hydrogen bonds
- Acids/bases protonate/deprotonate key areas, affecting hydrogen bonding and disrupting any ionic bonding
- Organic compounds destroy hydrophobic interactions by forming their own hydrophobic interactions with the protein
- Heavy metals disrupt ionic bonding and disrupt disulfide bonds
 - Shaking stretches polypeptide chains

◆ **Consequences of denaturation include:-**

- Loss of enzymatic activity (death)
- Destruction of toxins
- Improved digestibility
- Decrease in solubility
- Changes in texture
- Increase hydrophobicity

- Greater tendency to aggregate

2) Nucleotides and nucleic acids

◆ Nucleotides

- Energy transfer: energy currency in metabolic transactions
- Catalysis: Structural components of enzyme cofactors
- Signaling within and between cells
- Storage, decoding, and transmission of genetic information
 - Components of nucleic acids

Composed of :-

i) Nitrogenous bases - Heterocyclic rings

Properties include:-Aromaticity of the ring

- Electron rich nature of -OH & -NH₂
- Undergo Keto(lactam) -enol (lactim) tautomeric shift. Lactam form vastly predominates at neutral pH.
- Strong absorbance of UV light(The mutagenic effect of uv light results from its absorption by nucleotides in DNA with accompanying chemical changes.)
- Useful in quantitative and qualitative analysis of nucleotides and nucleic acids
 - Pyrimidines (cytosine, thymine and uracil)
 - Purines(adenine and guanine)-- pyrimidine ring fused with a five membered imidazole ring

❖ derivatives of purine:-xanthine and hypoxanthine(found rarely in nucleic acid)

-uric acid(never found in nucleic acid)

❖ Important functional groups of purine and pyrimidine:-

Amino groups of cytosine, adenine, & guanine

The **ring Nitrogens**: 3N of pyrimidines & 1N of purines

The strongly electronegative **oxygen atoms** attached at

- 4C of uracil and thymine
- 2C of cytosine,
- 6C of guanine

“SALVAGE REACTIONS” CONVERT PURINES & THEIR NUCLEOSIDES TO MONONUCLEOTIDES-this process requires less energy than de novo synthesis of nucleotides

ii) Pentose sugars - heterocyclic

- Ribose (RNA)

- Deoxyribose (**DNA**)

iii) Inorganic Phosphate group

-tribasic acid(has three dissociable protons)

Chemical bonds in nucleotides

I) Nucleosides (pentose sugar + base)

- ❖ Formed by N-glycosidic bond ---bonds are always of the β -configuration
- ❖ More water-soluble than the free bases
- ❖ Nomenclature(adenosine,cytidine,guanosine,uridine)
 - ✓ Inosine-uncommon nucleoside containing hypoxanthine
 - ❖ adding the ending **-idine** for pyrimidine
 - ❖ adding the ending **-osine for** a purine

II) nucleotide

- ❖ Formed by Phosphoester bond between sugar & phosphate:
- ❖ Nucleotide = Nucleoside+ Phosphate

<i>Base</i>	<i>Nucleoside</i>	<i>Nucleotide</i>
Purines		
Adenine	Adenosine	Adenylate
	Deoxyadenosine	Deoxyadenylate
Guanine	Guanosine	Guanylate
	Deoxyguanosine	Deoxyguanylate
Pyrimidines		
Cytosine	Cytidine	Cytidylate
	Deoxycytidine	Deoxycytidylate
Thymine	Thymidine or deoxythymidine	Thymidylate or deoxythymidylate
Uracil	Uridine	Uridylate

❖

★ Cyclic nucleotides

- Found in all cells
- Cyclized by enzymes
- Regulate cellular metabolism

★ Di or tri phosphate nucleotides

- They are polyprotic acids

- Form stable complexes with divalent cations such as Mg^{2+} (mostly because of abundance) and Ca^{2+} .
- Phosphoric anhydride linkages in NDPs and NTPs are readily hydrolyzed by acid, releasing inorganic phosphate
- NTP (nucleotide triphosphate) used as carrier of chemical energy
- NTP are Components of genetic materials
 - Used in energy metabolism and activation of metabolites
 - **ATP** – as energy currency
 - **GTP** – for protein synthesis
 - **CTP** – for phospholipid synthesis
 - **UTP** - Sugar activation in complex carbohydrate synthesis
- **3',5'-cAMP and c-GMP** - Serve as second messengers in signaling
- Acts as a donor of sulfate in sulfation reactions (phosphoadenosine-5-phosphosulfate (PAPS))
- Serves as the major methyl donor in transmethylation rxns - (SAM)
 - **Base pairing (Chargaff's rule)**
- DNA composition varies with species type
- Base composition of DNA in a given species does not change with age, nutritional state, or environmental change
- Bases commonly found in DNA do not occur in equimolar amounts
 - **[pyrimidines] = [purines]**
 - **[A]=[T] and [C]=[G]**
 - **So A-T base pair and C-G base pair---maximizes H bond formation**

III) **Nucleic acids**

- linear polymers of nucleotides
- Formed as 5'-nucleoside monophosphates are successively added to the 3'-OH group of the preceding nucleotide.
- The phosphates are acidic at biological pH and so the polynucleotide is a polyanion.

❖ **The two major classes are:-**

1. **RNA**- Polymers of ribonucleotides

- i. Messenger RNA (mRNA)
 - Carries information encoded in genes to the sites of protein synthesis
 - Synthesized during transcription along DNA single strand
 - Directs the synthesis of polypeptide during its translation
 - Eukaryotic mRNA is synthesized as large precursor molecule (heterogeneous nuclear RNA, or hnRNA)
- ii. Ribosomal RNA (rRNA)
 - Supramolecular complexes where protein is synthesized
 - Fold into secondary structures as a consequence of intramolecular hydrogen bonding
 - Contain a number of specially modified nucleotides, including ribothymidylic acid and methylated bases
 - Their different species can be referred based on their sedimentation coefficient which Roughly measure their relative size
 - Assembled with different types of proteins and form ribosomal subunits
Ribosomes are composed of two subunits
 - rRNAs + proteins = ribosomes
- iii. Transfer RNA (tRNA)
 - Serves as a carrier of amino acid residues for protein synthesis.
 - Fold into a characteristic secondary structure
 - Are the smallest type of RNAs
 - Aminoacyl-tRNAs are the substrates for protein biosynthesis.
 - Each amino acid has its own tRNA (one or more)
- iv. Small nuclear RNA (snRNA)
 - Found in eukaryotic cells mainly in nucleus
 - Are also containing modified nucleotides
 - form small nuclear ribonucleoprotein particles (snRNPs)
 - snRNPs are important in the processing of hnRNA into mature mRNA for export from the nucleus to the cytoplasm

Denaturation

- Solutions of isolated Native DNA is highly viscous at neutral pH and room temprature
- Extreme pH and high temperature may cause physical change
- Heat and extereme pH cause denaturation (melting) of DNA double helix
- Disruption of H-bonding between bases and of base stacking causes unwinding to single strand completely or partly (partial denaturation)
- G-C base pairs are more resitant than A-T base pair during denaturation

- No covalent bond in DNA is broken

2. **DNA – Polymers of deoxyribonucleotides**

- Contain information to make all functional macromolecules
- Found as single chromosome in virus and bacteria
- Found mainly as two copies in the diploid chromosomes
- Found also in mitochondria
- Remarkably flexible molecule and may result in different conformation

❖ **Watson and crick model of DNA**

- Two polynucleotide chains wind around a central axis to form a **right-handed helix**.
- The polynucleotide chains are **antiparallel** to each other.(if one strand begins from 5' end, the will begin from 3' end)
- The nitrogenous bases occupy the **core** of the double helix, while the sugar–phosphate chains are the **backbones** of DNA
- **Hydrogen bonding** between base pairs forming to form a planar base pair (like the rung of a ladder)-form double helix

❖ **Structurally classified as:-**

- **Primary Structure**
 - Describes the neucleotide sequence and its covalent bond
- **Secondary Structure**
 - Any regular, stable structure taken up by some or all of the nucleotides in a nucleic acid
- **Tertiary Structure**
 - The complex folding of large chromosomes within eukaryotic chromatin and bacterial nucleoids occurs in multiple copies and various forms

Group coordinator- Rediet Mekuria

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Eman omer
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